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INTERNATIONAL BANK FOR RECONSTRUCTION AND DEVELOPMENT

PROJECT APPRAISAL DOCUMENT

ON

PROPOSED IBRD PAYMENT GUARANTEES

IN THE AGGREGATE AMOUNT OF UP TO US$12 MILLION

TO THE REPUBLIC OF UZBEKISTAN

FOR A

SCALING SOLAR 2 INDEPENDENT POWER PRODUCER (IPP) PROJECT

FEBRUARY 10, 2023

Energy and Extractives Global Practice
Europe and Central Asia Region

CURRENCY EQUIVALENTS

Exchange Rate Effective January 11, 2023

Currency Unit =	Uzbek Sum (UZS)
UZS 11,305.28 =	US$1

FISCAL YEAR
July 1 - June 30

Regional Vice President:	Anna M. Bjerde
Country Director:	Tatiana Proskuryakova
Senior Global Practice Director:	Charles J. Cormier
Practice Managers:	Sudeshna G. Banerjee, Sebnem E. Madan
Task Team Leader(s):	Ferhat Esen, Maksudjon Safarov, Philip Lam

ABBREVIATIONS AND ACRONYMS

ADB	Asian Development Bank
ADSCR	Average Debt Service Coverage Ratio
AIIB	Asian Infrastructure Investment Bank
ASA	Advisory Services and Analytics
BESS	Battery Energy Storage System
Capex	Capital Expenditure
CCGT	Combined-Cycle Gas Turbine
CCS	Carbon Capture and Storage
CESMP	Constractor ESMP
CHSMP	Community Health and Safety Management Plan
CLO	Community Liaison Officer
COD	Commercial Operations Date
COVID-19	Coronavirus Disease 2019
DFI	Development Finance Institution
DPO	Development Policy Operation
DSCR	Debt Service Coverage Ratio
EBIT	Earnings Before Interest and Tax
EBITDA	Earnings Before Interest, Tax, Depreciation, and Amortization
EBRD	European Bank for Reconstruction and Development
ECCH	Emirates Centre for the Conservation of the Houbara
EHS	Environmental, Health, and Safety
EIA	Environmental Impact Assessment
EIB	European Investment Bank
EIRR	Economic Internal Rate of Return
ENPV	Economic Net Present Value
EPC	Engineering, Procurement, and Construction
EPRP	Emergency Preparedness and Response
ERR	Economic Rate of Return
ESAP	Environmental and Social Action Plan
ESHS	Environmental, Social, Health and Safety
ESIA	Environmental and Social Impact Assessment
ESMAP	Energy Sector Management Assistance Program
ESMP	Environmental and Social Management Plan
ESMS	Environmental and Social Management System
FM	Financial Management
GBV	Gender-Based Violence
GDP	Gross Domestic Product
GHG	Greenhouse Gas
GoU	Government of Uzbekistan
GSA	Government Support Agreement
IFC	International Finance Corporation
IFI	International Financial Institution
IFRS	International Financial Reporting Standards
IMF	International Monetary Fund

IPP	Independent Power Producer
IRR	Internal Rate of Return
ISA	International Standards on Auditing
ISCR	Interest Service Coverage Ratio
L/C	Letter of Credit
LCP	Least-Cost Power
MDB	Multilateral Development Bank
MIIT	Ministry of Investments, Industry and Trade
MIGA	Multilateral Investment Guarantee Agency
MoE	Ministry of Energy
MoEF	Ministry of Economy and Finance
NDC	Nationally Determined Contribution
NDS	National Development Strategy
NEGU	National Electric Grid of Uzbekistan Joint-Stock Company (Formerly "National Power Networks of Uzbekistan Joint-Stock Company")
NPV	Net Present Value
O&M	Operations and Maintenance
PAD	Project Appraisal Document
PDO	Program Development Objective
PLR	Performance Learning Review
PPA	Power Purchase Agreement
PPL	Public Procurement Law
PPP	Public-Private Partnership
PS	Performance Standards
PV	Photovoltaic
RFQ	Request for Qualification
SCP	Shadow Price of Carbon
SEP	Stakeholder Engagement Plan
SOE	State-Owned Enterprise
SPV	Special-Purpose Vehicle
TPP	Thermal Power Plant
UE	UzbekEnergo
UGT	UzGazTrade
UNFCCC	United Nations Framework Convention on Climate Change
UNG	UzbekNefteGaz
UPT	UzPowerTrade

BASIC INFORMATION

Is this a regionally tagged project?	Country (ies)	Lending Instrument
No	Uzbekistan	Investment Project Financing/Guarantee

[ ] Situations of Urgent Need or Assistance/or Capacity Constraints

[ ] Financial Intermediaries

[ ] Series of Projects

Approval Date	Closing Date	Guarantee Expiry Date	Environmental Assessment Category
March 7, 2023	July 1, 2025	September 30, 2045	B
Bank/IFC Collaboration			
Yes			

Proposed Development Objective(s)

The Project Development Objective (PDO) is to increase renewable energy generation through private-sector participation in Uzbekistan.

Components

Component Name	Cost (USD Million)
Construction and operation of two 220-MW solar power plants at each of Jizzakh and Samarkand, and purchase by the sovereign off-taker (NEGU) of electricity from these plants, supported through IBRD project-based payment guarantees	421

Organizations

Borrower: Republic of Uzbekistan

Implementing Agency: National Electric Grid of Uzbekistan (NEGU)

[ ] Counterpart Funding	[ ] IBRD	[ ] IDA Credit/Guarantee [ ] Crisis Response Window [ ] Regional Projects Window	[ ] IDA Grant [ ] Crisis Response Window [ ] Regional Projects Window	[ ] Trust Funds	[ ] Parallel Financing

Total Project Cost: USD 421.00 Million	Total Financing: USD 421.00 Million Of Which Bank Financing (IBRD): USD 12.00 Million	Financing Gap: USD 0.00

Financing (in USD Million)

Financing Source	Amount
Equity	193
USD-denominated Long-Term Debt	216
IBRD Guarantees	12
Total	**421**

Expected Disbursements (in USD Million)

IBRD Guarantee (unfunded)

INSTITUTIONAL DATA

Practice Area (Lead)

Energy & Extractives

Contributing Practice Areas

Climate Change, Public-Private Partnerships

Gender Tag

Does the project plan to undertake any of the following?

a. Analysis to identify Project-relevant gaps between males and females, especially in light of country gaps identified through SCD and CPF

No. WB guarantee operations are not considered for the Gender Tag.

b. Specific action(s) to address the gender gaps identified in (a) and/or to improve women or men's empowerment

No.

c. Include Indicators in results framework to monitor outcomes from actions identified in (b)

No.

SYSTEMATIC OPERATIONS RISK- RATING TOOL (SORT)

Risk Category	Rating
Political and Governance	Moderate
Macroeconomic	Moderate
Sector Strategies and Policies	Substantial
Technical Design of Project or Program	Low
Institutional Capacity for Implementation and Sustainability	Moderate
Fiduciary	Low
Environment and Social	Moderate
Stakeholders	Moderate
Other	Moderate
Overall	Moderate

COMPLIANCE

Policy

Does the project depart from the CPF in content or in other significant respects?

[ ] Yes [X] No

Does the project require any waivers of Bank policies?

[ ] Yes [ ] No

Have these been approved by Bank management?

[ ] Yes [ ] No

Is approval for any policy waiver sought from the Board?

[ ] Yes [ ] No

Safeguard Policies Triggered by the Project Yes X No

Legal Covenants

Guarantee Legal Covenants[1]			
Name	**Recurrent**	**Due Date**	**Frequency**
Usual and customary covenants for project financings of this nature	Yes	n/a	Ongoing

Description of Covenants

Usual and customary covenants for project financings of this nature (See term sheet in Annex 1).

The Project Companies will covenant that they will, among other actions:

a. comply with applicable laws (including environmental and social laws) and applicable environmental and social safeguard requirements, including the World Bank Performance Standards (PS), the World Bank Group (WBG) Environmental, Health, and Safety Guidelines, and other applicable requirements, including as to the use of forced labor;
b. provide annual audited financial statements and other reports to IBRD;
c. provide certain notices and other information to IBRD;
d. provide IBRD access to the Project;
e. not engage in (or authorize or permit any affiliate or any person acting on their behalf to engage in) any Sanctionable Practice[2] in connection with the Project;
f. comply with World Bank requirements relating to Sanctionable Practices regarding individuals or firms included in the WBG list of firms debarred from WBG-financed contracts; and
g. obtain IBRD's consent prior to agreeing to any change to any material Project-related transaction document to which the relevant Project Company is a party that would materially affect the rights or obligations of IBRD under the Guarantee Agreement.

[1] Please note that the legal documents are still under negotiation, which is regular practice in World Bank guarantee operations, even while proceeding with Board presentation of the guarantee.

[2] "Sanctionable Practice" means any fraudulent, coercive, corrupt, collusive, obstructive, or fraudulent practice, as defined in the World Bank's Anti-Corruption Guidelines for World Bank Guarantee and Carbon Finance Transactions.

Conditions

Guarantee Conditions[3]		
Source of Funds	**Name**	**Type**
IBRD Guarantees	Conditions precedent to effectiveness	Usual and customary conditions precedent to effectiveness of guarantees for project financings of this nature

Description of Conditions

Usual and customary conditions precedent to effectiveness of guarantees in support of project financings of this nature, among others:

a. Firm commitment for the financing necessary to complete construction of the Project, including satisfactory contribution of equity;
b. Execution, delivery, and effectiveness of all commercial and financing documents, including the Indemnity Agreements, the Project Agreements, and the Cooperation Agreements, in form and substance satisfactory to IBRD;
c. Payment in full of fees and expenses of IBRD's external counsel;
d. Provision of satisfactory legal opinions;
e. Payment in full of the Initiation Fee, Processing Fee (if invoiced), Front-end Fee, and the first installment/s of the Standby Fee and / or Guarantee Fee (if invoiced); and
f. Satisfactory integrity due diligence of the Project Companies (and related parties) and the guaranteed party.

PROJECT TEAM

World Bank Staff			
Name	**Role**	**Title**	**Unit**
Ferhat Esen	Team Leader (ADM Responsible)	Senior Energy Specialist	IECE1
Maksudjon Safarov	Team Leader	Senior Energy Specialist	IECE1
Philip Lam	Team Leader	Senior Infrastructure Finance Specialist	IPGFG
Manuel Berlengiero	Team Member	Lead Energy Specialist	IECE1
Rafael De Sa Ferreira	Team Member	Senior Energy Economist	IEEGK
Elcin Akcura	Team Member	Senior Energy Economist	IEEGK
Sameena Dost	Team Member	Senior Counsel (Guarantee Lawyer)	LEGAS
Koji Nishida	Team Member	Senior Energy Specialist	IECE1

[3] Please note that the legal documents are still under negotiation as per nature of the World Bank Guarantee products.

Bahodir Amonov	Team Member	Energy Specialist	IECE1
Gaurav D. Joshi	Team Member	Senior Environmental Specialist	SCAEN
Obaidullah Hidayat	Team Member	Senior Environmental Specialist	SCAEN
Gulru Azamova	Team Member	Social Development Specialist	SCASO
Elbek Yusupov	Team Member	Senior Financial Management Specialist	EECG1
Fasliddin Rakhimov	Team Member	Senior Procurement Specialist	EECRU
Hiwote Tadesse	Team Member	Operations Officer	IECE1
Shashank Shanker	Team Member	Infrastructure Finance Analyst	IPGFG
Tolmasbek Boltayev	Team Member	Social Development Specialist	SCASO
Veronika Pak	Team Member	Team Assistant	ECCUZ
Tamar Morchiladze	Team Member	Consultant (Guarantee Lawyer)	LEGAS
Georgiy Egamnazarov	Team Member	Consultant	IECE1
IFC Staff (Advisory)			
Zafar Khashimov	Country Officer	Senior Country Officer	CATUZ
Bernard Atlan	Advisory	Principal Investment Officer	CTAPU
Rustam Shamuradov	Advisory	Investment Officer	CTACA
Shukhrat Khojiyev	Advisory	Consultant	CTACA

UZBEKISTAN

SCALING SOLAR 2 INDEPENDENT POWER PRODUCER (IPP) PROJECT

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I. STRATEGIC CONTEXT

A. Country Context

1. **Uzbekistan's development goal is to become an industrialized upper-middle-income country by 2030**. With more than 34 million people as of 2021, it is the most populous among the Central Asian countries. The Central Asia region is adjacent to some of the largest and rapidly growing economies in the world, which presents an opportunity for the country to become a hub for economic growth, trade and energy exchange. Over the past decade, the country has maintained high and stable economic growth at 5.9 percent on average World Bank projections suggest that the national poverty rate fell from 22.8 percent in 2019 to less than 14.1 percent in 2022. The Government of Uzbekistan (GoU) announced, for the 2017–2021, a plan to radically open and transform Uzbekistan's economy, society, and public institutions and bring the previous model of closed economic developed to an end. The Reform Roadmap outlined the Government's economic reform priorities: (a) maintain macroeconomic stability; (b) accelerate the market transition; (c) strengthen social protection and citizen services; (d) strengthen government's role in a market economy; and (e) preserve environmental sustainability. The reform priorities within each pillar draw on lessons learned from the market transitions of other countries and are firmly based in Uzbekistan's unique context. Following the reelection of Uzbekistan's President for a second term in late 2021, the government announced a new Strategy for the Development of New Uzbekistan for 2022-2026 that will reduce the poverty rate by half by 2026 and enable the country to become reach an upper middle-income level in per capita gross domestic product (GDP) by 2030.

2. **Despite global uncertainties and external challenges, structural reforms and effective economic management in Uzbekistan have helped maintain macroeconomic stability and environment conducive for the next phase of reforms.** Reforms to liberalize trade, exchange rate, domestic prices, and the tax system have supported Uzbekistan's continued growth and the reduction of resource misallocations in the economy. As a result, notwithstanding the coronavirus disease 2019 (COVID-19) pandemic, Uzbekistan has maintained economic growth of 1.9 percent in 2020 and growth rebound to 7.4 percent in 2021 and the country continued reforms. The Russian Federation's economic crisis and the global and regional spillovers from the Russia's invasion of Ukraine have an overall negative impact on Uzbekistan's economic growth. Economic growth in 2022 is projected to moderate to 5.7 percent from 7.4 percent in 2021 and pre-crisis forecasts of 5.9 percent led by unexpectedly strong private money transfers from Russia, exports and investments. On the supply side, stronger growth in construction and agriculture partly offset slower growth in industry and services. The largest impacts in 2023 is expected from a slowdown in private consumption and investment growth due to a drop in remittances and weaker demand for Uzbekistan's textile, food, and machinery exports in Russia.

3. **A new generation of reforms will present more complex challenges** and requires a greater focus on converting high level policy agendas into tangible change for people, including in the face of climate change. The reform path in addition to navigating a difficult economic transition, also must successfully manage the growing risks**.** Uzbekistan has recently demonstrated increased commitment to climate initiatives. For instance, Uzbekistan signed the Paris Agreement and presented its first intended Nationally Determined Contribution (NDC) on April 19, 2017, which became the country's first NDC upon ratifying the Paris Agreement on October 2, 2018. A revised NDC was submitted in October 2021, which sees an

increase in ambition with a target of reducing CO2 emissions per unit of GDP by 35 percent below 2010 levels by 2030 (against the previous target of 10 percent reduction). Continued reforms can raise the prospect of new, green markets for the country that can contribute to improving the international competitiveness of Uzbekistan, while robust and inclusive growth can build resilience to climate change. At present, Uzbekistan's economy is characterized by reliance on natural resource use and minerals extraction and its export basket mostly consists of primary sector products such as natural gas and metals (gold and copper), and only these products make up almost half of the total merchandise exports. Yet these products offer limited scope for productivity growth and even more limited scope for job creation needed to reach higher income levels and poverty reduction. A range of key reforms, including in the areas of competition, energy, digital and physical connectivity, agriculture, trade, land, finance, public sector management, and state-owned enterprises (SOEs) will therefore be important to support both income growth and low carbon growth, providing a sustainable transition path going forward.

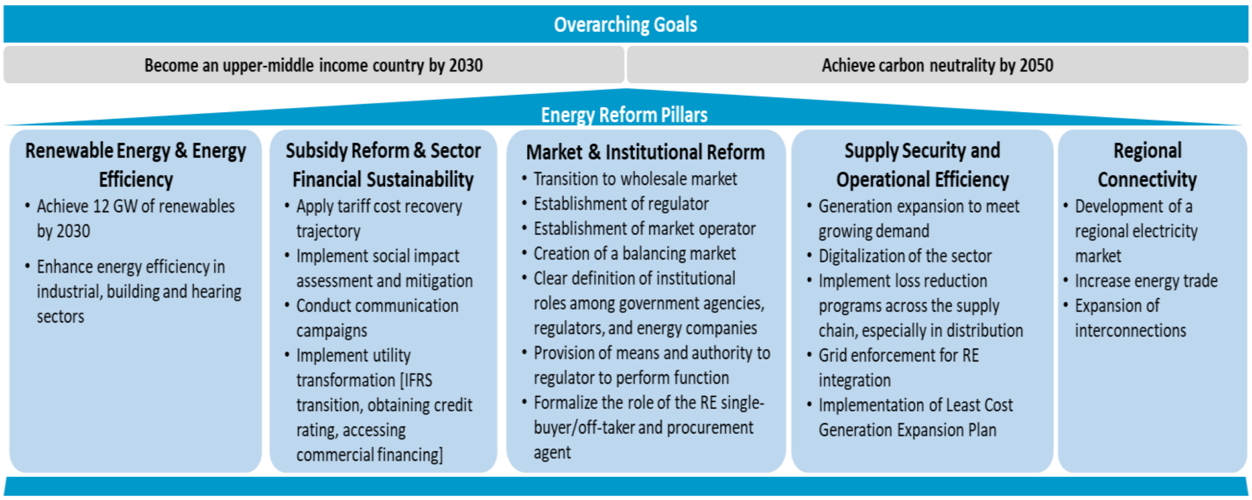
4. **Uzbekistan is vulnerable to climate change.** Over the past century, average annual air temperatures have risen steadily and significantly in Uzbekistan, coupled with more frequent heat waves, droughts, and flooding. As the threat of more extreme weather continues, energy supply will be negatively affected as the changing climate has the potential to cause more damages of power infrastructure and to change river flows, reduce hydro power generation and create large energy shortage during summer, mainly due to heat and drought. Furthermore, Uzbekistan's ageing physical infrastructure makes it more challenging to provide undisrupted power for the fast-growing domestic electricity demand. Without support to adapt and reduce disaster risks, climate change impacts are likely to be unequal, affecting Uzbekistan's poor and marginalized communities most. The adaptation priorities identified in Uzbekistan's National Communication to the United Nations Framework Convention on Climate Change (UNFCCC) include support to the understanding of climate change impacts across key sectors such as agriculture, water resource management, population health, disaster risk reduction, and energy.

B. Sectoral and Institutional Context

Uzbekistan has been pursuing energy sector reforms to enhance efficiency in service delivery to citizens.

5. **The GoU, with support from the Bank, has implemented an ambitious energy sector reform program since 2017** aimed at bringing in market principles, eliminating inefficiencies and promoting private sector participation. Recent reform measures included establishment of the Ministry of Energy (MoE) to consolidate the policy making function; creation of a public-private partnership (PPP) development agency to promote new investments; and unbundling of UzbekEnergo (UE, state-owned electricity company) and UzbekNefteGaz (UNG, state-owned oil and gas company) to strengthen corporatization and commercialization toward competition in the sector, and so on. In this context, National Electric Grid of Uzbekistan (NEGU) joint stock company (JSC) is currently a government-owned central utility responsible for the planning and operation of the power transmission system, as well as the single buyer of electricity in Uzbekistan, including the off-taker of electricity produced by PPP/ independent power producers (IPPs). The company is the operational backbone of a wholesale electricity market to be put in place in Uzbekistan. The Regional Distribution Network JSC (REN) manages and operates the entire electricity distribution business and is in charge of collecting the electricity bill in the country. The electricity access rate is at around 100 percent, and almost all customers are covered with advanced metering infrastructure and consequently collection efficiency is high.

Figure 1. Energy Sector Reforms in Uzbekistan



Note: IFRS = International Financial Reporting Standards.

6. **The next wave of energy sector reforms includes acceleration of solar and wind energy scale-up**, demand side energy efficiency, establishment of a separate energy sector regulator to consolidate all the regulatory functions, unbundling of the power transmission company to separate its transmission and single buyer functions, transitioning towards a competitive market, and carrying out of tariff reforms with the objective to achieve cost recovery by 2025. The reform program aims to accelerate the targeted clean energy transition with private sector participation and contribute to the Government's sustainable development and climate commitments. In this regard, the large-scale renewables, including the proposed Scaling Solar 2 Independent Power Producer (IPP) Project (Scaling Solar 2 IPP) represent an opportunity to further strengthen the security of supply, while sector reforms have now entered a more complex phase, also due to the international context and domestic political concerns, and domestic natural gas supply faces sustainability and shortage issues.

Uzbekistan's energy demand has been growing predominantly served by a largely fossil-fuel based energy system, and recent efforts have focused on a clean energy future.

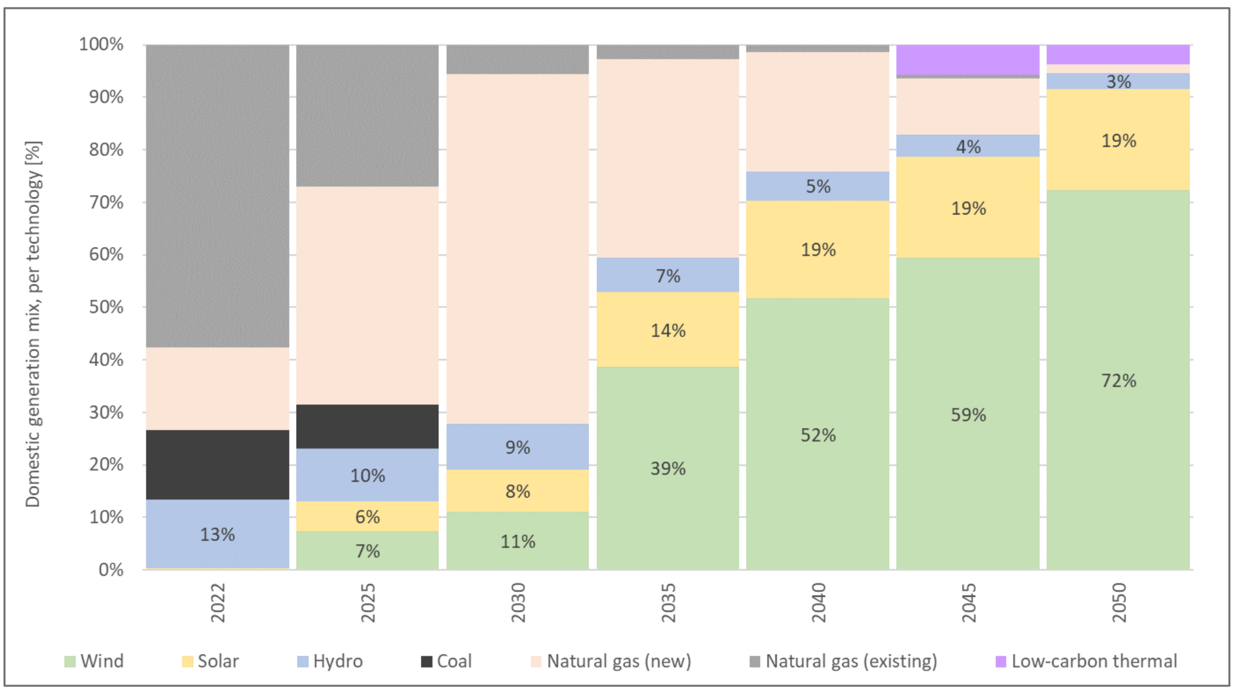
7. **The high level of energy intensity is common for all parts of the economy.** While Uzbekistan's energy intensity declined by about 45 percent over the last 15 years, the country's energy use per unit of GDP is 3.1 times higher than the average for the Europe and Central Asia region. Uzbekistan's total greenhouse gas (GHG) emissions are dominated by the energy sector, which accounted for more than 80 percent of total emissions. This is driven by the fact that the country's energy mix is dominated by fossil fuels, which account for around 90 percent of total power generation in the country (installed capacity of 14.2 GW as of the beginning of 2022). Natural gas accounts for more than 80 percent of the total primary energy consumption as well as of the power mix (followed by hydropower [10 percent] and coal [8 percent]).

8. **The demand for electricity is expected to continue growing steadily** in conjunction with economic growth, development trends, and changes in the structure of the national economy. The demand for electricity in Uzbekistan is primarily driven by industrial and residential customers, with a tight

supply-demand balance. The demand for electricity is expected to almost double to above 130 TWh in 2030, according to the World Bank-supported Least-Cost Generation Expansion Plan (least-cost power [LCP]; base case scenario). In terms of electricity consumption, the industrial sector currently represents the largest customer segment (41 percent), followed by residential (24 percent), agriculture (21 percent), commercial (11 percent), and others. Power generation growth rates from 2012-2019 were recorded at 2.6 percent per year on average. However, the demand for electrical power was not satisfied in full, with shortages averaging at about 9.4 percent of demand.

9. **Renewable energy scale-up is therefore the centerpiece of energy security and power sector carbon-neutrality by 2050, especially following the recent energy crisis in the country.** The GoU, with support from the World Bank, has recently developed an Energy Sector Decarbonization Pathways Assessment Study. This study assessed the least-cost decarbonization pathways for the power system, with the aid of an optimization-based expansion plan for a decarbonization scenario. In this context, a significant expansion of renewables capacity between 2022 and 2050 is part of the least-cost pathway for power sector decarbonization in Uzbekistan. Renewables, including hydropower, are expected to account for 95 percent of the electricity generation in 2050, in the Decarbonization 2050. Batteries, hydropower, and efficient thermal power plants, including plants with carbon capture and storage (CCS) and running on hydrogen in later years of the horizon, support the scaling-up of renewables from 2022 to 2050.

Figure 2: Power Generation Mix under Decarbonization Scenario 2050



Source: Energy Sector Decarbonization Pathways Assesment Study, Decarbonization 2050 scenario

10. **In addition to scaling up renewable energy, other measures will be required to achieve carbon neutrality in the power sector by 2050.** The GoU policy priorities include: (a) replacing inefficient old coal and gas equipment with low-carbon and gas-efficient modern technologies, while continuing large-scale deployment of clean energy resources; (b) reducing the carbon footprint (gas flaring and venting) of the gas value chain, which today accounts for about 30 percent of losses in the natural gas sector; (c)

deploying gas as a feedstock for hydrogen production in conjunction with CCS (blue hydrogen) to kick-start the commercial production of low-/zero-carbon fuels while transitioning to green hydrogen in the long run; and (d) leveraging private sector knowledge, technology, and financing in power and gas generation (exploration and production) and distribution segments. These actions will collectively result in diversification of the energy mix, while reducing the reliance on coal and inefficient gas, and subsequently towards reduced GHG emissions in the sector and the country, contributing to Uzbekistan's NDC. In parallel, substantial investments will also be required to strengthen the electricity grid system. On average, two-thirds of the transmission and distribution assets are beyond their useful lifetime. Uzbekistan's energy system is characterized by high losses and low reliability of supply, with transmission and distribution losses estimated at around 20 percent of net generation. This level is more than twice as high as electricity losses in high-income and some middle-income countries and in its current condition the energy system cannot sustain the planned high penetration of renewable energy.

11. **The government is actively seeking sustainable energy transition pathways also due to depleting natural gas resources.** Insufficient capital investment into new gas exploration and production and network modernization poses challenges to gas supply relying on domestic resources alone, though the country already considers regular gas imports from Turkmenistan. The price of natural gas for domestic consumption is kept at around 50 percent of the prevailing rate for Uzbekistan's regional gas exportation. The export price parity implies implicit subsidies to domestic gas consumers, including the electricity sector. The economic opportunity cost of gas with limited reserves is among the key drivers of the large-scale deployment of renewable energy generation and replacement of old inefficient gas-fired units, which will reduce demand pressure and renationalize gas consumption for power as part of the clean energy transition. In this context, the GoU plans to commission around 7.9 GW of new combined-cycle gas turbine (CCGT) projects by 2030, including the Syrdarya 2 CCGT IPP project (greenfield 1.6 GW), which has been competitively procured under the International Finance Corporation (IFC) Advisory support, with commercial close reached in April 2022.

12. **The GoU is also keen to develop the country's hydropower potential** to integrate higher volumes of intermittent generation from wind and solar and lower cost of electricity in the system. There are 49 operational hydropower plants (HPP) in Uzbekistan, including 11 large sized HPPs (>30 MW) with a total capacity of 1.82 GW (88.9 percent of the total capacity of HPPs). The utilization factor of the hydropower facilities of the country, whose effective generation output is mostly constrained by hydraulic constraints related to the use of water for agricultural purposes, remained comparatively low at only 27 percent[4]. According to the current GoU plans, the targeted hydro generation capacity is 3,4GW by 2030, in order to maintain 10 percent of current power mix for hydropower in future.

13. **Energy efficiency is part of the ambitious sector reforms in Uzbekistan.** The GoU declared its commitment to improve energy efficiency of the overall economy, including reduction of GDP energy intensity by 50 percent by 2030 (with 2015 as the baseline year), twofold increase in energy efficiency and decrease in carbon intensity per unit of GDP by 10 percent from the level of 2010. The commitments are stipulated in key strategy documents such as the Green Economy Strategy (October 2019) and the Low Carbon Energy Strategy (May 2020) further strengthened by laws and resolutions. In addition to the efforts on clean and efficient power generation, reliable transmission and distribution, and market liberalization, the GoU has been promoting the energy efficiency measures in the industrial, residential, public and social services and transport sectors to reduce energy consumption without lowering the

[4] Data from International Energy Agency Uzbekistan energy profile.

quality of service provided and result in lower GHG emission to contribute toward the decarbonization of the economy. In this context, among international financial institutions (IFIs), the World Bank has been playing a leading role through three major energy efficiency investments in the industrial, buildings and district heating segments with a total amount of US$483 million[5].

14. **The current GoU's renewable energy target includes development of solar (7,000MW) and wind (5,000MW) capacities by 2030**, in which the WBG is playing a significant role.

Table 1. GoU Renewable Energy Development Program *(competitive basis)*

Institution	Project	Capacity, MW	Status
WBG	Navoi Scaling Solar 1	100	Operational
WBG	Scaling Solar 2 (Samarkand, Jizzakh)	440	Financial close in 1Q 2023
WBG	Uzbek Solar 3 (Bukhara, Fergana and Khorezm) with a pilot BESS[6] component	500 + 62 (BESS)	Financial close in 2023
WBG	Uzbek Solar 4	1,000	Bidding in 2023
WBG	Scaling Wind	500	Bidding in 2023-2024
ADB	Solar Power Assignment	1,000	Financial close, Bidding and Preparation Stages
EBRD	Wind Power Assignment	1,000	Financial close, Bidding and Preparation Stages
	Total	**4,602**	

Sector financial sustainability presents a challenge towards renewable energy scale-up, but the Government has made strides in tariff reforms and creating enabling environment for private sector.

15. **NEGU has faced a challenging financial situation over the last two years** and failed to recover its operational costs in 2021, driven partially by impacts of the COVID-19 pandemic. In the absence of a regulated cost-plus tariff regime for transmission business, NEGU's revenues did not rise enough to offset the large increases in operating costs in 2021. The planned tariff increases were delayed due to the COVID-19 pandemic, and NEGU's cost of purchased electricity grew by around 19 percent, resulting in an earnings before interest, taxes, depreciation, and amortization (EBITDA) (operating) margin of -8 percent in 2021. Over the recent years, NEGU has received additional liquidity through on-lent loans from international finance institutions (IFIs), shareholder loans from the Ministry of Finance and Economy (MoEF), and accumulation of payables to state-owned generation companies. The financial support provided to NEGU demonstrates the critical role it plays in the country's energy sector, as well as the managerial linkages and strategic importance with the GoU.

[5] Uzbekistan Energy Efficiency Facility for Industrial Enterprises Project, Phase 3 (UZEEF, P165054; District Heating Energy Efficiency Project (DHEEP, P146206) and Clean Energy for Buildings in Uzbekistan Project (CEBU, P176060).
[6] BESS = Battery Energy Storage System

16. **The recent and planned developments in the country's energy sector are also expected to improve NEGU's financial standing**. The GoU has guaranteed all ongoing liabilities, including payments to IPPs, and debt obligations of NEGU. These developments also include some tariff increases amongst select non-residential customers in August 2022 that also positively affected transmission tariffs. Tariffs are expected to continue to rise, with cost recovery levels expected to be reached by 2025, while protecting the poor. The planned sector reforms and tariff initiatives will also be supported through the World Bank-financed pipeline "Innovative Climate and Carbon Finance For Energy Reforms Project" along with the subsequent Development Policy Operation (DPO) engagement. With this expected increase, NEGU's financial projections estimate a gradual improvement in operational profitability and financial sustainability in the medium term. Other initiatives such as smart metering will help maintain high bill collection efficiency rates (100 percent expected in 2022, further improved through advanced metering). As end-consumer tariffs become more cost reflective, NEGU's payables to state-owned generation entities are expected to come down in line with enhanced revenue collections. Lastly, the implementation of IPPs, most of which have a lower cost than Uzbekistan's current inefficient generation fleet, is also expected to improve NEGU financials.

17. **The undergoing sector reforms are expected to create the adequate enabling environment to foster private sector participation.** The GoU is implementing its ambitious clean energy transition agenda driven by private sector investment and technology. It has established an institutional framework for the development and management of PPPs given substantial investment needs in the electricity value chain and limited public finances. Various PPP units have also been set up within sector ministries such as energy and transport. The PPP Law enacted in June 2019 aims to provide a consolidated legal framework for PPP investments to reduce risks and increase clarity for investors. The successful implementation of PPP/IPP solar power transactions would hence help free up scarce public financial resources for other GoU priorities.

18. **While the private sector has enthusiastically embraced Uzbekistan, its participation is through both competitively tendered and bilateral deals.** In 2020-2021, the GoU signed approximately 6GW of bilaterally negotiated projects in the power generation sector with associated challenges such as pricing, risk allocation, and alignment with sector plans, to be mitigated through competitive processes and screening unsolicited proposals to avoid project and reputational risks and fiscal impact. The World Bank has been supporting the GoU/ Ministry of Economy and Finance (MoEF) through a Fiscal Impact Assessment of Energy PPP/IPPs using the analysis tool (PFRAM)[7] to better monitor risks and develop necessary mitigation. The World Bank's technical assistance project - Advancing Sustainable Uzbek PPP Agenda and Investments is also mobilized to provide the necessary advanced capacity building and technical skills, including PFRAM model update, to assess the PPPs and related risks.

The proposed guarantee continues the World Bank support towards competitive and affordable addition of renewable energy capacity, underpinned by a robust program of projects and analytics.

19. **The Scaling Solar 2 IPP project will benefit from the WBG support, including IFC Advisory Services and IBRD guarantees** as outlined in the Table 1 describing the WBG support to the GoU Renewable Energy Development Program (Page 14). Similar to the currently operational Navoi Scaling Solar (1) IPP (100MW with bid tariff of US¢ 2.67 per kWh) supported by the World Bank technical

7 The Public-Private Partnerships Fiscal Risk Assessment Model (PFRAM) developed by the International Monetary Fund and the WBG, is an analytical tool to assess fiscal costs and risks arising from public-private partnership (PPP) projects.

assistance and payment guarantee and IFC advisory and investment, the GoU requested the WBG products to support the project preparation and tender process, as well as to support NEGU's offtake obligations to the private sector. The WBG instruments increase the attractiveness of the renewable IPP investment opportunity in the country, which in turn helps mobilize quality international investors, resulting in increased competition and lower tariffs. Notably, the project tariffs for Jizzakh and Samarkand solar power plants are the lowest solar tariffs in the country and the broader Central Asian region. As the first major solar energy scale-up, the project will have a strong demonstration effect for future PPP/IPPs in Uzbekistan and its region.

20. **The project builds on the World Bank's energy program in Uzbekistan** by scaling-up of the private investment and commercial financing, diversification of power mix from domestic resources (solar), clean energy transition and decarbonization. The GoU requested the WBG to lead key themes for energy sector reform, including the institutional and structural reforms and renewable energy development, which are crucial for the security of supply, supporting the economic growth and well-being of Uzbekistan citizens, and are also an essential part of ensuring business continuity in the economy. In this context, the project contributes to the GoU's initiatives on clean energy transition, demonstration of benefits of competitive and transparent processes, leveraging private and commercial financing to meet the sector's significant investment needs, and institutional capacity building for designing and implementation of PPP projects, including during the post-pandemic economic recovery period. The GoU is also supported by the ongoing World Bank's ongoing Energy Strategy Programmatic Technical Assistance Program and investment project interventions such as the WBG Scaling Solar Initiative (Navoi IPP – Scaling Solar 1 IPP, operational; and Scaling Solar 2 (Project) and 3 requested by the GoU), Modernization and Upgrade of Transmission Substations (MUTS) and Electricity Sector Transformation and Resilient Transmission (ESTART) Project, and pipeline Innovative Climate and Carbon Finance for Energy Reform (ICCFER) to further accelerate the energy reforms, facilitate clean energy transition, diversify power mix with solar photovoltaics (PVs), improve the transmission grid reliability, facilitate integration of large-scale renewable/solar capacities and digitalize operations, reduce loss and strengthen regional connectivity and trade.

C. Higher Level Objectives to which the Project Contributes

21. **The proposed project is consistent with the new Country Partnership Framework (CPF)**[8] **for Uzbekistan (FY2022–FY2026) approved on May 24, 2022**. The CPF is focused on 10 objectives that will underpin three high-level objectives (HLOs) necessary to accelerate Uzbekistan's transition toward an inclusive and sustainable market economy and achieve its 2030 aspirations, which are to increase private sector employment (HLO1), improve human capital (HLO2), and improve livelihoods and resilience through greener growth (HLO3). The proposed project contributes to Objectives 1.1 (Expand competitive access to market), 1.2 (Enable private sector growth and investment), and 1.4 (Improve the infrastructure for competitiveness and connectivity) under HLO1, and Objectives 3.1 (Decarbonization and the greener development of industry and the economy), and 3.2 (More efficient use of natural resources under HLO3).

22. **Among the four development pathways described as priorities of the Systematic Country Diagnostic (SCD)**[9] **dated April 2022,** the proposed project covers (a) strong private sector response to

[8] Uzbekistan - Country Partnership Framework for the Period FY2022-FY2026, May 24, 2022
[9] Uzbekistan – Second Systematic Country Diagnostic (SCD) "Toward a Prosperous and Inclusive Future: The Second Systematic Country Diagnostic for Uzbekistan", April 2022.

encourage the emergence of a vibrant private sector as essential for the economic growth and employment ambitions of the Government and stakeholders; more room for private sector market entry and competition is needed; and (b) sustainable and resilient future requiring greener growth. Much of Uzbekistan's economic potential is underrealized because of the inefficient use of natural resources. There is wastage in use of water and energy, and the neglect of land management threatens livelihoods and future sources of growth. A focus on green growth can propel Uzbekistan toward a more sustainable future while creating new jobs in emerging high-skilled fields. Reforms are incomplete without a clear strategy to support such a sustainable and resilient structural transformation, including during the post-pandemic economic recovery period and regional conflicts in line with the WBG Global Crisis Response Framework, within which the project supports the Uzbekistan economy and energy sector through clean, diversified and sustainable electricity supply under Pillars 2, 3 and 4 as protecting people and preserving jobs; strengthening resilience; and strengthening policies, institutions and investments for rebuilding better, respectively.

23. **The proposed project will contribute to the World Bank Climate Change Action Plan** commitment to increase climate financing to 35 percent of total financing and the GoU's updated NDC target. The GoU has set the mitigation goal to reduce emission intensity by 35 percent in 2030 against the 2010 baseline. Deployment of renewable energy will be among the key drivers facilitating the climate change mitigation targets. Furthermore, the project will leverage private and commercial financing to meet huge financing needs for the clean energy transition.

II. PROJECT DEVELOPMENT OBJECTIVES

A. PDO

24. **The Project Development Objective (**PDO) is to increase renewable energy generation through private sector participation in Uzbekistan.

B. Project Beneficiaries

25. **The direct project beneficiaries are** (a) NEGU as the project off-taker, and (b) Nur Jizzakh and Nur Samarkand IPPs, special-purpose vehicles (SPVs) established by the IPP investor– Masdar (the Sponsor), as the Project Companies.

26. **NEGU** was established in June 2019 as a JSC with 14 regional transmission branches and a number of engineering, construction, and social subsidiaries and units. The sole shareholder of NEGU is the State Asset Management Agency (SAMA). The NEGU will benefit from the proposed IBRD payment guarantees under the Scaling Solar 2 IPP Project (a) for its payment obligations under the power purchase agreements (PPAs) signed with Masdar/Project Companies to cover the risk of NEGU non-payment of three months' equivalent of energy payments; and (b) through lower bid prices facilitated by the proposed WBG support which includes the IBRD guarantees.

27. **Masdar is a global developer in renewable energy** and is owned by Mubadala Investment Company PJSC, Abu Dhabi National Energy Company (TAQA) and Abu Dhabi National Oil Company (ADNOC), which are all strategically important companies of the Government of Abu Dhabi. Masdar was formed in 2006 to promote renewable energy and sustainable urban development. Masdar is active in

over 40 countries and has investments in clean energy projects with a combined value of US$20 billion and a gross capacity of over 15GW installed or under development. Masdar was announced as the winning bidder through the competitive tender, and as the project sponsor, it requested the World Bank payment guarantees as part of its bid submission.

28. **The ultimate project beneficiaries will be electricity consumers in Uzbekistan**, including households and private and public sector consumers, since the Scaling Solar 2 IPP project will contribute to diversifying the power mix and increase cleaner and sustainable power supply in Uzbekistan.[10] Additional power generated by the Scaling Solar 2 IPP will contribute to productivity and spur economic growth by assisting with conversion of domestic solar resources into improvements in the power supply, and with job creation, reducing poverty, and improving the prospects for shared prosperity.

C. PDO-Level Results Indicators

29. **The proposed PDO indicators** for the Scaling Solar 2 IPP Project are:

(a) Power generation capacity constructed (renewable/solar, MW);

(b) Electricity supplied by the generation project into the grid (renewable/solar, TWh);

(c) Private capital mobilized (equity/debt, US$); and

(d) Greenhouse gas emissions avoided (tCO_2/year).

The project's intermediate indicators are:

(a) Physical implementation progress in generation project constructed (percentage); and

(b) Project commissioning test completed (Y/N).

30. **The proposed PDO and indicators in this PAD relate to the Scaling Solar 2 IPP.** Similar PDO and intermediate indicators were used for the first project of the guarantee program (Navoi Scaling Solar 1 IPP, 100MW, P170598) and would also be developed for the subsequent projects being currently supported by IFC advisory services and considered for the World Bank guarantees such as Uzbek Solar 3 and 4 and Scaling Wind IPPs (additional 2,062MW).

Figure 3. Theory of Change

[10] The Project is estimated to produce clean renewables energy, enough to supply approximately 140,000 households.

Challenges	Project Activities	Project Outputs	Project Outcomes	Long term Outcomes
- Security of supply - Energy generation and supply inefficiencies - Energy sector decarbonization - Dependency on natural gas/fossil fuels - Demand growth - Operational and financial sustainability - Lack of adequate investment and private sector involvement	- GoU/NEGU securing competitively procured generation capacity - Clean and renewable/solar energy supply - Long-term financing mobilized by private developer with help of IBRD Guarantee	- Power generation capacity constructed - Electricity generated supplied by the Project into the grid - Private capital mobilized - Greenhouse gas emissions avoided	- Sector reform implementation - Diversification of the electricity supply - Electricity generation through private investment - Strengthening of private sector participation - First major renewable/solar capacity scale-up	- Clean energy transition and decarbonization - Support to reach national sector targets by 2030 - Strengthening of security of supply and reducing equivalent gas consumption

III. PROJECT DESCRIPTION

A. Project Components

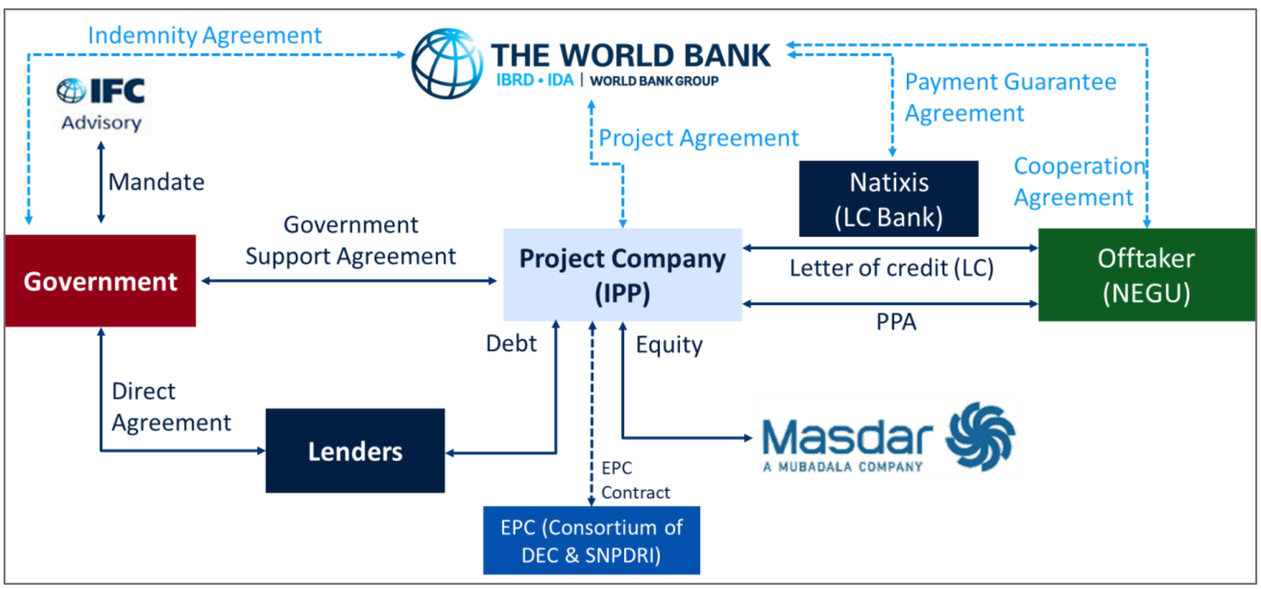
31. **The Scaling Solar 2 IPP Project presented for the World Bank Board's approval through this PAD has one component (supported by two IBRD payment guarantees) comprising the implementation of two separate solar PV plants,** each with a capacity of 220 MW in the Samarkand and Jizzakh Regions (Scaling Solar 2 IPP for a total of 440 MW), by two SPVs (the Project Companies), which will carry out the development, design, financing, construction, ownership, and operations, and maintenance (O&M) of the plants. The tender results for the solar PV projects were announced in May 2021. Overall, six and seven bids were received for the Samarkand and Jizzakh solar PV projects, respectively. Masdar, 100-percent-owner of the Project Companies and a leading United Arab Emirates-based sovereign-owned renewable-energy development company specializing in the development, financing, construction, and operation of renewable-energy projects[11] won both bids, with a tariff bid of US¢1.823 per kWh for Jizzakh and US¢ 1.791 per kWh for Samarkand, both under 25-year PPA terms. Due to disruptions and cost increases in the global solar market and logistics industry and in response to the sponsor's request, NEGU and Masdar amended the PPA tariffs and implementation periods in July 2022, helping to ensure project delivery, while maintaining positive value for money. The amended tariffs are still amongst the lowest in the region, and also compare favorably to the recently announced (December 2022) competitive tender results for Uzbek Solar 3, whereby the lowest received solar tariff was US¢2.88 per kWh. The electricity generated from the project will be sold to NEGU, the state off-taker, for the duration of the PPA. The Scaling Solar 2 IPP are expected to reach financial close in March 2023, following deadline extensions as a result of

[11] Since its establishment in 2006, Masdar has financed renewable-energy projects valued at more than US$20 billion and with a gross capacity of over 15 GW installed or under development.

COVID-19-related supply-chain disruptions and logistics uncertainties. The solar PV plants to be supported by the IBRD guarantees will increase the installed power-generation capacity by 440 MW, supplying an incremental 1.1 TWh per year of renewable electricity to the grid, avoiding GHG emissions of 109,306 metric tons per year on average, and mobilizing about US$408 million of commercial financing.

32. **The proposed project is the Scaling Solar 2 IPP Project developed under the WBG Scaling Solar Program.** The WBG Scaling Solar Program brings together a suite of World Bank, IFC, and Multilateral Investment Guarantee Agency (MIGA) services and instruments under a single engagement aimed at creating viable markets for grid-connected solar PV power plants. It is an open, competitive, and transparent approach that facilitates the rapid development of privately owned, utility-scale solar PV projects, previously in Sub-Saharan Africa and now in Uzbekistan. It is capable of rapid implementation and offers a 'one-stop-shop' package of advisory services, contracts, financing, guarantees, and political risk insurance. This enables governments and utilities to procure solar power transparently and at the lowest possible cost. The program was designed to ease replicability in similar countries while considering local specificities (for instance, site availability).

Figure 4. Summary Contractual Structure for the Scaling Solar 2 IPP Project



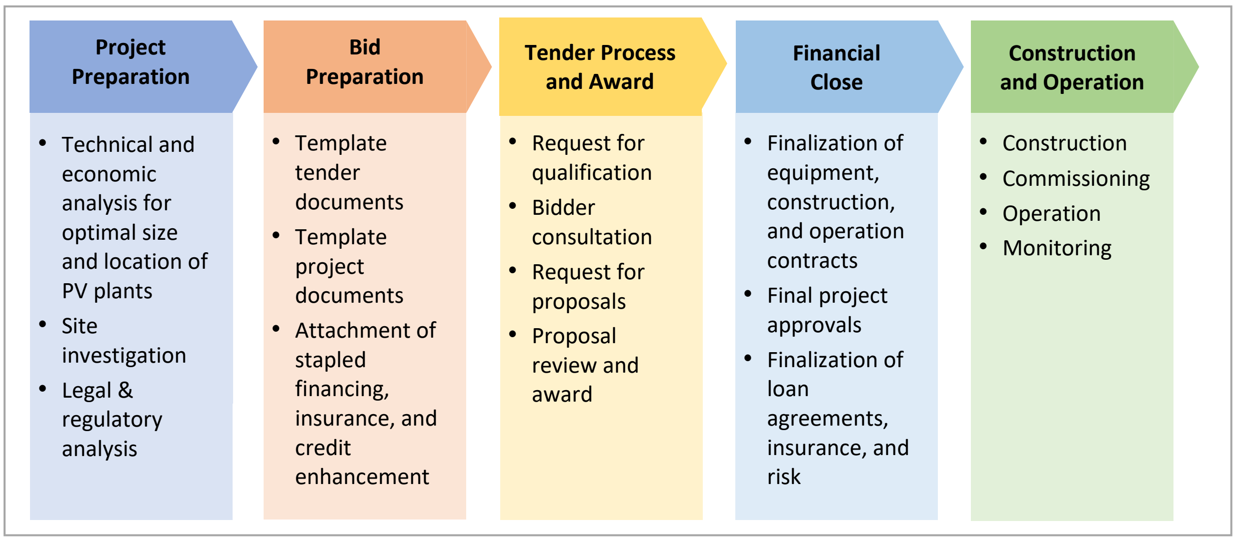
Note: EPC = Engineering, procurement, and construction; L/C = Letter of Credit

33. **As part of the WBG Scaling Solar Program's original design, IFC Advisory Services (IFC Advisory) supports governments in preparing a competitive and transparent solar tender** based on template documents and processes. Based on the bid package, IFC Investment Services (IFC Investment), World Bank, and MIGA then provide term sheets for financing, guarantees, and political risk insurance, respectively. Bidders can decide to use none, a combination, or all of these WBG instruments. Bidders who pass the technical and financial criteria are then ranked based on the offered tariff.

34. **The objective of the WBG Scaling Solar Program is to target markets with high perceived risk for the private sector, playing a catalytic role in relatively more challenging markets.** A typical Scaling Solar market would have the following characteristics: (a) a single-buyer electricity supply industry structure; (b) low credit quality off-takers; (c) governments with limited institutional capacity; and (d) nonexistent,

limited, or poor track record with IPPs. Solar PV technology for large-scale power generation solutions is now cost-competitive with other power-generation solutions and has the added advantage of swift rollout and a relatively long life. In the challenging markets that the program targets, the contractual framework is designed to address the single-buyer context with appropriate risk allocation and appropriate government support to backstop credit risks. Standardized documentation makes it easier for governments to adopt the project framework and achieve speedy implementation. Figure 5 shows the key steps for a scaling solar project.

Figure 5. Key Typical Steps for Scaling Solar Project



IBRD Payment Guarantees

35. **All WBG instruments are optional for investors under the Scaling Solar Program** and are designed to help mitigate risks perceived by potential long-term investors, foster higher participation in the bidding process, and improve the quality of bids and enhance competition and are ultimately expected to have a positive impact through more competitive and affordable electricity generation costs and electricity tariffs. This approach is in line with market precedents and World Bank experience in mobilizing commercial capital in similar circumstances where long-term investors (both international and domestic) would ask that government-related risks (including payment-default risk by a public off-taker or other public authorities) be mitigated, or otherwise addressed, with WBG support. In this case, the project will be supported by IBRD payment guarantees of up to US$6 million for each 220-MW project. This amount represents about three months of expected revenues, which is a notable improvement from the first Navoi project (in which the guarantee covered about six months), signifying a sector that has further developed. Masdar has not requested further WBG instruments to develop the project.

36. **The payment guarantee represents an effective instrument to support the early stages of Uzbekistan's IPP program.** Past international experience has showed that the WBG Scaling Solar approach and the World Bank payment guarantee are effective and efficient tools to kick-start renewable-energy development as they mobilize much-needed capital, technology, and management expertise from the private sector into frontier markets. Guarantees are particularly relevant in countries undertaking market reforms and that have a high degree of macro, regulatory, and institutional uncertainty. As the Uzbekistan

energy sector moves forward with its sectoral reform agenda, with regulations, institutional capacity, and payment track records that are not yet fully tested, the GoU and IBRD are considering the deployment of IBRD guarantees that are critical to mitigate (perceived) risks and to signal to international renewable-energy private investors the government's commitment to stabilizing and implementing further reforms. The principal risk factor of a call on the IBRD Guarantees is NEGU's ability to make timely payments to the IPPs, which is driven by NEGU and the overall power market's financial sustainability.

37. **Masdar requested payment guarantees as part of its bidding conditions.** This was mainly due to the limited track record of the GoU, NEGU and key government agencies handling IPPs, as well as ongoing energy-market reforms. The guarantees comprise an up-to-20-year L/C-based structure (counted from actual commercial operation date [COD] of the projects) committed upfront to mitigate the PPA payment risks, which is critical to the bankability of the transaction. The guarantees provided confidence to bidders and lenders to participate and provide a good opportunity for the GoU and the World Bank to continue assessing key risks preventing private investment on commercial terms and to devise appropriate action to improve the sustainability of the proposed project and any future IPP projects.

38. **The successful implementation of this transaction will be critical to the success of both the power-sector reform and the PPP agenda in Uzbekistan** as it will confirm the viability of the financial, transactional, and regulatory systems put in place under the reform program. The project will contribute to a low-cost base for new generation capacity while the GoU concurrently implements major sectoral reforms. In the longer term, as Uzbekistan's power-sector reforms progress, it is expected that the need for risk mitigation and transaction costs would decrease as electricity utilities (particularly NEGU) establish a track record of successful financial and operational performance, and the overall policy, institutional, and regulatory environment improves in line with progress in the implementation of power-sector reforms in Uzbekistan.

39. **The proposed IBRD payment guarantees backstop certain payment obligations undertaken by the PPA off-taker (NEGU)**. Under the Scaling Solar 2 IPP PPAs, NEGU will provide payment security for each project in the form of an L/C, issued through a commercial bank in favor of the relevant Project Company for the amount corresponding to three peak-monthly PPA payment obligations of the off-taker. Following a competitive procurement, NEGU selected Natixis (the corporate, investment, and financial services arm of Group BPCE, the second-largest banking group in France) as the L/C bank. The L/Cs may be drawn in the event NEGU fails to make timely PPA payments to the relevant Project Company, subject to certain grace periods. Following a draw under the L/Cs, NEGU/GoU[12] would be obligated under the Reimbursement and Credit Agreement (to be entered into between NEGU and the L/C bank) to repay the L/C bank any amounts drawn under the L/Cs (plus accrued interest) within a year. If NEGU/GoU repays such amount within that agreed period, the L/Cs would be reinstated to the amounts repaid. However, if NEGU/GoU fails to repay the L/C bank within this period, the L/C bank would have recourse to the relevant IBRD guarantee for the drawn amounts (plus any accrued interest) under the Guarantee Agreement (to be entered into between IBRD and the L/C bank). In case of such a call on the guarantee, the maximum L/C amount and the corresponding IBRD guarantee would be reduced by the amount of payment made by IBRD to the L/C bank under the guarantee. See Annex 1 (IBRD Payment Guarantee Term Sheet) for additional details.

[12] The GoU supports the PPA payment by committing separately through the GSA to fully replenish the L/C balance if the L/C is drawn by the Project Company within 44 days of receiving notice of such draw event.

40. **The proposed IBRD payment guarantees will be priced according to the approved IBRD pricing policy for guarantees (fee structure and levels)**. For the proposed Scaling Solar 2 IPP Project, IBRD's pricing applies and includes a guarantee fee ranging between 50 and 100 basis points per year depending on the guarantee average life, calculated on the highest annual guaranteed amount, and payable every six months in advance. In addition, there would be a standby fee (25 basis points), front-end fee (25 basis points on the maximum exposure of the guarantee), initiation fee (15 basis points on the maximum exposure of the guarantee or US$100,000, whichever is higher) and a processing fee (usually 50 basis points of the guaranteed amount) as well as the reimbursable expenses of each transaction. All such guarantee-related fees will be paid by the transaction Sponsor / Project Companies.

B. Project Cost and Financing

41. **The amount of equity to be provided to the Project Companies by the Sponsor is expected to be approximately US$193 million.** A total debt financing package of US$216 million will be provided by ADB, EBRD, EIB and AIIB. The table below reflects indicative project costs, the financing structure, and the IBRD guarantees for the project.

Table 2. Indicative Project Costs, Financing Structure, and IBRD Guarantees (US$ million)

	Jizzakh	Samarkand
Solar Power Plant Size	220 MW	220 MW
Estimated Project Costs incl. contingency*	206	202
Estimated Equity	96	96
Estimated Debt	110	106
Estimated Private Capital Mobilized	102	103
Estimated Payment Guarantees	6	6

* NOTE: Excludes IBRD payment guarantee of US$ 6 million for each project

C. Lessons Learned and Reflected in the Project Design

42. The key lessons learned from the WBG's energy-sector operations and project-finance (limited-recourse) transactions that are relevant to this project include the following:

a. **A comprehensive power-sector reform program must be initiated in advance of major new PPP investments.** This approach helps establish a sound legal and regulatory framework and underpins the financial viability and sustainability of the power sector and new investments. Achieving sound institutional and governance arrangements has been a key focal point of the reform program in Uzbekistan. The program consists of establishment of necessary institutional and legal frameworks, unbundling of energy SOEs, restructuring of sector institutions, and adoption of best practices in operations and financial management, which are also highlighted as COVID-19 risk mitigation factors to improve sector resilience. Implementation of reforms must be followed through as investments flow into the sector.

b. **Successful IPP development requires not only risk-mitigation instruments to attract private capital, but also a financially viable power sector that is able to pay for PPAs in a sustainable manner**. Past World Bank experience in developing renewable-energy IPPs in Zambia, Kenya, and Armenia affirm this requirement. A robust risk-mitigation package is necessary to attract private capital in newly opened markets that do not have a track record of IPPs; however, that package cannot guarantee the success of a project in the operation phase over 25-30 years if the power market is structurally unsustainable. For this reason, in addition to the risk-mitigation package developed to support the solar IPPs under this project, continued support to implementation of Uzbekistan's power-sector reform program is needed.

c. **Guarantee operations are best designed in parallel with other operations which support improvements in the financial situation of the power-sector value chain**. The IBRD guarantee operation is usually designed to enhance the creditworthiness of the off taker. At the same time, the underlying reason for the lack of creditworthiness of the off taker is not directly addressed in the operation. For Uzbekistan, the stabilization and improvement of NEGU's financials have been targeted under ongoing World Bank support through Programmatic Technical Assistance under implementation since 2018, a sequence of stand-alone Development Policy Operations (DPOs), and the Electricity Sector Transformation and Resilient Transmission (ESTART) Project. In the stand-alone DPOs approved by the World Bank Board in June 2018 and June 2019 (which also received supplemental DPO financing in June 2020), the prior policy-reform actions such as electricity-tariff increases, transparency measures in utility FM, adoption of a renewable-energy law, introduction of renewable energy into the power mix (that is, Navoi Scaling Solar IPP, presently in operation) were successfully implemented by GoU, demonstrating its commitment to the reform process. The World Bank's ongoing and future sector policy dialogue will be designed to further support the improvement of PPP design, the update and execution of the sector's LCP, NEGU commercialization, and other measures to strengthen sector financials.

d. **IPPs have made significant contributions to power-generation capacity and are linked to the spread of renewable energy in both the developing and developed worlds.** The risk of IPP mismanagement due to governance deficiencies are a lot more pronounced through unsolicited proposals and directly negotiated deals. In this context a transparent and competitive procurement process is strongly encouraged in Uzbekistan. It is further recommended that the Government should seek out best practice[13] to manage unsolicited proposals.

e. **Adequate sector planning is among the key factors in the success of renewable-energy deployment**. Comprehensive generation and transmission development planning should accompany large-scale renewable-energy scale-up to enhance the power/grid system to facilitate the integration of variable energy sources. With support from the WBG, the GoU finalized a Least-Cost Generation Expansion Plan in 2019. NEGU, with support from the World Bank, has also prepared a Long-Term Transmission Development Plan, which suggests around US$5 billion investment needs in the transmission network to ensure supply reliability and grid integration of 12GW of solar and wind energy as per the Government target. While the Government has committed to continue providing public financing (especially from IFIs) to NEGU program (given that transmission is the backbone-of the sector), NEGU

[13] Policy Guidelines for Managing Unsolicited Proposals in Infrastructure Projects: https://ppp.worldbank.org/public-private-partnership/library/policy-guidelines-managing-unsolicited-proposals-infrastructure-projects.

is planning to prepare a long-term investment and financing plan to prioritize strategic investments and identify financing sources for them. Upon the request from the GoU, WB will continue providing lead support to critical reforms and priority investments in the transmission segment.

f. **Competitive project selection, prudent structuring, and equitable contractual allocation of risks** among the various participating parties are critical ingredients to ensure a transaction's long-term sustainability. These principles are embedded in the WBG Scaling Solar Program, including in Uzbekistan, and have been followed in ensuring the selected projects are part of the GoU LCP and in the preparation process for the project.

g. **Readiness of core transaction agreements during project preparation is key to reaching commercial and financial close on time.** The WBG Scaling Solar Program embedded bankable transaction documents in the tender documents. The general acceptance of these documents by bidders allowed a smooth commercial close of the project following the tender results. The bankability of these documents is also expected to reduce the time needed between commercial and finance close.

h. **An effective and transparent PPP decision making process, especially in exceptional cases of subsequent amendments due to force-majeure events, is critical for the project sustainability, ensuring best value for money and preventing potential risks emerging from such cases.** In this regard, further strengthening the capacity of involved agencies and their PPP units and departments would be equally essential in order to improve the design of bankable PPP projects based on the lessons learned, including from this project, Navoi Scaling Solar 1 and other similar and hybrid projects. In this regard, the GoU has already introduced further measures in the design of pipeline Solar 3 project (500 MW in three sites and 62,5 MW/125 MWh innovative BESS) to enhance competition and considering generation concentration dimensions through provisions allowing for one developer to bid only for one site (out of three). While the guarantee documents will not include specific conditions pertaining to the price renegotiations, WBG will further support the GoU on improving the design and documents of future projects under the WBG Scaling Solar and Scaling Wind programs including based on the lessons learned from this project.

Box. Recent disruptions in global solar market and their implications[14]
Reduction of production of raw materials (polysilicon, glass, laminates, and other materials) and associated price hikes starting from 2021 have led to PV module prices increases to 40-50 percent. The price soaring was also observed in steel, aluminum and copper markets, which also put an upward pressure for prices of cables, inverters and other components of solar PV project. In addition, similar disruptions in logistics markets (freight price increase, increased demand for alternative transportation options, shortages of containers, cars, etc.) coupled with the implications of the war in Ukraine further added to the disruptions in the solar PV market. These developments collectively led to significant uncertainties posing huge risks on solar PV projects globally. As a result, heightened risks led to potential project delay, cost overruns and worsening project economics, declaration of force-majeure events by key developers, PPA renegotiations, and higher solar prices for new projects, among others, globally.

[14] IFC staff analysis.

IV. IMPLEMENTATION

A. Institutional and Implementation Arrangements

43. **The Scaling Solar 2 IPP companies, "Nur Samarkand Solar PV" and "Nur Jizzakh Solar PV" Foreign Enterprise Limited Companies, are SPVs incorporated and registered in Uzbekistan to develop, finance, build, own, operate, and maintain the project.** Masdar is bringing in long tenured and experienced staff to the board and management of the SPVs. The construction and operation of the solar power plants at the two sites will be implemented through an engineering, procurement and construction (EPC) contract and operations and maintenance (O&M) contract, respectively. The EPC contract will be procured competitively by Masdar and will be agreed and signed as a prior condition to financial close.

IBRD has not concluded any Non-Disclosure Agreements with respect to the project and will handle document disclosure in accordance with its Access to Information Policy.

44. **For the proposed project, the roles and responsibilities of the GoU (through its agencies and SOEs) and of the Governors (Khokimiyat) of the Jizzakh and Samarkand Regions are as follows**:

(i) The Ministry of Investments, Industry and Trade (MIIT) is the implementing agency for the Scaling Solar 2 IPP. It represents the GoU under the Government Support Agreement (GSA) signed with each Project Company and will also enter into the GSA Direct Agreement with the Project lenders.[15] Established[16] through the merger of the State Investments Committee and the Ministry of Foreign Trade, MIIT is the GoU agency responsible for implementation of the unified state investment policy; management of promotion of foreign investments, above all direct investments; and cooperation between IFIs and the GoU, as well as formation and management of the unified state policy in the field of foreign trade and international economic cooperation.

(ii) The MoEF, on behalf of the GoU, will enter into an Indemnity Agreement with IBRD (by which the GoU commits to reimburse IBRD for any payment under the proposed guarantees in case of a call on the guarantees). The MoEF has been involved in the preparation of the Scaling Solar 2 IPP with MIIT including, among other matters, in relation to the fiscal impact of the transaction and the GoU financial obligations arising out of the project contracts (such as early-termination payments pursuant to the GSA signed by MIIT).

[15] GoU (MIIT) also mandated IFC Advisory as transaction adviser to the GoU on the Navoi solar IPP project, and subsequently on an additional 900 MW of solar IPP projects (also under the WBG Scaling Solar Program) and on the 1,574 MW Syrdarya CCGT project.

[16] Pursuant to Presidential Decree No. 5643 dated January 28, 2019.

(iii) The MoE provides policy direction on sector development, and on planning and procurement of power-generation capacities.

(iv) NEGU, as the single purchaser of electricity from generation companies, including IPPs, is the off taker for the project's electricity and has entered into PPAs with the Project Companies. An independent engineering service will be jointly arranged by NEGU and the Project Companies to check the plant performance and monitor compliance with technical specifications under the PPAs.

(v) The Governors (Khokimiyat) of the Jizzakh and Samarkand Regions are the parties to the Land Lease Agreements signed with Masdar in relation to the long-term leases (for a duration equal to the terms of the PPAs and an additional six months) of the project sites.

B. Results Monitoring and Evaluation

45. **Overall monitoring of the project outcomes and results indicators and reporting to the World Bank will be undertaken by MIIT (implementing agency)** based on reporting by NEGU/the MoE and the Project Companies. IBRD will supervise the project with MIIT based on mandatory reporting by (a) the Scaling Solar 2 IPP Project Companies as required under the IBRD Project Agreements; (b) NEGU/the PPA off-taker as required under the Cooperation Agreement; and (c) the GoU as required under the Indemnity Agreement; as well as (d) through regular IBRD implementation support and supervision missions and field visits until the expiry of the proposed IBRD guarantees. Evaluation of results indicators will be part of regular IBRD supervision missions. Section VII below presents the Project Results Framework that defines specific outcomes and results with indicators to be monitored.

46. **As part of the supervision plan, the Bank's project team will pay special attention to NEGU's credit risk.** The ability of NEGU to make timely PPA payments to the Project Companies is highly correlated to the overall success of the power-sector reform, including, among other elements, achieving a cost-recovery tariff, successful implementation of the IPP program to establish a competitive cost of generation, reduction of technical and commercial losses, as well as maintaining the high collection rates achieved so far. Exogenous factors such as foreign exchange rates and power import/export prices will also drive NEGU's profit and cash position. GoU has demonstrated a commitment to further pursue sector financial-sustainability initiatives including gradual tariff increases to cost-recovery levels by 2026, formulation of a separate transmission tariff, loss-reduction programs, and promoting private-sector participation in generation and distribution. Understanding that envisaged broader sector reforms are critical to the project and sector sustainability, the WBG will continue to support priority sector reforms, including through the proposed project, ongoing Energy Strategy Programmatic Technical Assistance, DPOs, the ongoing MUTS, ESTART and ICCFER project under preparation. Before NEGU achieves full cost recovery, the GoU is expected to continue supporting the company's balance sheet through budget transfers, public credits, and cash injections into sector. The IBRD project team will monitor these key parameters of NEGU's financial position, payment discipline, and contract compliance, as well as sector-reform progress as part of its regular project supervision. NEGU will be required to supply key financial information to the Bank's project supervision team. During this transition period through 2025, the Bank's project team will also conduct regular meetings with MIIT, MoE, and MoEF to discuss the provision of sufficient government resources for NEGU under this project, as well as ongoing Programmatic Technical Assistance, the MUTS, ESTART and ICCFER projects, and potential DPO engagement.

C. Sustainability

47. **The project is financially sustainable on account of the low bids of US¢1.823 per kWh for Jizzakh and US¢ 1.791 per kWh for Samarkand.** This sustainability is owed in part to the Government's direct and indirect contributions to the project. The GoU has made public land available to the private developer, established a favorable investment regime through a presidential decree, and requested IBRD guarantees to mitigate project-specific risks. These measures, combined with the overall market opening, sector reform, and competitive terms of DFI financing, created the conditions for the lowest tariffs in the region.

48. **The Project will continue to scale the renewable-energy sector in Uzbekistan and build on the gains derived from the 100-MW Navoi Project.** As such, it will have a strong and continued demonstration effect for future IPPs, in Uzbekistan, Central Asia and beyond. The project alone represents a relatively small portion (c. 3%) of the current installed capacity in Uzbekistan and of the developer's global projects portfolio, but it is still an important achievement in initiating diversification of the power mix and attracting the private sector to take risks in entering a new market. The low-cost energy generation achieved through a competitive tender sets a good example for future PPP procurement. While, due to COVID-19 and supply-chain disruptions, the project's financial close has been delayed, the Project Sponsor remains fully committed to the project and is actively developing it.

Fiscal Impact Assessment

49. **The Scaling Solar 2 IPP Project will have limited fiscal impact.** Infrastructure projects developed in the form of PPPs generally create contingent liabilities for a country. For the Scaling Solar 2 IPP, the GSAs provide for the Project Companies' recourse to the Government in respect of termination payments and also include obligations for the Government's support to ongoing PPA payments. This Government support for ongoing PPA payments and termination payments creates contingent liabilities for the Government based on International Public Sector Accounting Standards[17]. Due to its limited size compared to Uzbekistan's energy system, the Scaling Solar 2 IPP Project will have a limited impact on the Government's contingent liabilities. It involves annual PPA payments of about US$ 25 million and peak termination payments of about US$ 408 million in certain scenarios.

50. **The GoU has an ambitious PPP agenda, including the development of about 14 GW of IPPs over 5 years.[18]** An impact assessment of these IPPs (solar, wind, and CCGT projects) has demonstrated that they could have a potentially significant impact on the Government's contingent liability and fiscal situation. Such a stress analysis on fiscal impact assumed certain project costs and tariff structures based on estimates and public information and was carried out under a project risk-allocation profile similar to the proposed Scaling Solar 2 IPP. The stress analysis indicates that if all contingent liabilities from every contract materialize (i.e., every PPA is terminated) in a particular year, the overall IPP pipeline will have a material impact on the Government's debt-to-GDP ratio. For instance, in 2026, if every PPA is terminated, total termination payments could be as high as US$ 11.3 billion, increasing the debt-to-GDP ratio by about 8 percent (from 29 percent to 37 percent). After 2026, because of GDP growth and amortization of project costs, the total contingent liabilities will have a less incremental impact.

[17] International Public Sector Accounting Standards (IPSAS) 32.
[18] The timeline is based on publicly available information and government's estimation.

51. **The stress analysis considers a highly unlikely scenario in which all PPA contracts would be terminated in the same year.** Historically, the default rate of project-finance loans was about an accumulative 10 percent over a 10-year period for all projects in low-middle-income countries,[19] and the likelihood of a default involving an actual PPA termination is even smaller. The scenario of every PPA in a country terminating in the same year (for example, national expropriation) is an even more remote event. Even under such a stress-case scenario, the GoU's debt-to-GDP ratio, while increasing significantly, would be within acceptable levels, partially due to its relatively low level of public debt today. Despite the comfort of the stress analysis result, the GoU, with the assistance of the World Bank, is developing tools and policies to institutionalize the monitoring of the contingent liabilities generated by its ambitious PPP/IPP program. In the next phase of IPP development, the GoU is focused on optimizing the risk allocation with the objective of reducing contingent liabilities and the probability of risk materialization while still attracting quality international developers and cost-effective financing.

D. Role of Partners

52. **In addition to the WBG engagement of 2,500MW clean renewable energy, ADB and EBRD have entered commitments with the GoU** to support development of 1,000MW of solar and 1,000MW of wind capacities on competitive basis, respectively. In this context, ADB has tendered the 457MW Sherabad Solar PV Park and EBRD has proceeded with the 100MW Karauzak Wind project (both at financial close stage) resulting in low tariffs, which demonstrate the benefits of private-sector participation and competitive processes as well as continuous IFIs support to accelerate the implementation of the GoU sector reforms and targets. In addition, ADB, EBRD, IFC and Japan International Cooperation Agency (JICA) have supported the Zarafshan Wind Project (500MW, at financial close stage as a negotiated project). On the efficient thermal power generation side, EBRD, AIIB, OPEC Fund and Multilateral Investment Guarantee Agency (MIGA) have supported the ACWA Syrdarya CCGT IPP (1,500MW, at early operation stage) on negotiated basis.

V. KEY RISKS

A. Overall Risk Rating and Explanation of Key Risks

53. **The overall risk of the project is Moderate, considering the following factors:** (a) it supports a utility-scale variable renewable/solar power generation (440-MW) investment in Uzbekistan with an environment in which institutional and technical capabilities are still under development; (b) subsequent solar PV project similar to the first project - Navoi Scaling Solar IPP; the project design and structuring hence will build upon this first renewable IPP and solar PV. It is also worth noting that the risks to the project and the World Bank guarantees are reduced as the PPA tariffs are lower than the expected generation tariff in the country and the size of the off-taker's PPA payments under the project is relatively small for the sector.

[19] Based on a Moody's study on Project Finance default
https://www.moodys.com/researchdocumentcontentpage.aspx?docid=PBC_1114036

54. **Sector strategies and policies** – Substantial**.** Uzbekistan is undertaking ambitious energy sector reforms, opening the hitherto state-dominated sector for private sector participation. High pace and sequencing of priority reforms could become a risk, if not managed well by the GoU going forward. Broader sector reforms currently remain of high priority and in progress, and the GoU is committed to further pursue such initiatives. The World Bank as a trusted long-term partner of the GoU for the design and implementation of the reform will continue to support the GoU in navigating the design and implementation stages of the reform to achieve transitions in a coherent manner. In this regard, the World Bank's active role in design and implementation of the GoU's ESRIP will be instrumental to strengthen the sector policies and strategies. On the financial viability side, further sector reforms are critical to improve long sustainability of the sector and NEGU, which is currently loss making on a standalone basis. The prevailing electricity tariff is still below cost recovery level and NEGU, as a recently established company, also does not have a long track record of dealing with IPPs. As part of ongoing World Bank support to the energy-sector reform, the Bank will continue to support the GoU in broader electricity sector reform, including on cost-recovery initiatives through implementation of the adopted electricity tariff-setting methodology and undertaking of tariff adjustments on a regular basis to be deployed jointly with social protection and cost mitigation measures for the poor and vulnerable as well as sound communication strategies. While these reforms are being implemented, it is expected that NEGU will benefit from external liquidity support as has been provided in the past. Additionally, it is expected that low-cost generation, such as that under the project, will improve the financial standing of the sector.

55. **Environmental and Social** – Moderate. The project will be prepared in accordance with the WB Environmental and Social Performance Standards, with a preliminarily assessed environmental and social risk rating of Moderate. During operations, the project will generate green energy and contribute to reduced carbon intensity of the Uzbek national grid, and a detailed carbon footprint and impact on climate change of project activities have been assessed as part of the due diligence on the project. The project location and landscape are vulnerable to climate and geophysical hazards; however, climate and disaster risk screening of project implementation has been conducted during the project preparation. The project will result in some economic displacement and impacts (loss of grazing and agricultural land, and assets, or limited access to assets, leading to loss of income sources or other means of livelihood) at the solar-panel installation sites and under the transmission lines, although the number of project-affected persons will not be significant. Other potential social risks and impacts relate to discrimination, sexual exploitation and abuse/sexual harassment, community conflicts and unrest, including solar-panel installation impacts on the neighborhoods in the aesthetic sense and lack of access to project benefits, labor influx, child and forced-labor risks among project workers, including solar panel supplier workers. These impacts are site-specific, with limited areas of influence, and easily identified, and can be addressed through the implementation of effective mitigation measures.

56. A preliminary screening for climate change and disaster risks was conducted for the project's main components, and the project's overall risk rating was found to be moderate. The identified risks included droughts and heatwaves, whose frequency and intensity may be increased by climate change, that may threaten water supply service. Additionally, the majority of Uzbekistan is at high risk of both river and flash flooding. Climate change is likely to increase the likelihood of floods.

57. The World Bank team notes, in light of the procurement of solar panels under the project, that there are allegations of risks of forced labor associated with polysilicon manufacturers (polysilicon being a key raw material in the solar-panel production chain). Accordingly, the following mitigation measures (among others) will be adopted for the project: The Project Companies (Jizzakh and Samarkand) will

(i) provide the World Bank contractual assurances that they have not used or engaged forced labor and will not use or engage forced labor in their operations; and (ii) require the same contractual assurances from the related entity charged with the procurement of solar modules for the project, which will, in turn, require the same from its primary (direct) supplier/s of solar modules.

58. **Other –** Moderate. Other risk comprises the potential impact of the war in Ukraine to the project. While the project marginally be impacted by potential supply chain and logistical disruptions caused by the ongoing war in Ukraine, investors and the GoU are aware of potential risks and have prepared potential mitigation measures, including through alterative supply routes. The Bank team will closely monitor the situation to provide timely support to the GoU to mitigate potential arising risks.

59. Furthermore, as evidenced by this project, disruptions in global solar market (PV module and associated commodities' price increases, etc.) and logistics industry may potentially lead to *inter alia* project implementation and procurement delays, contract renegotiations, force-majeure declarations. WBG will continue supporting the GoU in improving the PPP legal and regulatory framework, clarity and transparency of decision-making process, enhancing the design of pipeline PPP projects and strengthening institutional and human capacity of the involved stakeholders.

VI. APPRAISAL SUMMARY

A. Economic and Financial Analysis

Project Economic Analysis

60. **The economic rate of return (ERR) and net present value (NPV) of benefits are calculated using the WB's standard cost-benefit analysis methodology**.[20] The country-specific economic discount rate is at 8 percent per year following the World Bank guidelines for economic analyses. All monetary values presented in this section are expressed in constant US dollars of 2021 and this is the reference date used for the economic analysis. All discount rates are presented in real terms. A detailed economic analysis of the project was conducted with the assumptions described in Annex 3.

61. **Economic costs and benefits.** The economic cost of the Scaling Solar 2 IPP captures (a) EPC cost, (b) project development cost, and (c) O&M costs during the economic life of the plants. The main economic benefits are: (a) avoided operation costs of the power plants displaced by this project; and (b) avoided costs of greenhouse gas (GHG) emissions. Avoided operation costs are estimated assuming a simplified counterfactual scenario where the absence of the project results in increased dispatch of a CCGT running on natural gas with 61.7% efficiency that would deliver the same amount of yearly energy as the Jizzakh and the Samarkand plants. Avoided costs of GHG emission are estimated using: (a) the methodology of UNFCCC's "Tool to Calculate the Emission Factor for an Electricity System", based on the initial results of the Decarbonization 2050 scenario of the least-cost expansion plan available when this document was produced and complemented by the assumptions described in Annex 3, to estimate the amount of avoided emissions; and (b) the shadow prices of carbon (SPCs) of the *low* and the *high* scenarios

[20] Guidelines for Economic Analysis – Power Sector Investment Projects and Social Value of Carbon in Project Appraisal, 2014.

defined in the World Bank "Guidance Note on Shadow Price of Carbon in Economic Analysis". Detailed description of the methodology for benefits is provided in the Annex 3.

62. **Results of economic analysis.** Economic analysis confirms the viability of the Scaling Solar 2 IPP. The analysis indicates that the project is economically viable when avoided emissions are valued at the *low* and the *high* scenarios of SPCs, and also when avoided emissions are not valued at all. The economic rate of return (ERR) for the Jizzakh solar power plant is 12.4 percent per year if emissions are value at the *low* SPCs, 16.8 percent per year for the *high* scenario, and 9.8 percent per year if emissions are not valued at all. For the Samarkand solar power plant, the corresponding values of the ERR are 12.3, 16.6 and 9.7 percent per year, respectively. The ERRs are above the hurdle rate and demonstrates the positive economic returns of the project. Table 3 provides a summary of the economic analysis of the project, showing ERR and Economic Net Present Value (ENPV).

Table 3. Summary of Economic Analysis of Scaling Solar 2 IPP

		JIZZAKH	**SAMARKAND**
1	ERR (including GHG impact, low carbon price)	12.4%	12.3%
2	ERR (including GHG impact, high carbon price)	16.8%	16.6%
3	ERR (Excluding GHG Impact)	9.8%	9.7%
4	ENPV (including GHG impact, low carbon price)	US$92 million	US$88.1 million
5	ENPV (including GHG impact, high carbon price)	US$146.8 million	US$141.6 million
6	ENPV (excluding GHG impact)	US$36.7 million	US$34.3 million
7	Avoided GHG emissions during project lifetime, undiscounted	1.74 million tons CO_2	1.70 million tons CO_2

63. **Sensitivity analysis.** A sensitivity analysis was conducted for the following key cost-and-benefit drivers in the economic analysis: (a) discount rate; (b) O&M costs of the plants; (c) capital cost of the plants, and (d) generation output of the plants. The economic performance of the project is generally robust in all scenarios, details of the sensitivity analysis are presented in Annex 3.

64. **Avoided GHG emissions.** The project is expected to contribute to the reduction in total of about 3.4 million tons of CO_2 over its lifetime and an average of about 109,306 tons per year.

Project Financial Analysis

65. **A Project financial analysis** was conducted to assess its financial viability from (a) the Sponsor/SPVs' perspective and (b) a sector perspective.

66. **The analysis indicated that the projects are expected to generate sufficient cashflows to recover capital expenditure and meet operational and maintenance expenditures.** Furthermore, after meeting debt-service costs, the cash flows for both the projects allow for regular dividend payments, providing the equity shareholders with a reasonable return for a project of this nature. The lenders' base case further

confirmed an Average Debt Service Coverage Ratio (ADSCR) of 1.2x for the project (under P90 level of generation), indicating adequate comfort for the lenders comprising the debt syndicate.

67. **The Scaling Solar 2 IPP will contribute to power sector financial sustainability by reducing the average power purchase cost for NEGU as off-taker**. The project's tariffs are lower than the weighted-average electricity purchase cost of NEGU from state-owned power plants. The savings from the project are thus expected to deliver a positive NPV for NEGU and the sector at about US$40 million from 2025-30[21], which also contributes to the affordability of power supply for end-users.

NEGU Financial Analysis

68. **NEGU is a strategically important state-owned enterprise (SOE) fully owned by the GoU through the Ministry of Economy and Finance (MoEF).** As Uzbekistan's transmission-system operator and single electricity buyer/seller, NEGU operates as a monopoly and is therefore not expected to earn substantial profits, but rather operate on a cost-recovery basis, generating adequate revenues to meet operating expenses and capital expenses to the extent envisaged under its development plan. Given the critical role it plays in the country's energy sector, NEGU has significant financial, operational, and managerial linkages with the GoU.

69. **Historical financial performance of NEGU. NEGU has faced a challenging financial situation over the last two years, failing to break even on an operating-cost basis in 2021.** In the absence of a regulated cost-plus tariff regime, NEGU's revenues did not rise enough to offset the large increases in operating costs in 2021, primarily driven by cost of electricity purchased that grew by ~19 percent as a result of an increase in generation tariffs for domestic power plants. Currently transmission revenues are residual amounts remaining after regulated distribution, supply and generation revenues, leaving NEGU squeezed and sustaining operational losses. Operating costs represented 108% of total revenues in 2021, resulting in EBITDA of UZS -1,584 billion (~US$ -149 million) at an EBITDA (operating) margin of -8 per cent (see Annex 3 on sector financial analysis for more details). NEGU financed shortfalls through on-lent loans from IFIs, shareholder loans from the MoEF, and accumulation of payables to state-owned generation companies. Sovereign support is expected to continue while MoEF implements cost recovery measures in the near term.

70. **NEGU financial forecasts indicate improved EBITDA margins for NEGU over the next few years (2022-25) as sector reforms, particularly on tariffs, are implemented.** During these years, NEGU is not expected to generate adequate cash flows to meet interest costs or for reinvestments, and therefore would rely on sovereign support (of over US$ 350 million, based on forecasts). That said, from 2026 onwards, NEGU's operational performance will improve due to cost reflective tariffs, enabling it to generate adequate resources to not only service debt, but also channel to re-investments. During the forecast period from 2022-2030, the weighted-average cost of power purchase by NEGU from state-owned generation companies and IPPs increases by 1.5 times, with sales price increasing above purchase price starting in 2026. From that point, NEGU is forecasted to cover operational expenses and capital costs.

71. **Key risks**. Notwithstanding the strong support from the sovereign and recent positive developments, NEGU operates in a risky environment that could adversely impact its operational and financial performance. Key risks faced by NEGU are as follows: (a) delay in implementing tariff reforms;

[21] Assumes a discount rate of 8.8 percent, the current yield of the Eurobond issued by the GoU maturing in 2030.

(b) foreign exchange risk, as NEGU has sizeable liabilities in foreign currencies and (c) increase in purchase price of electricity.

72. **A sensitivity analysis has been conducted to evaluate the impact of key risks on NEGU's financial performance and sustainability.** Generally speaking, any materialization of these risks which are not offset through other means will result in additional liquidity required for NEGU to meet its operating and capital costs. This size of the liquidity gap depends on the extent the risk materializes during the impacted years. For detailed information on the sensitivity analysis, see Annex 3.

73. **Despite the current situation at NEGU and the risks highlighted above,** the risk of a draw on the WB-guaranteed LCs and therefore a call on the WB payment guarantees under the Scaling Solar 2 IPP is mitigated through the following:

a. Expected improvement in NEGU's financial performance: Scheduled commercial operation date for the Scaling Solar 2 IPP project is 2025, at which time it is expected that tariffs would have continued to rise towards cost-recovery levels. Based on the projections above, NEGU will be generating positive EBITDA by that time and will be further advanced on its path towards financial sustainability as demonstrated by healthy leverage and profitability ratios.

b. GoU commitment for NEGU's foreign-currency liabilities: GoU is committed to a growing electricity sector and recognizes the importance of ensuring on-time payments to IPPs. Currently, NEGU's cash waterfall characterizes IPP payments as the top priority only after external debt payments, and both are sovereign-guaranteed. In addition, MoEF is in the process of creating a fund capitalized by SOE-privatization proceeds to help backstop IPP payments. NEGU's financial condition is tied to the decision-making of MoEF (including in relation to tariffs), which is unlikely to allow such a vitally important entity to fail or fall short of its external obligations.

c. Project Participants: With the participation of Masdar as sponsor, which has over 1 GW of power projects in operation or under development in Uzbekistan, as well as the involvement of ADB, AIIB, and EBRD, the likelihood of unrectified payment issues is reduced.

d. Cure Period: A call on the WB guarantees would materialize only if NEGU (or its successor) fails to pay the L/C bank one year after the date on which the L/Cs are drawn. This one-year period is critically important to allow NEGU (and MoEF) to rectify the issue (and the Bank to support that process and help avoid a call on its guarantees).

74. **It is also worth noting that the WB guarantee exposure of up to US$ 12 million for the project** (vs. total project costs of over US$ 400 million) is relatively low, though still critical in ensuring that the project moves forward, and Uzbekistan remains on track to deliver on its energy-mix objectives.

75. **The GoU is committed to pursuing further reform initiatives, including cost recovery and sector sustainability.** These initiatives will be implemented as part of a broader reform program to sustain sector reforms: (a) tariff cost-recovery trajectory by 2026; (b) establishment of a sector regulator as well as a separate regulated tariff for NEGU; (c) implementation of loss-reduction programs, especially in the distribution segment, to improve sectoral operational efficiency; and (d) maintenance of state ownership over the transmission segment as the backbone of the sector, while promoting private-sector participation in generation and distribution, among others segments.

76. **As part of the ongoing policy dialogue, and upon request from the GoU, the World Bank will continue to support the GoU and NEGU reform initiatives** to (a) design and implement a cost-recovery

trajectory while protecting vulnerable households and designing communication strategies to strengthen citizen engagement on sector, subsidy and tariff reforms; (b) establish and operationalize a sector regulator as well as a separate regulated tariff for NEGU; (c) demonstrate, introduce, and prioritize, including through the proposed project, the benefits of transparent and competitive selection of private investors; and (d) improve the energy sector's operational efficiency, financial sustainability, and capacity. The World Bank support will be provided through the pipeline Innovative Carbon Finance for Energy Reforms Projects Technical Assistance, the ongoing MUTS, and ESTART projects, and series of DPO engagements. For details of the economic and financial analyses, including sensitivity scenarios, see Annex 3.

B. Technical

77. **The Project site assessments and designs are satisfactory.** The Jizzakh solar PV project will connect to a 220-kV substation via a transmission line which is 15 kms away, while the Samarkand solar PV project will require a 4.5-km transmission line. Financing of transmission lines is part of the project costs; the lines will be built by the Sponsor and later transferred to NEGU. Based on the lenders' technical advisor's reports on the two subprojects, the geotechnical, topographical, and hydrological studies are all adequate to assess the sites. While some additional geotechnical work remains to be done, the studies conclude that both sites are suitable for the construction of PV plants. In addition, technical designs of the subprojects are in line with market standards and the lenders' technical advisor does not envisage any design issues. All civil works will be designed to be capable of withstanding the effects of water, extreme winds, and other natural disasters, including the impacts of climate change, including extreme heat, droughts, floods, etc.

78. **The project will utilize bifacial monocrystalline dual-glass modules as the chosen technology with a horizontal single-axis mounting structure.** Currently four module manufacturers are being considered by the Sponsor: JA Solar, Jolywood, Longi Solar, and Suntech. Final selection will be done once commercial negotiations are concluded though the Lenders' technical advisor considers all proposed module suppliers to be well suited for the project. Monocrystalline technology is well established with high efficiency rates. The bifacial modules have a suitable track record and acceptable market-standard technical characteristics. They require relatively lower O&M resources and increase electricity generation due to bi-faciality. An independent engineer will be jointly appointed by the Project Companies and NEGU to ensure prudent practices are applied and relevant requirements met. The PV plants will operate under a standing dispatch instruction to deliver all electrical energy generated by the PV plants as per the PPA. Solar module suppliers provide performance warrantees for the entire length of the PPA.

79. **As required by the Uzbekistan Grid Code, the Project Companies will, at their own expense, install, test, and commission the metering systems at the delivery points,** which will be transferred to NEGU. The MoE and NEGU, with support from the World Bank, have initiated the preparation of a long-term Transmission Development and Renewable Energy Integration Plan to 2030. The plan will articulate medium- and long-term investment needs to ensure power-system reliability and smooth grid-integration of large-scale renewables and strengthen regional connectivity with neighboring countries. Furthermore, the GoU is considering development of a pilot battery energy storage system (BESS) coupled with one of the PV projects under development under the Solar 3 program to improve power-system flexibility and demonstrate BESS benefits and the impact of integrating PV solar projects to the grid.

80. **The selected EPC Contractor, China Dongfang Electric Group Co., Ltd (Dongfang Electric Group), has the competence and experience to carry out the work.** Dongfang Electric Group is a large conglomerate focused on engineering and construction, including the development of more than 80 GW of power plants globally. The works for the Scaling Solar 2 IPP are not particularly complex technically and there is no significant underground work. Close cooperation and effective interface management and supervision by the Owner's Engineer will be key to the successful execution of the works. The Owner's Engineer will coordinate with the EPC Contractor with the objective of ensuring that works are performed according to specifications and delivered on time. Construction is expected to start shortly after financial close for a total duration of about 27 months from initial mobilization and site preparation to commissioning of the entire PV plants.

C. Financial Management

81. **The Scaling Solar 2 IPP will benefit from payment guarantees.** In the event a guaranteed event takes place, and the L/C bank submits a valid demand notice to the Bank under the guarantees, the Bank will make the relevant payment to the L/C bank. There are no payments to the GoU and/or its agencies. On this basis, the provisions of the paragraph 7. Financial Management of the World Bank Policy on Investment Project Financing (October 2018) do not apply.

82. **Moreover, the overall FM of the project will be undertaken by the Project Companies per international best practices and national standards**. The Project Companies will hire dedicated financial staff supported by qualified accountants to perform duties including accounting, reporting, and planning, managing auditing and internal controls. Each Project Company's annual financial statements will be prepared in accordance with International Financial Reporting Standards (IFRS) and will be audited in accordance with International Standards on Auditing (ISA). The World Bank may request copies of the authorized audit reports and Management Letters within six months after the end of each reporting period.

D. Procurement

83. **The World Bank Procurement Regulations for IPF Borrowers** (November 2020) do not apply to this project given the Bank is providing guarantees, (IPF Procurement Regulations, paragraph 2.2(a)).

84. **The GoU (through the MoEF, MoE, and MIIT, as appropriate) has conducted a competitive bidding process for selection of an investor to design, finance, construct, and operate the 440-MW Scaling Solar 2 IPP.** IFC Advisory advised the GoU on structuring and tendering the project using the WBG Scaling Solar approach. The prequalification phase was initiated in June 2020. The Request for Prequalification stipulated specific prequalification criteria (including experience, and technical, financial, and legal requirements). The GoU prequalified 15 bidders. The bidding phase was initiated upon the issuance of the Request for Proposal in December 2020.[22] Overall, six and seven bids were received for the Samarkand and Jizzakh projects, respectively. Bidders were well-known global power developers and sponsors.

85. **On the basis of an evaluation of the technical (on a pass/fail basis) and financial proposals, Masdar was selected as the winning bidder for both projects**. The tariff bids were US¢1.823 per kWh for Jizzakh and US¢1.791 per kWh for Samarkand, both under 25-year PPA terms. Signing of the project documents (PPA and GSA) took place on July 12, 2021. In July 2022, NEGU and Masdar agreed and signed

[22] https://minenergy.uz/en/news/view/959.

PPA amendments to promote long-term sector sustainability and preserve value for money from the public perspective. The PPA amendments were approved by Presidential Decree in December 2022. Furthermore, the Bank task team has confirmed that the IBRD Articles of Agreement requirement of considerations of economy and efficiency in the use of the proceeds of any loan (here the L/C loans) guaranteed by the Bank has been met.

E. Environment and Social

86. **The proposed Scaling Solar 2 IPP Project is being prepared and will be implemented in accordance with the Bank's Operational Policy on Performance Standards for Private Sector Activities (OP 4.03)** with both environmental and social risk ratings of Moderate— Table 4 below specifies the applicable Performance Standards. Masdar has prepared site specific Preliminary Environmental and Social Impact Assessments (ESIAs), which indicate that the selected sites are appropriate for the planned solar project with no significant, unmitigable environmental or social impacts associated with the project. Masdar will ensure that the ESIAs are finalized, disclosed, consulted based on the agreed designs and that the findings of the ESIAs are adhered in the site-specific ESMSs will be used by the EPC contractors to prepare the contractors' ESMPs. The proposed Scaling Solar 2 IPP does not trigger the World Bank's Operational Policy on Projects on International Waterways (OP 7.50) or its Operational Policy on Projects in Disputed Areas (OP 7.60).

87. **Masdar has also prepared the following reports in relation to the Scaling Solar 2 IPP and submitted them to the World Bank for approval before appraisal:** Preliminary ESIAs; site specific Stakeholder Engagement Plans (SEPs); as part of the SEPs, Grievance Redress Mechanisms (GRM); site specific Social Due Diligence Reports on Land Acquisition, initial Livelihood Restoration Plans and an Environmental and Social Action Plan (ESAP). The World Bank will continue to work with Masdar to ensure, through the monitoring framework for the Scaling Solar 2 IPP, that the PV power plants comply with the World Bank Performance Standards throughout the life of the project.

Table 4. Applicable Performance Standards (PS)

Performance Standards	Yes	No
PS 1: Assessment and Management of Environmental and Social Risks and Impacts	X	
PS 2: Labor and Working Conditions	X	
PS 3: Resource Efficiency and Pollution Prevention	X	
PS 4: Community Health, Safety, and Security	X	
PS 5: Land Acquisition[a] and Involuntary Resettlement	X	
PS 6: Biodiversity Conservation and Sustainable Management of Living Natural Resources	X	
PS 7: Indigenous Peoples		X
PS 8: Cultural Heritage		X

Note: a. Long-term Land Lease Agreements, relating to the project sites and access rights to the sites, are in the process of being concluded between the Scaling Solar 2 IPP project Companies and the GoU (represented by the Khokimiyats (Governors) of the Jizzakh and Samarkand Regions).

88. **As indicated in Table 4 above, the following Performance Standards are applicable:** PS1 – Assessment and Management of Environmental and Social Risks and Impacts, PS 2 – Labor and Working

Conditions, PS 3 – Resource Efficiency and Pollution Prevention, PS 4 – Community Health, Safety, and Security, PS 5 – Land Acquisition and Involuntary Resettlement, and PS 6 – Biodiversity Conservation and Sustainable Management of Living Natural Resources. In terms of the magnitude of potential adverse environmental or social risks and impacts, this is a Category B project under IFC's Policy on Environmental and Social Sustainability. Specifically, the project is expected to have limited impacts that are site-specific and temporary. Those impacts can be avoided or mitigated through compliance with the applicable Performance Standards, and related procedures, guidelines, and design requirements.

89. **The Jizzakh and Samarkand PV plants will be built on government land. Until 2019, the Jizzakh plot was leased out to a single tenant, and the Samarkand plot to five tenants,** and both plots were being used for the grazing of animals. In 2018-19, the government informed the tenants on both plots of the planned changes in land use. The Jizzakh tenant surrendered his lease and returned the land to the government as it was apparently not productive, and he has no claims under national law. No physical displacement occurred, and the tenant has made no related complaints. The Samarkand tenants were required to partially or fully return their land to the government over the period from 2019 to 2020. Some of the Samarkand tenants were offered and accepted replacement plots; others agreed to a reduction of plot size. One Samarkand tenant is pending replacement land / compensation.

90. **Separately, where the transmission lines that the sponsor will be constructing for evacuation of the power generated under the project**, preliminary analysis of the relevant routes has identified 12 farms that will be affected, and GoU/the Project Companies will need to acquire an approximate total of three ha of land for this purpose between the two sites.

91. **The Bank's appraisal of the project considered the environmental and social management planning process and documentation for the project** and any gaps between this process and documentation and the requirements of the World Bank Performance Standards. Where necessary, corrective measures, intended to close these gaps within a reasonable period, have been identified. Through the implementation of these measures, the project is expected to be designed and operated in accordance with the Performance Standards. Preliminary ESIAs, and a complete Environmental and Social Review Summary (ESRS) with a detailed Action Plan (ESAP, set out in Annex 4), and Stakeholder Engagement Plans were disclosed on the World Bank website on February 1, 2023. The Developer (Masdar) has also developed the Construction E&S Management System (cESMS) for each site that identifies and structures of the key E&S procedures and plans to be implemented during the construction phase to ensure all risks and impacts are properly managed and handled. The ESMS will be updated for the operational and decommissioning phases of the project to manage the relevant E&S risks associated with operational and decommissioning activities, generating the ESMS and ESMS.

92. **The Project Companies will consider the national context in terms of gender-based violence** (GBV), including prevention and response to sexual exploitation & abuse (SEA) and sexual harassment (SH) at the workplaces and community sites. As part of the ESMS and ESMS, the Project Companies and/or EPC Contractors will implement and report regularly on the site-specific Gender Management Plans, including Sexual Exploitation & Abuse (SEA) and Sexual Harassment (SH) Prevention & Response Action Plans, Workers and Community Grievance Mechanisms with GBV/SEA/SH uptake channels, and Labor and Working Conditions Management Plans.

93. The World Bank team notes, in light of the procurement of solar panels under the project, that there are allegations of risks of forced labor associated with polysilicon manufacturers (polysilicon being a key raw material in the solar panel production chain). Accordingly, the following mitigation measures

(among others) will be adopted for the Project: Project Companies (Jizzakh and Samarkand) will (i) provide the World Bank contractual assurances that they have not used or engaged forced labor and will not use or engage forced labor in their operations; and (ii) require the same contractual assurances from the related entity charged with the procurement of solar modules for the project, which will, in turn, require the same from its primary (direct) supplier/s of solar modules.

94. **Finally, the Project Companies will develop Environmental and Social (E&S) Supplier and Vendor Management Plans** setting out the measures to be implemented to identify, manage, and monitor E&S (including health and safety) risks and track the related performance of key suppliers and vendors under the project.

F. World Bank Grievance Redress

95. **Communities and individuals who believe that they are adversely affected by a project supported by the World Bank may submit complaints** to existing project-level grievance redress mechanisms or the Bank's Grievance Redress Service (GRS). The GRS ensures that complaints received are promptly reviewed in order to address project-related concerns. Project affected communities and individuals may submit their complaint to the Bank's independent Accountability Mechanism (AM). The AM houses the Inspection Panel, which determines whether harm occurred, or could occur, as a result of Bank non-compliance with its policies and procedures, and the Dispute Resolution Service, which provides communities and borrowers with the opportunity to address complaints through dispute resolution. Complaints may be submitted to the AM at any time after concerns have been brought directly to the attention of Bank Management and after Management has been given an opportunity to respond. For information on how to submit complaints to the Bank's Grievance Redress Service (GRS), please visit *http://www.worldbank.org/GRS.* For information on how to submit complaints to the Bank's Accountability Mechanism, please visit *https://accountability.worldbank.org*.

VII. RESULTS FRAMEWORK AND MONITORING

Results Framework

UZBEKISTAN
Scaling Solar 2 Independent Power Producer (IPP) Project

Project Development Objectives

The Project Development Objective (PDO) is to increase renewable energy generation through private sector participation in Uzbekistan.

Project Development Objective Indicators

PDO Indicator Name	Core (Yes/No)	Unit of Measure	Baseline (FY20)	End Target (FY24)	Frequency	Data Source / Methodology	Responsibility for Data Collection
Indicator 1: Power generation capacity constructed (renewable/solar)	Y	MW	0	440	Annually	MoE Statistics, IPP SPV Reports	MoE
Indicator 2: Electricity supplied by the Project into the grid (renewable/solar)	Y	TWh/year	0	1.1	Annually	NEGU Statistics, IPP SPV Reports	NEGU
Indicator 3: Private capital mobilized	Y	US$ million	0	205	At financial close	MIIT statistics, IPP SPV Reports	MIIT

Indicator 4: Greenhouse gas emissions avoided	Y	$ktCO_2e$/year	0	109	Annually	MoE statistics, NEGU, IPP SPV Reports	MoE

Intermediate Results Indicators

Intermediate Indicator Name	**Core (Yes/ No)**	**Unit of Measure**	**Baselin e**	**End Target**	**Frequency**	**Data Source / Methodology**	**Responsibility for Data Collection**
Intermediate Indicator 1: Physical implementation progress in generation project capacity constructed	N	percentage	0	100	Annually	MoE/MIIT statistics, IPP SPV Reports	MoE
Intermediate Indicator 2: Project commissioning test completed	N	Y/N	N	Y	At commercial operation date	NEGU Statistics, IPP SPV Reports	NEGU

ANNEX 1: INDICATIVE TERMS AND CONDITIONS OF THE GUARANTEE

UZBEKISTAN
Scaling Solar 2 Independent Power Producer (IPP) Project
(Applies separately to each of Jizzakh and Samarkand Solar PV plants[23])

*This term sheet contains a summary of indicative terms and conditions of a proposed guarantee ("**Guarantee**") from the International Bank for Reconstruction and Development ("**IBRD**") for discussion purposes only and does not constitute an offer to provide a Guarantee. The provision of a Guarantee is subject, inter alia, to satisfactory appraisal by IBRD of the proposed Scaling Solar 2 Independent Power Producer (IPP) Project in Uzbekistan ("**Project**"), compliance with all applicable policies of the World Bank, including those related to environmental and social safeguards, review and acceptance by IBRD of the ownership, management, financing structure (including in connection with shareholders, suppliers, equipment, and Project design, and contracts proposed by the winning bidder), and project / transaction documentation, and the approval of the management and Executive Directors of IBRD in their sole discretion. Without limiting the generality of the foregoing, IBRD is highly selective with regard to the clients and beneficiaries it works with and is diligent about Know-Your-Customer requirements for all project participants (equity investors, ultimate shareholders, lenders, contractors, advisors) and will undertake an appraisal of the Project and the Project Company including an assessment along these parameters.*

*This term sheet is based on the executed versions of the Jizzakh and Samarkand Power Purchase Agreements ("**PPAs**") dated July [ ], 2021, the related amendments dated September [ ], 2022, and the Jizzakh and Samarkand Government Support Agreements ("**GSAs**") dated 12 July 2021, and is designed to support obligations set out in those documents.*

IBRD Guaranteed Letter of Credit ("L/C")	
L/C Applicant:	**National Electric Grid of Uzbekistan JSC** (a GoU-owned company, as Purchaser under the PPA)
L/C Beneficiary:	**Project Company** (a privately-owned company responsible for the implementation of the Project)
L/C-Issuing Bank:	A commercial bank acceptable to IBRD (as Guarantor under the Guarantee Agreement), the L/C Applicant, and the L/C Beneficiary. An acceptable L/C-Issuing Bank will be selected by the L/C Applicant, the GoU, and IBRD following a competitive selection process. *[Note also the PPA minimum requirements (including in relation to credit rating) for L/C-Issuing Bank and Acceptable Bank.]*
Maximum L/C Amount:	The maximum amount available for draw under the L/C shall not exceed USD [6] million[24]. The Maximum L/C Amount may be reduced from time to time in accordance with the terms of the L/C and the Guarantee Agreement.

[23] A separate L/C will be issued for each of the Jizzakh and Samarkand project companies, and a separate IBRD guarantee will backstop NEGU repayment of any draws on each L/C.
[24] Based on (i) the requirement under the PPAs for a payment security equivalent to up to three (3) months of payments under the PPAs; and (ii) the winning bidder's projected cash flows for the Project.

L/C Effective Date:	[Date][25]
L/C Validity Period	[Term to be inserted once agreed with Masdar.]
Guaranteed L/C:	Revolving standby irrevocable L/C issued in favor of the L/C Beneficiary (Project Company) by the L/C-Issuing Bank at the request of the L/C Applicant (Purchaser) to backstop the monthly payment obligations of the L/C Applicant under the PPA following the occurrence of a Guaranteed Event (as defined below). Any amounts drawn by the L/C Beneficiary (Project Company) under the L/C that are repaid by the L/C Applicant to the L/C-Issuing Bank within the L/C Reimbursement Period (as defined below) would be reinstated. The obligation of the L/C Applicant to repay the L/C-Issuing Bank amounts drawn under the L/C would be guaranteed by IBRD up to the Maximum Guaranteed Amount. Any amounts drawn by the L/C Beneficiary under the L/C that are subsequently repaid by IBRD (as Guarantor under the Guarantee Agreement) to the L/C-Issuing Bank under the Guarantee would not be reinstated. That is, any principal amount repaid by IBRD would be deducted from the Maximum L/C Amount.
Guaranteed Events (Permitted Drawdown under L/C):	Failure by the Purchaser to pay any amount due and payable to the L/C Beneficiary pursuant to a monthly invoice according to the terms of the PPA. Failure by the GoU to make the same payment (in lieu of the Purchaser) according to the terms of the Government Support Agreement (GSA).
L/C Fees:	To be payable by the L/C Beneficiary to the L/C-Issuing Bank. Level of L/C Fees must be acceptable to the L/C Applicant and IBRD.
L/C Reimbursement and Credit Agreement (RCA)	
Borrower (of the L/C loan):	**L/C Applicant** (National Electric Grid of Uzbekistan JSC (NEGU), as Purchaser under the PPA)
Lender (of the L/C loan):	**L/C-Issuing Bank**, as guaranteed lender
L/C Reimbursement Period:	Following a draw under the L/C by the L/C Beneficiary following the occurrence of a Guaranteed Event, the amount drawn becomes a loan from the Lender (L/C-Issuing Bank) to the Borrower (Purchaser). The Borrower would be obligated to repay the Lender such L/C loan, together with accrued interest thereon, within a period of twelve (12) months (the "**L/C Reimbursement Period**") from the date of each draw (loan), pursuant to the RCA. In the event of a timely repayment of the L/C loan, the L/C will be reinstated by the amount of such repayment. In the event of a non-payment of the L/C loan by the due date, the L/C-Issuing Bank would have the right to call on the Guarantee for principal amounts plus accrued interest due from

[25] A date that (a) follows the date of commercial close (signing of the PPA) and the date of satisfaction of the conditions precedent to the issuance of the L/C, but (b) precedes the date of financial close.

	the L/C Applicant under the RCA.
Interest Rate charged by the L/C Issuing Bank:	An appropriate spread above [SOFR or other] acceptable to the L/C-Issuing Bank, the L/C Applicant and the GoU and agreed by IBRD. The maturity of the selected [SOFR or other] base rate should ideally be one (1) month.
IBRD Guarantee Agreement	
Guarantor:	**IBRD**
Guarantee Beneficiary:	**L/C-Issuing Bank**, as guaranteed lender
Guarantee Face Value:	Up to USD [6] million
Guarantee Support (Scope):	IBRD (as Guarantor) will backstop the payment obligations of the L/C Applicant (Purchaser) under the RCA to the extent that (i) the said obligations result from a Permitted Drawdown under the L/C and (ii) the L/C Applicant has failed to repay the L/C-Issuing Bank in accordance with the RCA; and (iii) the GoU has also failed to repay the L/C-Issuing Bank pursuant to the terms of the GSA. That is, if the amount remains unpaid after the expiry of the L/C Reimbursement Period, the L/C-Issuing Bank would have the right to call on the Guarantee for the principal amount (equal to the amount drawn under the L/C) plus accrued interest due from the L/C Applicant.
Maximum Guaranteed Amount:	Maximum Guaranteed Principal plus accrued interest thereon in accordance with the RCA. IBRD may cover compound interest, but will not cover penalty interest, default interest, or charges of a similar nature.
Maximum Guaranteed Principal:	The Guarantee Face Value, i.e., USD [6] million. Any principal amount paid by IBRD to the L/C-Issuing Bank under the IBRD Guarantee would be deducted from the Maximum Guaranteed Principal and those amounts would <u>not be reinstated</u>.
Maximum Guarantee Period:	The L/C Validity Period plus 14 months.
IBRD Financial Exposure Limits:	The average life of the financial exposure of IBRD under the Guarantee will not exceed 20 years, and the Maximum Guarantee Period will not exceed 35 years. Financial exposure of IBRD under the Guarantee will start on the L/C Effective Date.
Signing:	If the Guarantee-related legal agreements are not signed within 24 months following approval by the Board of Executive Directors of IBRD, IBRD may withdraw the offer of the Guarantee.
Exclusions, Withholding, Limitation/Suspension & Termination Events:	Standard exclusion, withholding, limitation / suspension and termination events for transactions of this nature.
Substitution of	If IBRD exercises remedies against the L/C-Issuing Bank under the Guarantee Agreement for reasons attributable to the L/C-Issuing Bank, IBRD may enter into a new Guarantee Agreement

Guarantee:	with a substitute L/C-Issuing Bank on substantially the same terms and conditions as the Guarantee Agreement and for the remaining term of the Maximum Guarantee Period.
Conditions Precedent to Effectiveness of the IBRD Guarantee:	Usual and customary conditions for financing of this type, including but not limited to the following: (a) Firm commitment for sufficient financing to complete the construction of the Project, including satisfactory contribution of equity; (b) Execution, delivery, and effectiveness of all Project and financing agreements, in form and substance satisfactory to IBRD, including the Indemnity Agreement and the Project Agreement; (c) Delivery of all relevant host-country environmental approvals required for the operation of the Project, and compliance with all applicable World Bank requirements relating to Sanctionable Practices and environmental and social safeguards, including the World Bank Performance Standards and as to use of forced labor; (d) Effectiveness of all required insurance; (e) Satisfaction of all conditions precedent to the first disbursement under the financing documents, save for any condition that requires the effectiveness of the Guarantee Agreement to have occurred; (f) Provision of satisfactory legal opinions; (g) Payment in full of the Initiation Fee, Processing Fee (if invoiced), Front-end Fee, and the first installment/s of the Standby Fee and / or Guarantee Fee (if invoiced)); (h) payment of fees and expenses of IBRD's external legal counsel; and (i) Satisfactory integrity due diligence of Project Company (and related parties) and guaranteed party.
Subrogation:	If, and to the extent IBRD makes a payment under the Guarantee, IBRD will be subrogated immediately, to the extent of the unreimbursed amount of such payment, to the L/C Bank's rights under the RCA.
Governing law:	English law or New York Law.
Indemnity Agreement	
Parties:	IBRD and Uzbekistan (the "Member Country")
Indemnity:	The Member Country will reimburse and indemnify IBRD on demand, or as IBRD may otherwise direct, for all payments under the Guarantee and all losses, damages, costs, and expenses incurred by IBRD relating to or arising from the Guarantee.
Covenants:	Usual and customary covenants included in agreements between member countries and IBRD. Any specific covenants will be determined during the Guarantee documentation phase.
Remedies:	If the Member Country breaches any of its obligations under the Indemnity Agreement, IBRD may suspend or cancel, in whole or in part, the rights of the Member Country to make withdrawals under any other loan, credit, or grant agreement with IBRD or IDA, or any IBRD loan or IDA credit to a third party guaranteed by the Member Country, and may declare the

	outstanding principal and interest of any such loan or credit to be due and payable immediately. A breach by the Member Country under the Indemnity Agreement will not, however, discharge any guarantee obligations of IBRD under the Guarantee.
Governing Law:	The Indemnity Agreement will follow the legal regime, and include dispute settlement provisions, usual and customary for agreements between member countries and IBRD.
Project Agreement	
Parties:	**IBRD** and the **L/C Beneficiary** (**Project Company**, a privately-owned company responsible for the implementation of the Project)
Representations and Warranties:	The L/C Beneficiary will represent, among other standard and Project-specific provisions, as of the effective date, that: (a) it is in compliance with applicable environmental and social laws and the applicable World Bank environmental and social safeguard requirements, including the World Bank Performance Standards and other applicable requirements, including as to the use of forced labor; and (b) neither it (nor its direct and indirect shareholders and any other relevant Project participants, as determined by IBRD), nor any of its affiliates has engaged in any Sanctionable Practices in connection with the Project. Note: "**Sanctionable Practices**" include corrupt, fraudulent, collusive, coercive, or obstructive practices, as defined in IBRD's Anti-Corruption Guidelines.
Covenants:	The L/C Beneficiary will covenant, among other things, that it will: (a) comply with applicable laws, including environmental and social laws, and the applicable World Bank environmental and social safeguards requirements, including the World Bank Performance Standards and other applicable requirements, including as to the use of forced labor; (b) provide annual audited financial statements and other reports to IBRD; (c) provide certain notices and other information to IBRD; (d) provide IBRD access to the Project; (e) not engage in (or authorize or permit any affiliate or any other Person acting on its behalf to engage in) any Sanctionable Practices in connection with the Project; (f) comply with World Bank requirements relating to Sanctionable Practices regarding individuals or firms included in the World Bank Group list of firms debarred from World Bank Group-financed contracts; and (g) obtain IBRD's consent prior to agreeing to any change to any transaction document which would affect the rights or obligations of IBRD under the Guarantee Agreement or any other Guarantee-related agreement.
Payment of Fees to IBRD:	Payment of fees due to IBRD is the obligation of the L/C Beneficiary. [However, if the L/C Beneficiary fails to pay any installment of the fees due to IBRD in full or when due, [Other Party – Bidder may propose another party, subject to acceptance by the GoU and IBRD] may elect to

	pay the unpaid amount of the fees and seek reimbursement from the L/C Beneficiary.]
Initiation Fee:	15 bps of the Guarantee Face Value (but not less than USD 100,000)
Processing Fee:	50 bps of the Guarantee Face Value. (In exceptional cases, projects may be charged over 50 bps of the Guarantee amount.)
Front-end Fee:	25 bps of the Guarantee Face Value
Standby Fee:	25 bps per annum, charged periodically and applied to that portion of the guaranteed amount that IBRD has contractually committed and for which IBRD does not yet have financial exposure under the Guarantee. The IBRD standby fee is normally charged semi-annually and accrues sixty (60) days after the date of signing of the agreement providing for IBRD's Guarantee. The Standby Fee must be paid in advance on regular payment dates.
Guarantee Fee:	[50-100] basis points per annum (depending on the Guarantee average life – refer to separate pricing table below). The IBRD Guarantee Fee is charged on that portion of the guaranteed amount that IBRD has contractually committed and for which IBRD has financial exposure under the Guarantee (Maximum Guaranteed Principal). The Guarantee Fee must be paid in advance semi-annually on regular payment dates. The Guarantee will terminate in the event of nonpayment of any installment of the Guarantee Fee.
External Legal Costs:	Fees and expenses of IBRD's external legal counsel will be borne by the L/C Beneficiary.
Cooperation Agreement	
Parties:	**IBRD** and **National Electric Grid of Uzbekistan JSC** (NEGU, a GoU-owned company, as Purchaser under the PPA, and also as L/C Applicant)
Cooperation agreement:	NEGU will covenant, among other things, that it will: (i) comply with all its obligations under the transaction documents; (ii) obtain IBRD's consent prior to agreeing to any change to any transaction document which would materially affect the rights or obligations of IBRD under the Guarantee Agreement or any other transaction document; (iii) provide certain notices to IBRD; (iv) take all action necessary on its part, in accordance with and as required by the terms of the Project-related agreements to which it is a party, to enable the L/C Beneficiary to perform all of the L/C Beneficiary's obligations under the Project Agreement and other relevant transaction documents; and (v) cooperate with IBRD and furnish to IBRD all such information related to such matters as IBRD shall reasonably request; and promptly inform IBRD of any condition which interferes with, or threatens to interfere with, such matters.

Guarantee fee applicable to IBRD Guarantees per average maturity

IBRD Guarantee Average Life	IBRD Guarantee Fee

Up to 8 years	50 bps
From 8 to 10 years	60 bps
From 10 to 12 years	70 bps
From 12 to 15 years	80 bps
From 15 to 18 years	90 bps
From 18 to 20 years	100 bps

ANNEX 2: PROJECT DESCRIPTION

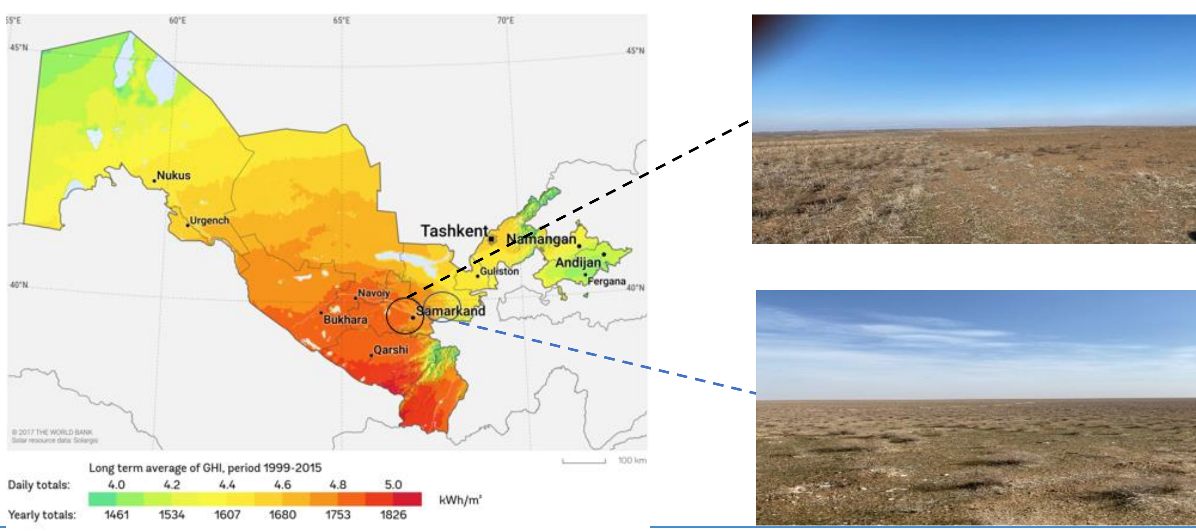
UZBEKISTAN
Scaling Solar 2 Independent Power Producer (IPP) Project

Following the success of the Navoi Scaling Solar IPP Project (P170598), the GoU has indicated a keen interest in proceeding with the Scaling Solar 2 IPP Project on competitive basis. This second phase under the Scaling Solar Program comprises the development by Masdar of two solar power generation plants in the Jizzakh and Samarkand regions with a capacity of 220 MW each. Notably, the Jizzakh and Samarkand regions are among the most dynamic in the country. With populations of 4 million and 1.5 million, respectively, gross domestic product of the region (GDPR) of Samarkand and Jizzakh reached 53.7 trillion (~ US$ 4.9 billion) and 23.4 trillion (~ US$ 2.1 billion) Uzbek Soums, respectively, contributing 7.3% and 3.2% of GDP formulation as of 2021.

Samarkand Scaling Solar IPP: The Samarkand plant will be located in the Kattakurgan district of the Samarkand region at a 427-Ha site and will have an estimated 220-MW capacity and 560GWh/year-electricity production on average across its lifetime. The plant will be connected to the nearest 220-kV Ishtihan transmission substation by a 4-km 220-kV transmission line. The transmission line will be constructed by a private developer and transferred to NEGU at commercial operation date (COD). The site has access to reasonable road and logistic facilities in good condition, and its north side can be reached via the M37 road and its south side via the 4P46 road (Both asphalt roads are in good condition).

***Jizzakh Scaling Solar IPP**:* Another plant with an estimated capacity of 220 MW will be located in the Gallaorol district of the Jizzakh region to produce an estimated 577 GWh/year on average, on a 601-Ha site. The plant's grid connection will be secured through the 220-kV Saribazar electrical substation and a 15-km transmission line to the PV site, to be constructed by a private developer and transferred to NEGU at COD. The site is accessible from a dirt road of about 0.5 km that connects the west side with the R-42 road (an asphalt road in good condition that goes from Gallaorol to Karakchi).

Figure 6. Scaling Solar 2 IPP Project: Jizzakh and Samarkand Sites



On May 20, 2021, following the opening of financial proposals for Scaling Solar 2 IPP, Masdar was awarded the project in the Samarkand region with a bid to supply solar power with at a total energy charge of US¢1.791 per kWh. Masdar also awarded the project in the Jizzakh region with a bid to supply solar power at a total energy charge of US¢1.823 per kWh. NEGU and Masdar amended the Scaling Solar 2 PPAs in July 2022.

The design of the PV plants, equipment, and materials and their installation, construction, and testing will be in accordance with local, national, and international codes and standards such as International Organization for Standardization (ISO), International Electrotechnical Commission (IEC), and will consider the anticipated impacts of climate change in the area. All civil works will be designed to be capable of withstanding the effects of water, extreme winds, and other natural disasters. The drainage system is designed such that buildings, equipment, and internal roads are protected. The plant monitoring and control system will perform supervisory control and data acquisition functions.

The selected EPC Contractor, China Dongfang Electric Group Co., Ltd. (Dongfang Electric Group), is competent and experienced to do the work. Dongfang Electric Group is a large conglomerate focused on engineering and construction, including the development of more than 80 GW of power plants globally. The works for the Scaling Solar 2 IPP are not particularly complex technically and there is no significant underground work. Construction is expected to start shortly after financial close for a total duration of about 27 months from initial mobilization and site preparation to commissioning of the entire PV plants.

For the construction of the PV project, four module manufacturers are being considered at this stage: JA Solar, Jolywood, Longi Solar, and Suntech. At this stage of a project, it is common to consider several different manufacturers for the project, and the final one will be selected at a later stage (i.e., after full Notice to Proceed to the EPC contractor and solar supplier negotiations). The Lenders' technical advisor considers all module suppliers to be well suited for the Project.

ANNEX 3: ECONOMIC AND FINANCIAL ANALYSIS

UZBEKISTAN
Scaling Solar 2 Independent Power Producer (IPP) Project

The economic and financial analyses presented in this annex are only relevant for the Scaling Solar 2 IPP PV Plants at the Jizzakh and Samarkand sites (2 x 220 MW) The analyses are conducted on the basis of certain assumptions described below.

Methodology

The economic feasibility of the proposed Project is assessed using a standard cost-benefit analysis. Net economic benefits of the Project are calculated using the total costs (excluding taxes and financial costs) and total benefits. The economic cost for the Scaling Solar 2 IPP Project captures the (a) EPC cost, (b) Project development cost, and (c) O&M costs during the economic life of the plant. The total economic benefits are: (a) avoided operation costs of the power plants displaced by this Project; and (b) avoided costs of greenhouse gas (GHG) emissions, except for scenarios described as "excluding GHG impacts".

Avoided operation costs are estimated assuming a simplified counterfactual scenario where the absence of the project results in increased dispatch of a CCGT running on natural gas with 61.7% efficiency that would deliver the same amount of yearly energy than the Jizzakh and the Samarkand plants. Avoided costs of GHG emission are estimated using: (a) the methodology of UNFCCC's "Tool to Calculate the Emission Factor for an Electricity System" (version 7.0), to estimate the amount of avoided emissions; and (b) SPCs of the *low* and the *high* scenarios defined in the World Bank "Guidance Note on Shadow Price of Carbon in Economic Analysis".

When estimating avoided emissions with aid of UNFCCC's methodology indicated above, the following assumptions and input data were used:

(i) The *operation margin* (OM) is defined using the "Average OM" method, based on initial results of the Decarbonization 2050 scenario of the least-cost expansion plan developed by the World Bank. Disaggregated data for each power plant in the system were not available. Rather, projections of domestic generation were only available per classes of assets defined by fuel type. The generation shares per fuel class were used to estimate the OM, and conservative assumption were adopted on average efficiency and heat rate for each class. Emissions per unit of fuel consumed, and global warming potentials of methane and nitrous oxide are defined in accordance with IPCC guidelines.
(ii) As detailed data to determine the *build margin* (BM) based on past capacity additions were not available, the simplified procedure defined in section 6.6.2 of UNFCCC's tool was used for this analysis. The year in which the share of renewable generation capacity was expected to surpass 20% was defined based on initial results of the Updated Least-Cost Generation Expansion Plan available when this document was produced.
(iii) The standard weights applicable to solar photovoltaic plants were used to estimate the combined margin as a weighted average of the OM and BM.

This procedure leads to a conservative estimation of avoided emissions, in comparison with the situation where baseline emissions had been estimated assuming the dispatch of a CCGT running on natural gas for the entire horizon. On the other hand, using the avoided dispatch (i.e., only variable production costs) of a gas fired efficient CCGT to estimate avoided generation costs in the counterfactual scenario is also a conservative choice, in comparison with a situation where the full marginal expansion and operation costs were used. These conservative choices were made because full output information of the power system analyses to allow the determination of the marginal technology mix were not available when the analyses were prepared in early November 2022.

Emissions associated with the construction of the transmission system connecting the wind plants to the bulk grid are negligible in comparison with emissions in the baseline scenario, due to the short length of the connection circuits (4.5 km and 15 km) and considering that a connection with a double circuit at 230 kV would typically require a right-of-way with approximately 300 ft in width. Emissions due to technical losses in connection circuits with these features are also negligible, at 0.15% and 0.50% of the transported energy even under a very conservative calculation.

The country-specific economic discount rate used was 8 percent per year. This means that the Project is categorized as economically feasible when the ERR exceeds the hurdle rate of 8 percent. As per the Guidance Note on Shadow Price of Carbon in Economic Analysis, results are reported for *low* and *high* shadow price of carbon, as well as without valuing avoided emissions at all (ERR without GHG impact).

Details of the counterfactuals, assumptions, results, and sensitivity analysis of the Project are described below.

A. Project Economic Analysis

A comprehensive economic analysis was carried out for the Uzbekistan Scaling Solar 2 IPP Project, which consists of two 220 MW of solar photovoltaic (PV) plants at Jizzakh and Samarkand sites each (440 MW in total). The input data and assumptions used correspond to the best information available at the time of project appraisal.

The economic viability of both the Samarkand and the Jizzakh solar projects are assessed using a standard cost-benefit analysis which are evaluated in terms of the NPV and ERR from total economic costs and benefits attributable to the Project. The economic costs and benefits are expressed in real price terms in constant US$2021. The economic analysis is exclusive of any taxes and duties that might be applicable to the Project inputs and outputs.

As this is an economic analysis, fuel prices are valued at opportunity costs for Uzbekistan. Opportunity costs of natural gas are assumed to be defined by imports. The values are estimated based on a conservative estimate of gas prices at the neighboring border with Turkmenistan, from which Uzbekistan imports, and estimates of transport costs within Uzbekistan. Estimates of prices at Turkmenistan's border are defined by net-backing projections of gas prices in China, and in turn assuming that the percent changes in price (in constant US$ of 2021) at which Turkmenistan will be able to sell gas to China roughly follow movements of the real spot price in the Japanese market. The projections for the spot price in the Japanese market are based on the October 2022 World Bank Commodity Markets Outlook.

The project technical assumptions and inputs (installed capacity, capacity factors, grid connection cost) are based on the technical parameters received from the Sponsor, for the *reference scenario of assumptions*. Sensitivity analyses were carried out for key parameters such as capex, opex, to attest that the economic performance remains satisfactory even if the actual costs depart from sponsor projections that may be optimistic. The most significant departure from the sponsor's assumptions in the *reference scenario* is to assume that the project starts early operations in 2024, allowing at least 12 months for implementation of the plants. The sponsor's original assumption was that operations would start in October 2023.

A summary of the assumptions and the results for the economic analysis is presented in table 3.1.

Table 5. Summary of Economic Analysis for Scaling Solar 2 IPP Project

ASSUMPTIONS			
		Jizzakh	**Samarkand**
	Social Discount rate	8%	8%
1	PV nominal capacity	220 MW	220 MW
2	Implementation period	2023	2023
3	Commercial operation year	Jan/2024 (1 year and 6 months ahead of target contractual date, July/2025).	Jan/2024 (1 year and 6 months ahead of target contractual date, July/2025).
4	Lifetime	31.5 years	31.5 years
5	Capacity factor	29.94% (average value across lifetime)	29.06% (average value across lifetime)
6	Capex cost per watt	US$0.73	US$0.71
7	Average annual degradation of PV output[26]	0.21%	0.26%
8	Capacity credit of solar	0.0%	0.0%
RESULTS			
9	EIRR (including GHG impact, low carbon price)	12.4%	12.3%
10	EIRR (including GHG impact, high carbon price)	16.8%	16.6%
11	EIRR (excluding GHG impact)	9.8%	9.7%
NET PRESENT VALUE			
12	NPV (including GHG impact, low carbon price)	US$92.0 million	US$88.1 million

[26] Degradation is affected by local weather and environmental conditions. The parameters indicated in this table were received from the sponsor and are assumed to take local conditions into account.

13	NPV (including GHG impact, high carbon price)	US$146.8 million	US$141.6 million
14	NPV (excluding GHG impact)	US$36.7 million	US$34.3 million
AVOIDED EMISSIONS ACROSS PROJECT LIFETIME (SIMPLE SUM)			
15	Lifetime GHG emission, undiscounted	1.74 million tons CO_2	1.70 million tons CO_2

The capacity factors reported above are average values across the project lifetime, resulting in an average yearly generation of 1,137 GWh/year, or 1.1 TWh/year as reported for the corresponding PDO indicator.

Net economic benefits of the project are calculated using the total system costs (excluding tax and financial costs) and the total benefits from the avoided costs. Avoided operation costs are estimated assuming a simplified counterfactual scenario where the absence of the project results in increased dispatch of a CCGT running on natural gas with 61.7% efficiency that would deliver the same amount of yearly energy as the Jizzakh and the Samarkand plants. Avoided costs of emissions are estimated using the methodology and inputs described in the "Methodology" section of this Annex.

The analysis takes a conservative approach to developing the counterfactual scenario: the capacity credit for the solar power plant is assumed to be zero (0) percent (as the winter peak in the country is at or around 7 pm per the hourly dispatch in Uzbekistan), and hence it is assumed that the energy produced by the solar power plant will replace only the fuel cost from the thermal (gas and coal) power plants. It is assumed that the transmission and distribution losses for the solar power plant and those of the alternative sources are the same. The average electricity generation of the Project is estimated at around 577 GWh per year for Jizzakh solar power plant and 560 GWh per year for the Samarkand solar power plant over their Project life of 31.25 years (P50).

The analysis indicates that the Project is economically viable with and without factoring in the GHG impact. The analysis indicates that the Project is economically viable when avoided emissions are valued at the *low* and the *high* scenarios of SPCs of the abovementioned guidance note, and also when avoided emissions are not valued at all. The economic rate of return (ERR) for the Jizzakh solar power plant is 12.4 percent per year if emissions are value at the *low* SPCs, 16.8 percent per year for the *high* scenario, and 9.8 percent per year if emissions are not valued at all. For the Samarkand solar power plant, the corresponding values of the ERR are 12.3, 16.6 and 9.7 percent per year, respectively. The economic rate of return for both solar power plants are above the hurdle rate and demonstrates the positive economic returns of the Project. Table 3.1 displays other evidence of the satisfactory economic performance of the project.

Avoided GHG emission. The amount of avoided GHG emissions were estimated based on the methodology described at the beginning of this Annex. The Project in total is expected to contribute to a reduction of about 3.4 million tons of CO2 over its lifetime and an average of about 109,306 tons per year. This includes emissions avoided during the 1 year and 6 months of operation in advance of the target delivery date specified in the PPA[27], during which the project receives pre-operational payments as early generation, per the sponsor's schedule.

C. Sensitivity Analysis (of Project Economic Analysis)

A sensitivity analysis was conducted for the following key cost-and-benefit drivers in the economic analysis: (a) discount rate; (b) O&M costs of the plants; (c) capital cost of the plants, and (d) generation output of the plants. The economic performance of the Project is generally robust in all scenarios, as indicated by the following results:

(i) *When avoided GHG emissions are valued at* low *SPCs:* ENPV turns negative when the social discount rate exceeds 17 percent. ERR drops below hurdle rate when the capex exceeds US$1.04 per watt, which is relatively high compared to recent market trends.
(ii) *When avoided GHG emissions are valued at* high *SPCs:* ENPV turns negative when the social discount rate exceeds 24 percent. ERR drops below hurdle rate when the capex exceeds US$1.27 per watt.
(iii) *When avoided GHG emissions are not valued at all:* ENPV turns negative when the social discount rate exceeds 11 percent. ERR drops below hurdle rate when the capex exceeds US$0.83 per watt.

B. Project Financial Analysis

Results of financial analysis. The analysis arrived satisfactory positive IRRs, indicating that the projects are expected to generate sufficient cash flows to recover capital expenditure, and meet operational and maintenance expenditures. Furthermore, after meeting debt-service costs, the cash flows for both the projects allow for regular dividend payments, providing the shareholders with a reasonable return for a project of this nature. The lenders' base case further confirmed an Average Debt Service Coverage Ratio (ADSCR) of 1.2x for the Project, indicating adequate comfort for the lenders comprising the debt syndicate.

The Scaling Solar 2 IPP will contribute to power-sector financial sustainability by reducing the average power purchase cost for NEGU as off taker. The proposed Project's tariffs are lower than the weighted-average electricity-purchase cost of NEGU from state-owned power plants. The Scaling Solar 2 IPP is thus expected to deliver a positive NPV for NEGU and the sector at about US$ 40 million from 2025-30 (when applying a discount rate of 8.8 percent, the current yield of the Eurobond issued by the GoU maturing in 2030).

D. Sector Financial Analysis

Financial performance of NEGU. NEGU has faced a challenging financial situation over the last two years, failing to breakeven on an operating-cost basis in 2021. In the absence of a regulated cost-plus tariff regime, NEGU's revenues did not rise enough to offset the large increases in operating costs in 2021,

[27] The project is expected to commence operations in advance of the target contractual delivery date by 1 year and 3 months. This early operation is expected due to the advanced stated of project preparation arrangements, including licensing and land use, and due to construction and montage times of solar power plants recently experience in Uzbekistan. This would result in operations commencing in the beginning of 2024, and the total technical lifetime being extended to 31 years and 3 months. This lifetime would be made feasible due to prudent operational expenditures across the project life, including a refurbishment in 2037. There is, however, some deterioration in energy output across the lifetime. The refurbishment costs were considered in the economic and financial analyses.

primarily driven by the cost of electricity purchase, which grew by ~19 per cent as a result of an increase in generation tariffs for domestic power plants. NEGU's revenues witnessed a muted increase of 7% in 2021 due to factors including a) pause in end-consumer tariff hikes towards cost recovery to help the economy recover from the COVID-19 pandemic, b) softer increase in domestic electricity demand than expected, and c) decrease in export demand leading to fall in export revenues by 16%. Notably, operating costs represented 108% of total revenues in 2021, resulting in EBITDA of UZS -1,584 billion (~US$ -149 million) at an EBITDA (operating) margin of -8 per cent.

Negative profits have adversely impacted NEGU's ability to ensure recovery of capital expenditure and debt-service costs from operations. NEGU has been unprofitable for the last two years, reporting net losses of UZS 113 billion (US$ 11 million) and UZS 2,344 billion (US$ 220 million) in 2020 and 2021, respectively. In 2020, NEGU registered a positive EBITDA margin of 4%, generating enough revenues to cover electricity cost and (debt) interest costs, while not enough for re-investment. The situation worsened in 2021, as NEGU's EBITDA margins were negative, and it experienced a shortfall of UZS 380 billion (US$ 36 million) in the full recovery of capital and debt-service costs. NEGU financed this shortfall through on-lent loans from IFIs, shareholder loans from the MoEF, and accumulation of payables to state-owned generation companies.

NEGU is yet to access commercial debt markets, and its existing debt is primarily from development banks, including the World Bank, in the form of on-lent loans through GoU's MoEF and guaranteed by the sovereign. As of end-December 2021, NEGU's outstanding debt stood at UZS 2,868 billion (US$ 265 million), of which the World Bank (IDA&IBRD) accounted for the largest share at 50%, followed by ADB at 23%. NEGU's current loans are in foreign currencies, which could entail substantial foreign-exchange risk, especially in the context of increasing volatility expected across global financial and currency markets in the short/medium term. In this context, the World Bank's ongoing Strategy Programmatic ASA and pipeline Electricity Transmission Modernization and Market Development (ETMMD) project will support NEGU to develop robust debt-management plans to mitigate foreign-exchange volatility risk.

Due to sustained losses since its establishment in June 2019, NEGU's equity reached minus UZS 581 million (US$ 54 million) at the end of December 2021. Negative equity stock is usually a cause for concern, although given NEGU's full state ownership, strategic position in the country's energy sector, and strong support from the sovereign in the form of shareholder loans, equity infusions, and explicit guarantees of NEGU's loans, NEGU will be able to sustain operations while the MoEF implements cost-recovery tariffs.

The GoU is aware of NEGU's current challenging financial situation, and has guaranteed all ongoing liabilities, including payments to IPPs, and debt obligations of NEGU. Recent developments in the country's energy sector are also expected to improve NEGU's financial standing. These include tariff increases amongst some categories of non-residential consumers (who pay higher tariffs than residential consumers) in July 2022 that also positively impacted transmission tariffs, and initiatives such as smart metering that will help maintain high bill collection efficiency rates (100% expected in 2022, further improved through advanced metering). As end-consumer tariffs become more cost-reflective, NEGU's payables to state-owned generation entities are expected to come down in line with enhanced revenue collections. Lastly, the implementation of IPPs, which have a lower cost than Uzbekistan's current inefficient generation fleet, is also expected to improve NEGU's profitability.

NEGU financial performance forecast. NEGU's financial projections have been prepared based on feedback received from key GoU stakeholders, including MoEF, Ministry of Energy (MoE) and NEGU, that highlighted improvement in NEGU's financial performance in the medium/long term as sector reforms are implemented. The following key assumptions around electricity demand, tariffs, investment plan, and other key parameters, form the basis of NEGU's financial forecast:

(i) As per Least Cost (LC) Generation Expansion Plan and dispatch efficiency analysis for Uzbekistan, baseline electricity demand for the country is expected to rise at a compound annual growth rate (CAGR) of 6.5% to reach c. 132 TWh by 2030, from c. 80 TWh in 2022;

(ii) NEGU's financial performance is expected to gradually improve from 2022 onwards, and it is expected to reach full cost recovery by 2026 on the back of upward revisions of end-consumer tariffs by GoU/MoEF in accordance with cost-plus tariff methodology. During the transition period from 2022-25, it is assumed that NEGU's tariff will increase to enable it to improve cost-recovery linearly from 94% in 2022 (vs 88% in 2021) to 100% in 2025;

(iii) Implementation of over 10 GW of IPP projects that are planned to be operational over the next 5 years. By 2030, these IPP projects are expected to meet c.50% of Uzbekistan's electricity demand. Notably, these IPPs have lower electricity-generation costs by ~12% on average compared to Uzbekistan's current fleet of old inefficient gas plants. Therefore, increasing the share of IPPs in its electricity mix will help NEGU reduce operating costs, and in turn, improve financial performance over the medium/long-term;

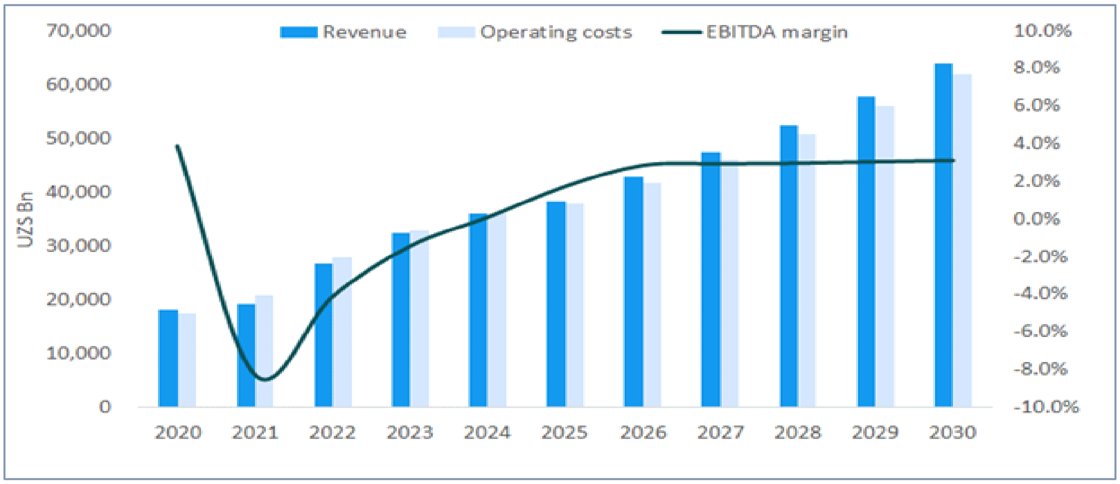
(iv) In an effort to absorb substantial variable renewable energy (VRE) into the country's energy grid, and modernize the grid, NEGU is expected to incur capital expenditure at US$ 100 million equivalent in 2023, increasing gradually to US$ 200 million equivalent by 2030. It is useful to note that NEGU's capital expenditure stood at US$ 81 million and US$ 65 million equivalent in 2021 and 2020 respectively;

(v) NEGU is expected to receive continued support from GoU in the form of equity financing, shareholder loans and on-lent loans from IFIs, including from the World Bank, to finance its capital expenditure program;

(vi) NEGU is able to stabilize its working capital situation and receive near-term liquidity support from GoU as needed to repay past dues of state-owned power producers;

The financial forecasts indicate improved EBITDA margins for NEGU over the next few years (2022-25) as sector reforms, particularly in relation to tariffs, are implemented. During these years, NEGU is not expected to generate adequate cash flows to meet interest costs or for reinvestment, and therefore would rely on sovereign support (of up to US$350 million, based on forecasts). That said, from 2026 onwards, NEGU's operational performance will improve due to cost-reflective tariffs, enabling it to generate adequate resources to not only service debt, but also channel to re-investment. During the forecast period from 2022-30, the weighted-average cost of power purchase by NEGU from state-owned generation companies and IPPs will increase by 1.5 times from UZS 325 per kWh in 2022 to UZS 441 per kWh in 2030. On the other hand, NEGU's average sale tariff is expected to be between UZS 316 per kWh and UZS 469 per kWh in 2022 and 2030, respectively, increasing above purchase price in 2026, following the same trend in later years, thus enabling NEGU to cover operational expenses and capital costs from 2026. The chart below provides a snapshot of NEGU's revenues, operating costs, and EBITDA margin for the historical (2020 and 2021) and forecast years (2022 onwards).

Figure 9. NEGU's revenues, operating costs and EBITDA margin (2020-30)



NEGU's overall leverage is expected to rise in the coming years as NEGU is not expected to generate resources from operations for its development program and would need to rely on IFI loans for financing 80% of its capital expenditure to modernize and expand its transmission network. Due to this, NEGU's debt is expected to reach a peak of 37% of total assets by 2025, from 28% in 2021. In the absence of a cost-plus tariff regime during the forecast years of 2022-25, and amid rising debt levels, NEGU will need to rely on enhanced GoU support to maintain comfortable cover ratios, i.e., Interest Service Coverage Ratio (ISCR) and Debt Service Coverage Ratio (DSCR). Such additional liquidity requirement is expected to be about UZS 4,554 billion (US$ 377 million equivalent) from 2022-25, averaging UZS 1,138 billion per year from 2022-25.

Additional liquidity requirements signify NEGU's enhanced reliance on sovereign support to ensure its creditworthiness, including in the form of provision of explicit government guarantees for all of NEGU's debts. Furthermore, GoU support to NEGU may also come in the form of equity, shareholder loans (as in the past), and/or tariff adjustments earlier than planned.

Notwithstanding the strong support from the sovereign and recent positive developments, NEGU operates in a risky environment that could adversely impact its operational and financial performance. Key risks faced by NEGU are described below.

(i) Delay in implementing tariff reforms: Cost-recovery tariffs are critical for NEGU to reach operational and financial sustainability. The current (interim) tariff structure arrives at NEGU's tariff by deducting generation and distribution tariffs from end-consumer tariffs, entailing a heightened risk of adverse impact on NEGU's revenues to accommodate rising generation and distribution costs.

(ii) Foreign-exchange risk: NEGU has sizeable liabilities in foreign currencies. These primarily include debt service obligations on outstanding IFI debt, and dollar-indexed payments to IPPs under Power Purchase Agreements. Such foreign-currency liabilities of NEGU are expected to increase due to reliance on IFI debt to finance its transmission-network development program, coupled with the rising share of IPPs in the country's energy mix. In the event such increased foreign-exchange losses are not passed onto consumers through higher tariffs, NEGU's financial situation could be negatively impacted.

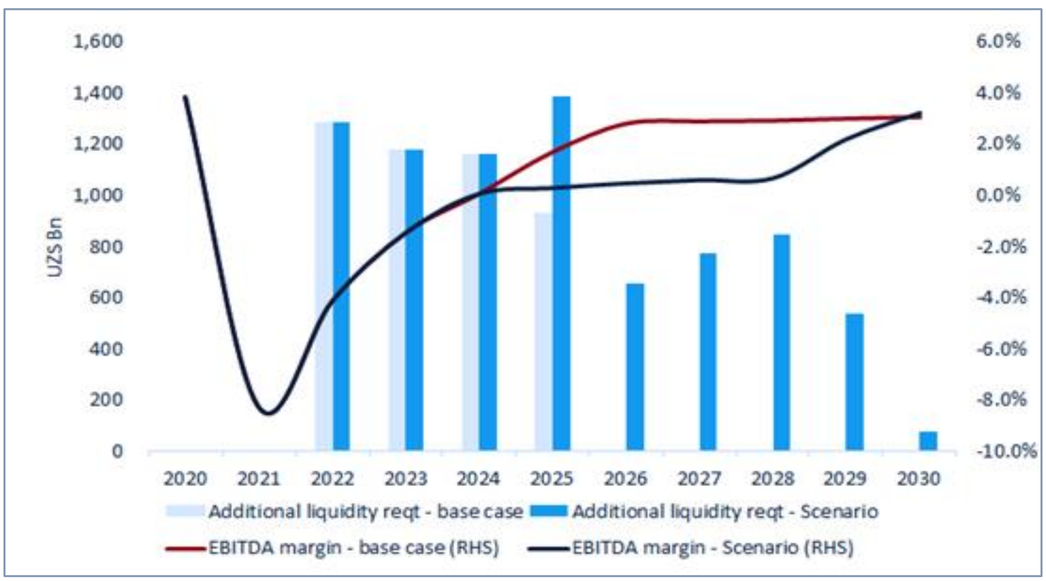
(iii) Increase in purchase price of electricity: Since NEGU is expected to continue to rely on gas-based electricity generation to meet the majority of the country's demand in the short-medium term, there is risk of a rise in power-generation costs due to higher gas prices not fully subsidized by the GoU. Furthermore, NEGU's balance sheet may be directly exposed to gas-price risks under its tolling arrangements with IPPs if such risks are not fully transferred to entities best placed to manage it (gas suppliers). Additionally, NEGU's electricity purchase price may also increase if it makes deemed-energy payments to IPPs under 'take-or-pay' contracts. In the event such incremental generation costs are not passed onto end-consumers, NEGU's financial situation could be negatively impacted.

Sensitivity analysis. A sensitivity analysis has been conducted to evaluate the impact of key risks on NEGU's financial performance and sustainability. The analysis presents the following two scenarios:

Delay in implementing tariff reforms: Under this scenario, the implementation of tariff reforms committed to by the GoU will face a four-year delay, reaching full cost-recovery levels only by 2030, as compared to the base case of 2026. This scenario could also include delays in regulatory approval of NEGU's cost components, resulting in a time lag in recovering costs.

While the cost of generation, and accordingly power purchase, will be growing as projected under the base-case scenario, the tariff increases will be lower, resulting in lower EBITDA margins. Net profit margin would range from –5.7% in 2022 to -1.4% in 2028 to 1.0 percent in 2030 (year of full cost recovery). Due to the delays in tariffs reaching cost-recovery levels, it is expected that NEGU will need additional liquidity support during the 'delay-impacted' years from 2026-29 to cover operating expenses and debt-service payments and will incur capital expenditure as it will be unable to generate adequate resources for re-investment due to non-cost-reflective tariffs. The chart below compares the EBITDA and additional liquidity requirements under this scenario with the base case.

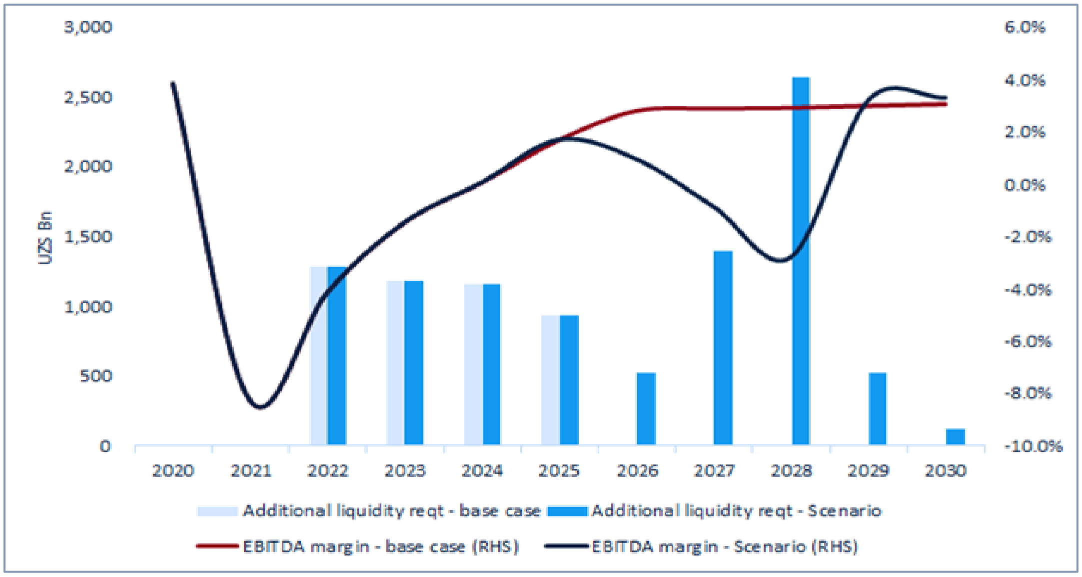
Figure 11. EBITDA margin and additional liquidity requirements - Base case vs delay-in-tariff-reform scenario



Foreign-exchange risk: Under this scenario, the Uzbekistani Som is assumed to depreciate at a higher rate of 8% per annum during 2026-28 as against 3% assumed in the base case for these years. The years 2026-28 are chosen for the scenario analysis since, by 2026, the planned 10GW+ IPPs (that have USD-indexed tariffs) are expected to become operational, and NEGU's debt-service payments in hard currency are also expected to rise substantially to an annual average of UZS 630 billion (US$ 45 million), thus entailing significant foreign-exchange risk for NEGU. It is further assumed that increased foreign-exchange losses are not passed onto consumers through higher tariffs.

While NEGU's debt-payment obligations (which are in foreign currencies) and IPP costs (which are USD-indexed) increase in UZS Som terms due to higher depreciation of the UZS Som vis-à-vis the USD during the period 2026-28, its revenues, which are primarily in UZS Som, are assumed to stay equal to the base-case scenario. Therefore, NEGU's EBITDA margin will drop sharply during 2026-28, the years during which the local currency is assumed to depreciate more against the USD. This results in additional liquidity requirements for NEGU to meet the higher debt-service costs and IPP power-purchase costs, and to fill the financing gap created due to lower cash generation from operations. The chart below compares the EBITDA and additional liquidity requirements under this scenario with the base case. It is useful to note that NEGU requires higher EBITDA in later years (2029 and 2030) to offset the increase in interest costs from usage of a higher proportion of debt to finance its capital expenditure.

Figure 12. EBITDA margin and additional liquidity requirements - Base case vs higher-USD-UZS-depreciation scenario during 2026-28



Despite the current situation at NEGU and the risks highlighted above, the risk of a draw on the WB-guaranteed LCs is mitigated by the following:

(i) *Expected improvement in financial performance:* Scheduled commercial operation date for the Scaling Solar 2 IPP is 2025, at which time it is expected that tariffs would continue to rise towards cost-recovery levels. Based on the projections above, NEGU will be generating positive EBITDA by that time and will be further advanced on its path towards financial sustainability as demonstrated by healthy leverage and profitability ratios.

(ii) *GoU commitment for NEGU's foreign-currency liabilities:* GoU is committed to a growing electricity sector and recognizes the importance of ensuring on-time payments to IPPs. Currently, NEGU's cash waterfall characterizes IPP payments as the top priority only after external debt payments, and all of these are sovereign-guaranteed. It is estimated that by 2030, debt and IPP payments will comprise up to 27% of NEGU's revenues. In addition, MoEF is in the process of creating a fund capitalized by SOE privatization proceeds to help backstop IPP payments. NEGU's financial condition is tied to the decision-making of MoEF (including tariffs), which is unlikely to allow such a vitally important entity to fail or fall short of its external obligations.

(iii) *Project participants:* With the participation of Masdar as sponsor, which has over 1 GW of power projects in operation or under development in Uzbekistan, as well as the involvement of ADB, AIIB and EBRD, the likelihood of unrectified payment issues is reduced.

(iv) *Cure period:* A call on the WB guarantees would materialize only if NEGU (or its successor) fails to pay the L/C bank one year after the date on which an L/C is drawn. The one-year period is critically important to allow NEGU (and MoEF) to rectify the issue (and the Bank to support that process and avoid a call on the guarantees).

It is also worth noting that the WB guarantee exposure of US$ 12 million for the Project (vs. total Project costs of approximately US$ 380 million) is relatively low, although still critical in ensuring that the Project moves forward and Uzbekistan remains on track to deliver on its energy-mix objectives.

The GoU committed to pursuing further reform initiatives, including cost recovery and sector sustainability. These initiatives will be implemented as part of broader reform program to sustain sector reform: (a) implementation of a tariff cost-recovery trajectory by 2026; (b) establishment of a sector regulator as well as a separate regulated tariff for NEGU; (c) implementation of loss-reduction programs, especially in the distribution segment, to improve sector operational efficiency; and (d) maintenance of state ownership over the transmission segment as the backbone of the sector, while promoting private-sector participation in generation and distribution, among other segments.

As part of the ongoing policy dialogue, and upon request from the GoU, the World Bank will continue supporting the GoU and NEGU reform initiatives to (a) design and implement a tariff cost-recovery trajectory while protecting vulnerable households; (b) establish and operationalize a sector regulator as well as a separate regulated tariff for NEGU; (c) demonstrate, introduce, and prioritize, including through the proposed Project, the benefits of transparent and competitive selection of private investors; and (d) improve energy-sector operational efficiency, financial sustainability, and capacity. World Bank support will be provided through the pipeline Innovative Carbon Finance for Energy Reforms Projects, Programmatic Technical Assistance, the ongoing MUTS and ESTART projects, and series of DPO engagements.

ANNEX 4: Environmental and Social Review Summary

UZBEKISTAN
Scaling Solar 2 Independent Power Producer (IPP) Project

Disclaimer. *This Environmental and Social Review Summary (ESRS) is prepared and distributed in advance of the World Bank (WB) Board of Directors' consideration of the proposed transaction. Its purpose is to enhance the transparency of WB's activities, and this document should not be construed as presuming the outcome of the Board of Directors' decision. Board dates are estimates only. Any documentation attached to this ESRS has been prepared by the Project Sponsor, and authorization has been given for public release. WB has reviewed this documentation and considers that it is of adequate quality to be released to the public but does not endorse the content.*

1. Project Description. The World Bank's support to the proposed Project comprises two IBRD payment guarantees, each one to support the development of a solar photovoltaic (PV) plant in Uzbekistan, each with a capacity of 220 MW in Samarkand and Jizzakh regions (for a total of 440 MW) with private-sector entities as Independent Power Producers (IPPs). The electricity generated from the Project will be sold to the Uzbekistan state-owned power utility, National Electric Grid of Uzbekistan JSC (NEGU), under the Power Purchase Agreements (PPAs). The Project is being developed by Abu Dhabi Future Energy Company PJSC ("Masdar"), a UAE-based private renewable energy and sustainable urban development company owned by Abu Dhabi National Oil Company (ADNOC), Mubadala Investment Company PJSC ("Mubadala"), and Abu Dhabi National Energy Company PJSC (TAQA). The Project will be supported by two payment guarantees in the aggregate amount of US$12 million from IBRD. The Project's solar PV plants are expected to be completed within 27 months under turnkey EPC contracts, with construction activities commencing in Q2/Q3 2023. The EPC contractor will also provide Operations & Maintenance ("O&M") services for the first two years following the commercial operation date (COD), to be subsequently replaced by an affiliate of the sponsor, Masdar. The plants are anticipated to be operational for 31.25 years. The Project is the second one in Uzbekistan under the World Bank Group's Scaling Solar Program and builds on the successfully developed and operational first project, the Navoi Scaling Solar (1) IPP project (P170598; 100 MW).

2. **Project Sites.** Samarkand and Jizzakh are among the most dynamic regions in the country. With populations of 4 million and 1.5 million, respectively, gross domestic product of the region (GDPR) of Samarkand and Jizzakh reached 53.7 trillion (~ US$ 4.9 billion) and 23.4 trillion (~ US$ 2.1 billion) Uzbek Soms, respectively, contributing 7.3% and 3.2%, respectively, of GDP as of 2021.

Samarkand Scaling Solar IPP: One plant will be established in Kattakurgan district of Samarkand region at a 427-ha site and will have an estimated 220 MW of capacity and 560 GWh/year of electricity production on average. The plant will be connected to the nearest 220-kV Ishtihan transmission substation by a 4-km 220-KV transmission line. The transmission line will be constructed by the private developer, Masdar, and transferred to NEGU at COD. The site has access to reasonable road and logistic facilities in good condition, and the north side can be reached via the M37 road and the south side via the 4P46 road (Both asphalt roads are in good condition).

Jizzakh Scaling Solar IPP: A second plant, also with an estimated capacity of 220 MW, will be in Gallaorol district of Jizzakh Region and will produce an estimated 577 GWh /year of electricity on average, on 600.24 ha of land. The proposed plant is a medium-size utility-grade grid-connected solar-PV power system consisting of approximately 300,000 PV panels, string inverters, power conditioning units, 220-kV step-up power transformers, and grid-connection equipment. The plant's grid connection will be secured through the 220-kV Saribazar electrical substation and a 15-km transmission line to the PV site, to be constructed by the private developer, Masdar, and transferred to the state off-taker, NEGU, at COD.

3. Overview of Potential Environmental and Social Risks and Impacts.

The environmental risks are assessed to be moderate. Physical works envisaged under the Project are of a small to medium scale and the expected environmental impacts are site-specific and easily identifiable, and mitigation is readily identifiable. Impacts associated with the proposed construction and operational phases of the project may include increased pollution due to improper care, handling, and storage of construction material and waste, generation of excessive noise, and dust levels, occupational health and safety risks of workers during construction and operation, management of community health and safety risks and impacts and biodiversity conservation of natural habitat.

The social risks are also assessed to be moderate. The social risks and impacts associated with the proposed construction may include land acquisition and economic impacts, restriction of access to footpaths leading to grazing sites, discrimination and sexual exploitation and abuse/sexual harassment, community conflicts and unrest, labor influx, child and forced labor, including forced-labor risks associated with polysilicon manufacturers (in light of the procurement of solar panels for the Project).

The two Project Companies (Jizzakh and Samarkand) have prepared, consulted upon, and disclosed site-specific draft E&S instruments, including Environment and Social Impact Assessments (ESIAs), Social Due Diligence Reports on Land Acquisition, Livelihood Restoration Plans (LRPs), and Stakeholder Engagement Plans (SEPs) during Project preparation. E&S Management Systems (ESMSs) have been finalized and will be incorporated in the final ESIAs. The Project companies disclosed the Project's E&S instruments on December 19, 2022 (*http://www.masdar.ae*) and consulted on them with Project stakeholders. The Project companies revised these instruments and disclosed the revised versions on January 17, 2023.

The World Bank's existing guarantee project in Uzbekistan, Navoi Scaling Solar (1) IPP Project, (100-MWac / 130-MWp PV plant in Navoi region), is being implemented by the same sponsor, Masdar, and is currently under supervision. The World Bank has rated the Navoi Project Company's E&S performance satisfactory to date.

(a) Identified Applicable Performance Standards

The following World Bank Performance Standards (WB PSs) apply to the Project:

PS 1 – Assessment and Management of Environmental and Social Risks and Impacts
PS 2 – Labor and Working Conditions
PS 3 – Resource Efficiency and Pollution Prevention
PS 4 – Community Health, Safety and Security
PS 5 – Land Acquisition and Involuntary Resettlement

PS 6 – Biodiversity Conservation and Sustainable Management of Living Natural Resources

Based on the Bank's review of the Project, PS7: Indigenous People is not applicable as there are no Indigenous Peoples in the Project's area of influence. PS8: Cultural Heritage is not applicable as the Project is not located in areas of known historical or cultural significance and does not impact any known cultural heritage. However, the Project Companies will nonetheless develop a chance-finds procedure to be applied in the event of the identification of cultural/archeological resources in the Project area.

The Bank will periodically review the Project Companies' ongoing compliance with the World Bank Performance Standards.

(b) Overview of World Bank's Scope of Review

The Bank carried out appraisal of the Project, including the Project sponsor and the Project Companies, through a desktop and virtual process (conducted via a series of video / tele conferences). The Bank's appraisal, which is an ongoing process in guarantee operations, and was initiated in 2021, included a desktop review of available information, including an Environmental and Social Scoping Report developed during the IFC Scaling Solar Uzbekistan advisory project, the draft E&S instruments prepared for each site, including ESIAs, SEPs, Social Due Diligence on Land Acquisition Reports, LRPs, as well as the Companies' responses to a series of ESHS questionnaires. The Bank's appraisal focused on the Companies' capacity to manage ESHS risks and compliance with Uzbek regulatory requirements, the Bank's Performance Standards, and the WBG's EHS Guidelines. Specific items reviewed included information relating to: (i) the Companies' and the EPC contractors' capacity to manage the ESHS risks under the Project; (ii) the Companies' HR policies and labor management procedures, especially in relation to working conditions and terms of employment; (iii) construction-related occupational health and safety of the Companies' staff and contractors, including any primary supply-chain labor issues associated with migrant and/or seasonal workers, i.e., forced and child labor; (iv) water resource availability; (v) community health and safety and security; (vi) provision of adequate alternate land to previous land owners; and (vii) early engagement with surrounding communities and other stakeholders as to the Project.

(c) Environmental and Social Categorization and Rationale

This is a Moderate Risk Project under WB's Operation Policy 4.03 Performance Standards for Private Sector Activities. The proposed investment is expected to have limited number of temporary, site-specific environmental and social impacts may result that can be avoided or mitigated by adhering to generally recognized performance standards, guidelines, and design criteria. The Project's key E&S impacts are associated with the following aspects of the Project, namely, water use for cooling and panel cleaning during operation, land use, use of hazardous materials in manufacturing, labor and working conditions, and contractor management during construction. However, these impacts are expected to be limited, generally Project-specific, and with limited areas of influence, and can be addressed through the implementation of CESMPs and good international and industry practices.

Further, it is possible to design and implement engineering and management measures to mitigate potential adverse impacts during both construction and operation. The E&S risks are rated as being moderate (as per IBRD classification), and the sponsor's E&S management system and the E&S Action Plan (ESAP) provide appropriate mitigation for those that have been identified.

(d) PS1 - Assessment and Management of Environmental and Social Risks and Impacts

The E&S risks are both rated as being moderate, as mentioned above. To address these risks, the sponsor, Masdar, has prepared the following draft instruments for each subproject site: (i) a detailed ESIA; (ii) a SEP; (iii) a Social Due Diligence Report on Land Acquisition; and (iv) an LRP. Prior to construction, the Project Companies will (i) prepare site-specific E&S instruments as part of their ESMS; (ii) finalize and implement the LRPs; (iii) conduct public consultations on all of these instruments; (iv) submit them to the Bank for clearance; and, following such clearance, (v) disclose them, and authorize the Bank to do so as well.

The Project is expected to benefit the local and regional economy during construction due to the direct procurement and supply of materials and services from the area. Construction of the solar PV plants with the proposed installed capacity is expected to employ 700 to 1,000 personnel at each site between technicians and lower-skilled personnel. Experience from similar projects has indicated that, whilst local employment is a positive impact, significant adverse impacts can occur if the local recruitment process is not adequately managed, risking poor acceptance of the project by local communities, and possible related grievances. To mitigate this risk, the Project Companies will prepare Local Recruitment and Employment Plans (LREPs) setting out their recruitment strategies and processes, including for the promotion of equal opportunities, including preferential treatment of vulnerable groups, women, and Project Affected Persons (PAPs). In addition, the Grievance Mechanism was designed for each site to provide prompt feedback and remedial action, with procedures on a platform that is free from manipulation, interference, coercion, intimidation and retribution, and readily accessible to all affected parties. Any grievances are to be addressed in a timely and responsive manner.

Environmental and Social Policy and Management System

At the corporate level, the Project Companies have developed Quality, Health, Safety, and Environment (QHSE) policies that outline their commitment to quality, the prevention of injury and ill-health, limiting pollution of the environment, and compliance with applicable statutory and regulatory requirements. At the Project level, the Project Companies will develop Project-specific ESHS policies, building upon their corporate-level policies and defining their ESHS objectives and principles in alignment with applicable laws and regulations and consistent with the objectives of the WB PSs and World Bank Group Environmental, Health and Safety Guidelines. These policies will also indicate the departments / units in the respective Project Company that will be responsible for their implementation. The Companies will communicate the policies to all Project employees and will require their contractors to develop their own ESHS policies aligned with these.

At the corporate level, the Project Companies have developed a QHSE Management System (MS), detailed within a QHSE MS Manual, which is aligned with the general requirements of ISO 14001, ISO 9001, and ISO 45001. The Companies are yet to establish mechanisms to align Project-level and corporate-level QHSE MS.

The Project Companies have developed site-specific ESMSs (E&S Management Systems) for both the Jizzakh and Samarkand projects. The ESMSs comply with the requirements of PS1 and are in line with the objectives of ISO 14001 (Certification is not required) applicable to this Project. The Companies' ESMSs set out processes to be implemented to provide adequate management and supervision of their contractors, including the engagement of Project Management Consultants (PMCs).

The ESMSs outline the process developed by the Companies to identify environmental and social risks and impacts and detail the specific measures to be implemented to manage these. The Companies will require their EPC and O&M contractors to develop ESMSs to identify, manage, and control risks related to these contractors' activities and those of their sub-contractors. The ESMS will be updated for the operational and decommissioning phases of the Project to manage the relevant E&S risks associated with operational and decommissioning activities, generating the operational ESMS and decommissioning ESMS.

Identification of Risks and Impacts

During the inception phase of the Project, i.e., Scaling Solar advisory project, IFC commissioned a Site Suitability Report and an E&S Scoping Report to provide Project bidders with a basic understanding of environmental and social conditions in the Project area and assess the general suitability of the sites.

Following the Project award, the Project Companies developed detailed site-specific ESIAs in accordance with good international practice. The purpose of the ESIAs was to conduct assessment of impacts based on existing information, to be supplemented by additional surveys. The ESIAs also identify the type and extent of further assessments, including residual impacts if required. The findings of ESIA and the site-specific E&S instruments will be adopted, consulted, and disclosed as a condition of the first disbursement under the financing. Further site surveys will be carried out as part of the full ESIAs when circumstances allow. The full ESIAs will be finalized prior to construction.

In addition to the above, the Project Companies are required to develop Environmental Impact Assessments (EIAs) in accordance with national legislation and seek regulatory approval of these EIAs prior to the commencement of construction activities.

The Companies have engaged a local consulting firm to assist in the development of the EIA documents as per regulation of applicable country law. To date, Stage I – Concept Statements of Environmental Impact have been completed in accordance with local requirements and submitted to the regulators. The

Project sites are on pre-existing government land, and the change of lease contracts entails economic resettlement of (i.e., livelihood impacts on) land leaseholders and users. The Companies have identified the potential impacts in site-specific Social Due Diligence Reports developed during Project preparation. The Livelihood Restoration Plans (LRPs) prepared for each site and for the areas of the connecting overhead transmission line describe the measures to be taken by the responsible government agencies to compensate Project-affected persons. If these measures do not meet the relevant requirements of PS5, the Companies will develop a Resettlement Action Plan for each site to complement government action. This may include additional compensation for lost assets, and additional efforts to restore lost livelihoods where applicable. Civil works may also have accidental damages to private property or livelihood that will need to be mitigated by the Project Companies. Any social issues to be raised related to handing back of the leased land will also need to be addressed by the Government Administrations of Jizzakh and Samarkand regions.

Management Programs

The ESIAs include project specific ESMP matrices, which will provide summaries of the E&S risk management procedures, monitoring requirements, and mitigation measures applicable to both the construction and operation phases of the Project. As per ESAP Actions Items 3 and 4, the Companies will also develop Construction E&S Management Plans (CESMPs) and Operation E&S Management Plans (OESMPs). The CESMPs and the OESMPs will be based on the ESMPs and will be reviewed and approved by the Project lenders prior to adoption and disclosure.

The CESMPs will consist of a suite of sub-plans, which will address: waste management; pollution prevention (including emissions, erosion, spill response, etc.); water management (including supply, treatment, disposal, etc.); handling of hazardous materials; biodiversity management (including fauna/flora, topsoil, ground disturbance, invasive species, etc.); emergency preparedness and response; community health, safety, and security; road safety and traffic management; accommodation management; local recruitment and labor management; labor influx management; cultural heritage (including chance finds); environmental monitoring; stakeholder engagement (including grievance management); and contractor management.

The OESMPs will consist of a suite of sub-plans, which will address waste management; pollution prevention (including emissions, erosion, spill response etc.); water management; handling of hazardous materials; emergency preparedness and response; community health, safety, and security; biodiversity management; environmental monitoring; and stakeholder engagement (including grievance management).

The Companies will require their contractors to develop, implement, and ensure continued compliance with these contractors' ESMPs and associated procedures aligned with Project requirements and the contractors' scope of work. These documents will be reviewed and approved by the LTA and / or Project lenders prior to adoption.

Construction- and operation-phase E&S management and monitoring responsibilities will be shared between the Companies and their contractors. Both the EPC and O&M contractors will be contractually bound to adhere to the requirements for which they are indicated as being the responsible party in the ESMPs.

As per ESAP Action Item 5, the Companies will develop, and will require their EPC & O&M contractors to develop, OHS MSs aligned with (but not necessarily accredited under) OHSAS 18001 / ISO 45001, and elements of ESMSs to identify, assess, manage, and control risks.

As part of the OHS MSs, the Companies will develop OHS Management Plans (ESAP Action Item 6) setting out the Project's health and safety management framework and establishing OHS requirements / standards for the Project, including to address the risk of COVID-19 infection in the workforce.

The Companies will require their EPC & O&M contractors to develop a series of plans, procedures, and systems to adequately manage occupational health and safety risks associated with their respective scope of work. The contractors will establish comprehensive risk identification and management processes and permits for the working system.

The EPC and O&M Contracts are key tools to ensure compliance with the Companies' ESHS requirements. As per ESAP Action Item 7, the Companies will develop and include ESHS and labor provisions (including in alignment with the WB PSs) and compliance conditions in their EPC and O&M contracts, which will provide contractors and third-party service providers (including security agencies) clear guidelines as to performance requirements.

<u>Organizational Capacity and Competency</u>

The Companies have QHSE and Corporate Social Responsibility (CSR) Managers at the corporate level, who will have oversight of EHS issues under the Project. The sponsor has demonstrated the capacity to manage a range of environmental, social, and occupational health and safety issues under renewable-energy projects for which IFC has provided funding.

The Companies' project managers will have overall responsibility for environmental, social, health, safety, and security management for the Project. The Companies will establish Project-level boards which will include ESHS representation and require ESHS reporting. The Project boards will provide a means for ESHS issues to be raised and escalated as appropriate to senior management. A mechanism to link the Project-level boards and the corporate-level boards should also be established.

The Companies will appoint qualified professionals to be responsible for the environmental, social, health, and safety aspects of the Project. The Companies will engage PMCs to provide additional support in Project supervision, including ESHS oversight.

As per ESAP Action Item 8, the Companies will appoint, at site level, suitably qualified stakeholder-liaison managers, and, through the PMCs, suitably qualified Project E&S and OHS managers as well as human resource management support. At the corporate level, the Project Companies will appoint qualified E&S and OHS representatives to be the lenders' focal points throughout the duration of both the construction and operation phases of the Project.

The appointed individuals will have adequate qualifications and experience, including knowledge of international requirements and best practice. These individuals must have sufficient authority and resources to fulfil their responsibilities as required by the site-specific E&S instruments and the SEPs, lenders' requirements, the ESAPs, and local regulation.

As per ESAP Action Item 9, the Companies will require their PMCs and their EPC and O&M contractors (including these contractors' sub-contractors) to appoint suitably qualified environmental, labor, and OHS teams to manage their scope of work. These individuals must be aware of, and fully understand, the obligations and responsibilities placed upon them by the site-specific ESMSs.

Emergency Preparedness and Response

The Companies and their contractors will develop Emergency Preparedness and Response Plans (EPRPs) in accordance with World Bank PS1 and PS4.

These plans will cover preparedness and responses to a range of potential emergency scenarios, including medical emergencies (including pandemic-type outbreaks), fire, earthquakes, extreme weather conditions, transport incidents, and major hydrocarbon spills. Security-related incidents, such as acts of sabotage/vandalism, terrorism etc., will be addressed through separate Security Management Plans (ESAP Action Item 21).

The EPRPs will include communication protocols to alert local authorities and communities as appropriate, as well as the management teams, in addition to specific response and evacuation procedures.

Regular drills and emergency exercises covering the different emergency scenarios will be conducted by the EPC and O&M contractors.

Monitoring and Review

The Companies will be responsible for reviewing and formally auditing their contractors in relation to ESHS performance and compliance with agreed Project standards and national and lender requirements.

As part of their ESMSs, the Companies will establish procedures and allocate resources to monitor and measure the effectiveness of their and their contractors' E&S management plans / programs and compliance with relevant Uzbek regulatory requirements.

E&S / OHS monitoring requirements of the EPC and O&M contractors in relation to their activities and those of their subcontractors will be set out in their CESMPs, and their implementation will be closely followed by the Project Companies.

The EPC and O&M contractors will provide, as a minimum, monthly E&S / OHS reports as part of their general reporting on the Project. These reports will include coverage of the contractors' E&S / OHS performance in accordance with reporting requirements outlined in the site-specific ESIAs & ESMSs. Clear key performance indicators (KPIs) will be developed as part of the reporting.

The Companies will appoint independent auditors to undertake reviews of the effectiveness of CESMPs and OESMPs quarterly during the construction phase, and annually in the first three years of the power plants' operation as noted in ESAP Action Item 10.

(e) PS2 – Labor and Working Conditions

The Project is expected to employ approximately 140 workers at each site at the start of construction, up to 900 workers at each site during peak construction, and up to 120 workers at each site at the end of construction. During operation, around 20 workers, working in shifts, will be employed at each site. The EPC contractor's site management teams are expected to comprise around 30 individuals at each site during the construction period.

The precise number and gender of local and national workers expected to be employed are yet to be fully established by the Companies, although it is likely that approximately 400 technicians and low-skill personnel will be required at each site. These workers could be sourced either locally or nationally, depending on the skill sets available.

<u>Human Resources Policies and Procedures</u>

At the corporate level, the sponsor has an existing Code of Conduct (Mubadala Code of Conduct) which sets out principles related to sponsor's core values and ethics. These principles will be applied throughout Project execution.

The Project Companies will develop site--specific human resources policies in line with the WB PS2, International Labor Organization (ILO) guidelines, and national labor laws. The HR policies will contain an enforceable Code of Conduct applicable to all project workers. As per ESAP Action Item 11, the HR Policy must include a commitment to (i) non-discrimination, equal rights, and equal pay; (ii) prevention of child labor and forced labor; (iii) freedom of association and right to collective bargaining; (iv) terms of employment, including hours of work, overtime arrangements and overtime compensation, and the right to refuse overtime requests; (v) zero tolerance for any proven case of gender-based violence (GBV) and workplace abuse and harassment, including sexual exploitation and abuse/sexual harassment (SEA/SH).

All Project workers will be informed about the HR Policies in their language(s). Induction training on the HR Policies will be provided to all newly hired workers.

The Project Companies will ensure that all contractor and sub-contractor workers, including site-security personnel understand and sign the Code of Conduct prior to the commencement of works.

As required under PS1, the Companies will develop and implement contract-management plans which will set out mechanisms to monitor and enforce contractors', sub-contractors', and service providers' (including private security agencies') compliance with labor management procedures and human resources requirements.

<u>Non-Discrimination and Equal Opportunity</u>

The Companies will commit to avoid all forms of discrimination against their employees based on age, gender, sexual orientation, health, race, nationality, political opinions, or religious beliefs. The principles of non-discrimination and equal opportunity will be applied to all contractors and subcontractors as part of the Companies' contractual obligations.

<u>Recruitment</u>

As per ESAP Action Item 12, the Companies will supplement existing publicly available local workforce statistical data by undertaking further stakeholder engagement to determine likely local and regional workforce availability and gender and skill levels to guide workforce planning.

The Companies will engage with local community leaders (including the Deputy *Khokims* (Governors) of Samarkand and Jizzakh regions, Deputy Heads of the Investment Departments of Samarkand and Jizzakh Region *Khokimiyats* (governments), senior members of the local Women's Committees, and the chairmen of the *mahalla* (neighbourhood) committees in order to establish adequate workforce profiles. The Companies will discuss the skills required and provide indicative job specifications during the engagement. All necessary measures will be taken to appropriately manage community expectations.

This engagement will assist the Project Companies to prepare Local Employment and Recruitment Plans (LEPs) setting out their recruitment strategies and processes, including for the promotion of equal opportunities. The LEPs will describe how women and Project Affected Persons (PAPs) will be selected on a preferential basis, along with other residents from the two most affected communities, for recruitment and training in advance of the start of construction. The LEPs should include an analysis of local workforce skills against required worker profiles and numbers, a description of engagement with regional vocational training centers that could be used (potentially with the support and technical assistance of the Project Companies) to provide vocational training, and employment targets for women. The Companies will be required to maximize the inclusion of women in the workforce as much as reasonably practicable and investigate different options such as working with local NGOs to assist in achieving this aim. Roles and responsibilities associated with local recruitment between the Companies and their appointed contractor(s) should be clearly defined.

The Project Companies' recruitment strategies will be required to include a tiered approach under which recruitment campaigns will focus on recruiting suitably skilled individuals from Project-affected communities as a priority. Should such suitable individuals not be available, the recruitment campaigns will expand to the local region, and subsequently other regions of Uzbekistan as necessary. Employment of international workers would be the last resort.

The Companies will ensure that their contractors develop LEPs aligned with the requirements set out in the Companies' LEPs.

<u>Working Conditions and Terms of Employment</u>

As per ESAP Action Item 13, the Companies will ensure that all employee contracts are consistent with local labor codes and ILO and WB PS2 requirements.

All employees will be provided a copy of their contracts (in a language they understand), and these will stipulate the terms of employment, such as working conditions (including health and safety requirements), wages and benefits, hours of work, overtime arrangements and overtime compensation, annual and sick leave, maternity and paternity leave, and vacation and holidays.

All construction-phase worker contracts will clearly describe the short-term nature of the Project and provide an indication of likely employment duration.

The Companies will ensure that their contractors provide all workers, including those of sub-contractors, written documentation setting out the terms and conditions of their employment as per the above. This requirement will also extend to any personnel engaged via labor-hire companies.

The Companies will ensure that all workers have contracts and background checks, including references from their most recent employers.

Workers' Grievance Mechanisms

As per ESAP Action Item 14, the Companies will develop confidential grievance reporting, referral, and support systems for workers. These workers' grievance mechanisms (WGMs) will be consistent with local labor codes and WB PS2 requirements.

The WGMs will involve an appropriate level of management (including designated staff and accountability, and the establishment of an appeals panel) to address concerns promptly and an understandable and transparent process that provides timely feedback to those concerned without any retribution. The WGMs will include specific considerations for grievances related to harassment / gender-based violence.

The Companies will provide specific training to grievance officers and general awareness to employees on harassment and bullying and will engage with women employees as to their concerns regarding transportation and safety.

The Companies will ensure that their WGMs are available to their contractors, sub-contractors, and service providers' personnel as required. The PMCs will be responsible for the operation and management of the WGMs during the construction phase of the Project. It is anticipated that the PMCs will identify named individuals who will take the lead on this, but these individuals will be supported by panels which will consider appeals. Final arbitration will sit with each Company's Project Management. During operations, the O&M contractors will establish and operate WGMs aligned with the Companies' WGMs (and including necessary training and awareness programs).

The existence and availability of the WGMs will be clearly communicated to all employees through employment contracts, HR plans, and site induction processes.

Workers' Accommodation

The worker accommodation camps will not be located on sites; rather, the workers will stay in existing accommodation at nearby settlements. It is not anticipated that there will be any displacement as a result of worker accommodation. If any such displacement is identified, it will trigger the need to update, or develop an addendum to, the LRPs.

The final ESIAs will include an assessment of likely risks and impacts associated with workers' accommodation and will propose necessary mitigation measures.

As per ESAP Action Item 15, the Project Companies and the EPC contractor will prepare an Accommodation Management Plan for each site in line with IFC and EBRD's "Worker's Accommodation: Process and Standards" Guidance Note. These plans will incorporate social risks posed and impacts caused to nearby communities by Project workers, including required mitigation measures.

Consideration of the need for workers' accommodation must take account of the COVID-19 pandemic and the potential impact of the construction workforce on the local communities. The Companies' and the EPC contractor's Accommodation Management plans must assess the health risks to the workforce and put appropriate measures in place to protect the workforce and the local communities.

The Project Companies will conduct safety audits to identify specific settings under / affected by the Project that might carry an increased risk of sexual exploitation and abuse/sexual harassment (SEA/SH), such as workplaces and labor camps. For instance, the Project Companies will consider whether adequate measures have been put in place to manage interaction points with communities, such as truck stops.

Every effort will be made to provide safe, secure, and separate living spaces for male and female construction workers, including adequate lighting and segregated washing facilities.

Workers' Organizations

It is unlikely that workers' unions will be involved in the Project as the main concentration of workers will be on-site only during short construction periods; as noted above, the operations staff will be small. The Companies will not prevent workers from seeking to join unions or other workers' organizations in any way; this will be specified in the Companies' labor policies and procedures.

Child and Forced Labor

No child or forced labor will be used under the Project at any time. Proof of identification and age verification will be required at the time of employment. The Project Companies will ensure that appropriate provisions are included in the contracts with their contractors/suppliers.

Workers' Occupational Health and Safety

Key OHS risks for a solar PV project include slips and falls, hazards from on-site moving machinery and heavy load lifting, traffic accidents, electric shocks and burns, and safety issues relating to PV-module assembly.

As mentioned under the section on PS1, the Project Companies will develop, and will require their EPC and O&M contractors to develop, OHS MSs aligned with OHSAS 18001 / ISO 45001. The OHS MSs will include Project-specific OHS plans / procedures for the construction and operations phases. These procedures will cover, among others, the following issues: hazard identification and assessment; construction-site safety (barricades, safety nets, access control, clear demarcation of different areas, provision of safety information to visitors, etc.); specific procedures for hazardous works; workers' safety and training plan; personnel qualification, limitations and equipment needs (e.g., personal protective equipment); site supervision and audit procedures; incident-reporting system and intervention measures (first aid, etc.). The procedures will be specific to the solar PV sector (in terms of industry-specific hazards) and the Project sites. The OHS procedures will also link to the Project-specific EPRP, which will include fire risk assessment and control systems, fire alarm systems and drills, and emergency preparedness and planning as part of OHS procedures for both the construction and operation phases.

The Project Companies, in coordination with their contractors, will also develop and implement a training program under the Project. As per ESAP Action Item 15, the EPC and O&M contractors will develop and implement a Training Needs Analysis and a Training and Competency Plan to ensure that all workers are appropriately trained, skilled, licensed / permitted, and competent to undertake the tasks required of them in their roles.

Supply Chain

As per ESAP Action Item 16, the Project Companies will develop site-specific E&S Supplier and Vendor Management Plans setting out processes to be implemented to identify, manage, and monitor environmental, social, and health and safety risks and track the related performance of key suppliers and vendors under the Project.

The processes will include initial screening and due diligence of key potential suppliers and vendors with a focus on OHS performance, licensing / permitting, and human resources risks (child and forced labor), and an assessment of gender and safety risks in the bidding process for contractors.

The Project Companies will also require the primary suppliers to identify forced-labor risks. If any forced-labor cases are identified, the Project Companies will require the primary suppliers to take appropriate steps to remedy them. Ultimately, if remedy is not possible, the Project Companies will, within a reasonable period, shift the Project's primary suppliers to suppliers that can demonstrate that they meet the relevant requirements of PS2.

(f) PS3 – Resource Efficiency and Pollution Prevention

Resource Efficiency – Greenhouse Gases

Potential pollution issues during the Project's construction phase will be those typically associated with construction sites such as dust, noise, oil spills, and traffic. All these impacts are limited and temporary and will be mitigated with standard mitigations measures which are covered in the site-specific ESMSs. The Project is expected to have limited impact in terms of pollution and its implementation will result in a significant positive impact due to its nature and lower GHG emissions during the operation phase. Significant use of materials and resources, including water and fuel, is not anticipated given the moderate scale of the infrastructure development for the solar parks and transmission lines, and their construction and operation stages. Nevertheless, risks and impacts related to potential water usage, the release of pollutants, waste generation, the management of disposal materials and hazardous wastes, impact on the community, and resource-use efficiency are assessed and identified in the course of the ESIA process, and mitigation measures and risk management plans are proposed for the solar park infrastructure development.

The Project will apply measures set out in the WBG ESHS Guidelines and other good international industry practices for the effective use of raw materials and for optimizing energy use to the extent that they are technically and financially viable. The Project is expected to generate approximately 270 gigawatt hours (GWh) of electricity per year, resulting in a predicated annual GHG reduction of 156,000 tCO_2eq/year.

Resource Efficiency – Water Consumption and Availability

The Companies are yet to fully quantify the water-consumption requirements for the Project. Water required for construction has been estimated as 3,600m^3 per site excluding accommodation requirements, and 10,658m^3 including accommodation requirements. The main water requirement during construction is likely to be for dust suppression, concrete production, and workers' domestic use. It is yet to be determined whether concrete-batching activities will be conducted on-site, although initial calculations have assumed that 0.2 m^3 of water will be required for every 1 m^3 of concrete. Construction-phase water consumption requirements will be fully quantified by the Companies once they have made a decision concerning concrete-batching arrangements and worker-accommodation requirements.

Water requirements during operation and maintenance will be largely focused on solar-panel cleaning. The Companies have indicated that due to high levels of ambient dust, the Project design considered wet cleaning of the PV modules as the base case. The Companies have estimated that this requirement would be approximately 4,620 m^3 per annum per site. The Companies will consider the contextual risk of water scarcity in the region as a guiding principle for selecting a water-efficient cleaning-technology option. The Companies will refine the cleaning regime during the operations phase to avoid unnecessary cleaning and use of water.

The Companies are currently assessing different water supply options available to the Project, including groundwater abstraction, tapping into an existing water pipeline in close proximity to the Project and trucking in water from Samarkand and Jizzakh. It was noted during stakeholder-engagement sessions held with local communities that water scarcity/availability is a primary concern of the local population. The Companies have indicated during the Project's virtual appraisal that water will most likely be sourced from Samarkand/Jizzakh and trucked to the sites. If this is to occur, it is estimated that the water demand would be equivalent to one truck every two days.

As required under ESAP Action Item 17, the Companies will conduct Alternatives Analyses for water use considering the environmental, social, and technical aspects of the Project, will then develop a Water Management Plan for each site that will quantify water requirements for the construction (including non-potable water for mixing concrete and for dust control) and operation phases of the Project and assess potential impact on the availability to local communities (including farmers and herders) of water resources.

If ground water is planned to be used, the Companies will first undertake hydrological studies to determine the availability and suitability of groundwater for construction water and for panel-washing during operation. Groundwater will not be used for potable water unless subject to appropriate treatment.

The Companies will obtain all required permits / licenses / permissions related to water abstraction, consumption, and treatment etc.

Pollution Prevention – Waste

The overall volumes of both solid and hazardous waste generated by the Project during both construction and operation are expected to be low. The Companies are yet to develop a comprehensive estimation of their anticipated waste streams and volumes for the Project or undertake an evaluation of waste treatment and disposal options.

It is anticipated that the Project will produce both non-hazardous waste, such as paper, wood, plastic, scrap metal, and glass, and a limited quantity of potentially hazardous materials such as transformer oils, paints, batteries, and other electronic waste.

As per ESAP Action Item 18, the Companies and their contractors will develop a Waste Management Plan for each Project site aligned with local legal requirements, WB PS3, and WBG EHS General Guidelines.

These Waste Management Plans will commit the Companies to the reduction of waste to the extent possible whilst maximizing the re-use and recycling of materials and will set out processes for appropriate waste storage, segregation, transportation, and disposal/treatment.

The Companies will develop a waste inventory and will undertake an assessment of disposal and treatment facilities/options. Project waste will only be disposed of and treated at appropriately licensed facilities. The Companies will adhere to the principle of "duty-of-care" in waste management.

An assessment of the suitability of each licensed disposal facility will be undertaken prior to use to ensure Project waste is disposed of / treated in a manner that is safe for human health and the environment.

Pollution Prevention – Hazardous Materials

Hazardous materials likely to be required during the construction and operation phases of the Project include hydrocarbons, oils, lubricants, and paints. The Companies and their contractors will establish and implement a hazardous material management plan and spill prevention and response plan for each site that are commensurate with the potential risks. These management plans will address the protection of the workforce and the prevention and control of releases and accidents.

A limited number of waste PV modules are expected to require disposal during the construction phase, and they will be returned to the PV manufacturer for recycling.

Pollution Prevention

During construction, a minor and insignificant amount of pollution to air, noise, water, and soil is anticipated that can be easily mitigated through standard pollution prevention and control measures which will be set out in the respective EMP. During the operational phase, no environmental pollution impacts are anticipated except for wastewater, primarily from panel cleaning, however, the cleaning agent mostly consists of a small portion of vinegar, which is not hazardous to health or the environment, and regular housekeeping waste generation. As a result, no significant water pollution risks for both surface and ground water are sought if disposed naturally. The Companies will require their EPC contractor/O&M operator to implement pollution prevention measures in accordance with national law, WB's Performance Standards, the ESIAs prepared for the Project, and the ESAP for the Project agreed between the Companies and the WB.

(g) PS4 – Community Health, Safety and Security

The project is not expected to lead to significant safety and health impacts on the community. Increased traffic associated with the project, in particular during the construction phase, may pose some safety risks to the community. Before the start of construction, the Companies will assess risks and impacts to host communities of the subprojects. The construction phase may generate noise, dust, solid waste, plastic sheets, and the influx of workers. Based on the findings of the assessments, the Companies and their contractors will develop Community Health and Safety Plans and procedures to protect public health, safety and security on issues that include: road safety and traffic management; potential emergencies and required responses; security measures to prevent unauthorized access during construction; measures to prevent GBV/SEA/SH including the sexual exploitation of children; and measures to prevent the spread of disease, including COVID-19 and HIV/AIDS.

As per ESAP Action Item 19, the Companies will inform local stakeholders about the construction program and any potential associated hazards including expected increases in traffic around the sites. The Companies must actively consult with local stakeholders and authorities regarding the number of workers that will be on-site during the construction period, the potential for employment, worker accommodation, and worker welfare and health. The Companies must also listen and respond to any concerns raised by local stakeholders or authorities relating to COVID-19 and the potential impact on the health of the local community and the capacity of local medical services.

As required under PS 2, the Companies will maintain Codes of Conduct for Workers', which will be applicable to all Project employees, including contractors and subcontractors. It will address non-discrimination, GBV/SEA/SH, and set out rules for interaction with the local population.

The Project Companies will ensure that the community and project workers are aware of the Grievance Mechanisms (GMs) protocols and procedures to enable them to access them. The GMs will be streamlined based on the SEA/SH needs and will be strengthened with the confidential and victim centric approach and information made available to communities and project workers. The referral pathways would need to be developed and services providers identified prior to implementation of any construction activity.

Road Safety and Traffic Management

The ESIAs have identified access to the Project sites for both the Jizzakh and Samarkand sites. The sites are well connected by an active road; therefore, no specific access road is foreseen as being required at this stage. However, necessary road safety and traffic management requirements will be set out in the CESMPs.

The Companies have indicated that it is likely that solar panels required by the Project will be transported by rail to the nearest rail head and trucked to a site. As required under PS1, the Companies / contractors will develop and implement a site-specific Road Safety & Traffic Management Plan (RSTMP) which will set out traffic management and accident prevention measures, including needed infrastructure, signage, and training, to mitigate any impact on local communities. The RSTMPs will define road transport routes to be used by the Project, including routes to be used during equipment deliveries. The transport routes will be based on risk assessments undertaken to evaluate road conditions and to minimize impacts on local communities.

Community Exposure to Disease

As required under PS1, the Companies/contractors will develop a Community Health and Safety Management Plan (CHSMP) for each project site, which will include necessary mitigation measures to minimize the potential for community exposure to water-borne, water-based, water-related, and vector-borne diseases, and other communicable diseases that could result from Project activities. The CHSMPs will avoid or minimize transmission of communicable diseases that may be associated with the influx of temporary Project labor.

The Companies and their contractors will implement stringent measures to manage risks related to COVID-19 and will ensure that adequate plans and procedures are developed to minimize, as much reasonably possible, its transmission. Planning of the Project workers' accommodation will be carried out considering these risks.

Community Emergency Preparedness and Response

As required under PS1, the Companies and their contractors will develop emergency preparedness and response plans (EPRPs) for the construction and operational phases of the Project sites.

The EPRPs will outline the Project's emergency preparedness and response activities, resources, and responsibilities, and will disclose appropriate information to affected communities, relevant government agencies, and other relevant parties as appropriate. The EPRPs will set out engagement processes and ensure that local communities are informed of any emergency which may impact them.

<u>Security Personnel</u>

The entire perimeter of the Project sites will be fenced and entry to the Project sites will be regulated. In the event the Project Companies benefit from state support in ensuring the safety of their workforce and construction sites, the Companies will assess and document risks arising from the Project Companies' use of government security personnel. The Companies will seek to ensure that security personnel will act in a manner consistent with the requirements of PS4 and encourage the relevant public authorities to disclose the security arrangements for the Project facilities to the public (subject to overriding security concerns).

As per ESAP Action Item 20, if the Project Companies additionally employ private security contractors, the Companies will ensure that site security personnel understand and sign the Code of Conduct prepared in line with the requirements of PS2, PS4, and the Voluntary Principles of Security and Human Rights and which will request that public security adhere to the same standards when working with the Companies and supporting their security personnel. The Companies will ensure that their security contractors' personnel are appropriately screened, trained and competent for their scope of work. Security personnel will be made aware that the use of force is explicitly forbidden, and the security guards must not carry arms of any kind (including batons).

Both the Companies and the government agree to notify each other immediately and in writing if they suspect the other has committed a breach of human rights or the laws of Uzbekistan.

(h) PS5 - Land Acquisition and Involuntary Resettlement

The types of land impacts include a) Permanent land acquisition; b) Temporary land take; c) Restrictions in land use (no trees above 6 meters); and d) Loss of public right of way through the sites.

Permanently affected areas include the Solar PV Area and the OTL footprints. These areas of land will be required by the project during construction and operation and will only become available to the community following the decommissioning of the project. Temporarily affected areas are required during the construction phase and include the tower assembly areas adjacent to each tower footprint and an 18m wide right of way along the OTL required for the suspension of power cables. Access to each tower will be obtained via this right of way. The sterilization zone includes land 32m meters either side of the OTL. The land inside the sterilization zone will not be occupied by the project however users of this land will be required to abide by safety conditions, in particular structures or trees over 6m tall cannot be present in this area. All other farming activities can continue inside this zone. The primary routes for the OTL were designed in consultation with affected leaseholders along the line route to avoid impacts on structures, high value crops or other social receptors, where possible.

The PV plants will occupy approximately 1,000 hectares (10 km^2) of land, all of which is government owned. The sites of the PV plants overlap with land that the government had leased out previously to individuals who were using it for livestock-grazing (a single tenant on the Jizzakh site, and five leaseholders on the Samarkand site). In 2018-19, the government informed these tenants of the planned change in land use, and has compensated or/will compensate them for any economic loss

In Samarkand region, five historical leaseholders are affected by the PV solar construction site. According to the Livelihood Restoration Plan, the following measures have been/are to be taken: 1) a tenant farmer at the north-western boundary has voluntarily returned 7 ha, and has 3 ha left in his lease contract; 2) a tenant (two brothers) will be compensated for land tenure due to inappropriate location of replacement land offered; 3) a farmer has returned 40 ha of leased plots, and received alternative land (20 ha) in another area, stating no need for additional land; 4) a farmer has agreed to relocate to a new plot of land 4 kms from the Project site on the condition that he be compensated for the cost of bringing the new plot of land up to the quality of the previous one. This requires installation of a drilling well, power supply, and levelling of the land. The farmer expects financial support from the district government to carry out these works; and 5) a farmer has agreed to return 40.6 ha of land for the Project, and the remaining 15.8 ha is apparently enough to support his family. Leaseholders experienced a reduction of land ranging from 66 to 100 percent. It is important to note however that the level of impact is not symmetrical to the quantity of land lost due to the varying qualities of land taken and investment made into that land. Historic leaseholders 1, 3 and 5 have stated that they were able to complete their harvest before returning the land and so did not lose any crops. These leaseholders also did not invest in their land and therefore will not require compensation for these entitlements. Historic leaseholders 2 and 4 should be considered significantly impacted by the project as both households heavily invested in their land and received no compensation for their investments and were put into debt by the project. It is agreed during the PAP consultations that compensation for historical farmers 2 and 4 will be paid within four months of the site fencing.

Other land impact of the Project comprises restriction of access to a footpath crossing the Project site once fencing is installed. The Solar PV area in Samarkand as well as large areas to the north and east, are currently used by community herders to graze sheep, goats and cattle. In total, 42 herder households participated in the census and 218 project affected people (individuals) were recorded as part of the herder households. 7 out of 42 herders are professional herders, whereas the rest herded on behalf of the community. Unlike the solar site at Jizzakh, community grazing land at Samarkand is not abundant and there is already conflict between herders and farmers over space which was mentioned in interviews with the historical leaseholders. The impact on the professional herders should be considered significant as the Project will directly interfere with their primary source of income. Impacts will also be experienced, albeit to a lesser extent, by all households in the community who own the herds who are taken to the project area by community herders or by the professional herders. During discussions with the design team and interviews with herders, Mahalas, Khokimiyat and other community members possible livelihood restoration measures were discussed. The following options were discussed and ruled out for reasons explained below:

- Maintaining access to for herders to graze inside the Solar PV Area, between the panels was discussed with the design and engineering team however the health, safety and liability risks were too high for this option to be considered feasible. It was however concluded that allowing selected community members inside to collect feed manually, without bringing in any livestock could be possible.
- Transitional support will commence as soon as the site is fully fenced and access to herders is denied during construction.
- Creating vertical wheat feed banks was discussed as a potential option with herders as this is something that has been practiced in the community. This option was however considered less favorable than collecting feed from within the project area. Vertical wheat feed banks could

however be considered as a backup option if access to the Solar PV becomes unviable in the future.

Preliminary analysis of the route has identified the current land uses which could potentially be affected by the construction and operation of the Overhead Transmission Line (OTL). This analysis is based on the following assumptions: a) a clearance corridor of 32m on each side of the OTL will be required as a wayleave; b) a clearance area of 5 m² has been defined around the required towers; and c) a total number of 19 towers will be pre-designed. In summary, the OTL will affect four farms and will require the acquisition of approximately 0.678 ha of land. No structures were identified along the route of the OTL, and all the fields that will be affected grow seasonal crops.

In Jizzakh region, there are currently no leaseholders in the solar PV area. There was, however, a historic leaseholder that was identified as part of the social impact assessment process. The Jizzakh subproject site is located at the top of a hilly plateau that stretches from northwest to southeast. The area is used primarily for livestock grazing. The local population uses the land without any official permits. The Jizzakh subproject area was zoned as a poultry farm, but in 2018 the party leasing the land returned it to the government as it was apparently not productive. There are no physical structures, including any wells, on the subproject site. Community members and herders interviewed during site visits pointed out the importance of the community pathway to the north of the site used by local villagers (mainly by people who live in the Sayfin Ota community) to commute for herding activities as well as to reach the farming area located to the north of the Project site.

The proposed 15-km OTL will connect the Project sites to the existing Saribazar substation. An analysis of the route for the OTL has identified the current land uses which could potentially be affected by the construction and operation of the line: a) A clearance corridor of 32m on each side of the OTL will be required as a wayleave; b) a clearance area of 5 m² has been defined around the required towers; and c) a total number of 66 towers have been pre-designed. Farmers will be able to continue to use the land under the OTL and will not be impacted by the construction; however, they will no longer be able to grow trees that exceed 6m in height or build structures in this area. Estimates show that construction of the transmission line will require the acquisition of approximately 2.54 ha of farmland. The farmland and crops of eight leaseholders will be affected along the route of the OTL. However, the permanent impact is below 5%, so cash compensation will be provided for loss of income.

In accordance with PS5, restoration of livelihood losses for PAPs is the responsibility of the government and will be carried out by the Regional Administrations (*Khokimiyiats*) in the form of alternative land on lease and/or compensation to the previous leaseholders and current land users, prior to the fencing-off of the sites. While the current users of the land, the farmers whose use it for animal grazing, may not be leaseholders, they should also be provided with alternative land for animal grazing as they will likely experience adverse livelihood impacts due to the Project.

As required under ESAP Action Item 21, as the Project sites are on pre-existing government land, and the change of lease contracts entails economic resettlement of (i.e., livelihood impacts on) land leaseholders and users, the Companies have identified the potential impacts in site-specific Social Due Diligence Reports developed during Project preparation. The Livelihood Restoration Plans (LRPs) prepared for each site and for the areas of the connecting overhead transmission line describe the measures to be taken by the responsible government agencies to compensate Project-affected persons. If these measures do not meet the relevant requirements of PS5, the Companies will develop a Resettlement Action Plan for each site to complement government actions. This may include additional compensation for lost assets, and additional efforts to restore lost livelihoods where applicable.

There is a potential difference in recognition of compensation between national legislation and Project Lenders requirements, since national legislation refers to market value, which may not be perceived to reflect full replacement cost. Masdar will provide compensation for the loss of any crops or infrastructure based on full replacement value which includes market value in addition to elements such as transaction costs, interest accrued, transitional and restoration costs and other applicable payments. There is no requirement to provide development benefits under the national legislation. However, it is best practice to ensure the local community benefits from the project. To enable this, Masdar will offer employment benefits in the construction and operational phase of the Solar Farm development. Masdar will also develop a Community Development Plan during the construction phase.

Special resettlement assistance will be required for vulnerable people because they are less able to cope with economic displacement compared with others. The LRPs have identified seventeen (17) vulnerable households and individuals. Vulnerable people are defined as those who might be disproportionally negatively impacted by the project or who might not be able to take full advantage of project benefits. Vulnerable households will be subject to additional focused monitoring during implementation of the LRPs. If the household wishes to spend Livelihood Restoration packages on supporting the disabled family member rather than on investment in land, this will be permitted. Transitional support (available to herders until livelihood restoration is underway) may be extended for vulnerable households in financial difficulty. The project will support the households to access existing government disability benefits by liaising with the local Khokimiyat. The project will offer critically vulnerable households' preference in employment opportunities. Specific livelihood restoration measures will be targeted at women and land enhancement grants (for historical farmers) will be split between farmer and spouse.

Farmers along the OTL lines and their spouses expressed an interest in training on business and finance, increasing agricultural yields, veterinary training and also meat, milk and wool processing (land enhancement grants were not an option presented to farmers along the OTL due to the very small impact).

The final LRPs will include a Social Audit of the sites and surrounding areas, as well as a clear explanation of the land compensation and livelihood restoration undertaken for the Project sites.

(i) PS6 – Biodiversity Conservation and Sustainable Management of Living Natural Resources

The PV plants will occupy approximately 1,000 hectares (10 km2) of land and will be enclosed by a 2.5-meter-high fence. The land is currently used for the rough grazing of livestock and is generally flat with few features of interest. Previous attempts to cultivate arable crops were made, but a combination of poor soil quality and lack of water made this unviable. The regions are semi-arid, and the sites predominantly covered by scrub grassland. Although the sites are used for grazing by domestic animals, it is largely natural habitat, although localized areas within the northern and eastern boundaries of the sites have been modified by historic cultivation. Therefore, the Natural Habitat requirements of PS6 apply to the Project.

The Project sites are located to the eastern edge of the Central Asian Southern Desert Ecoregion, with the Alai-Western Tian Shan Steppe Ecoregion to the south. Both ecoregions are characterized by desert vegetation, which varies according to soil type. The Project is not located in the vicinity of a cultural-heritage zone or a Protected Area as defined in PS6. The nearest Key Biodiversity Area (KBA) is the Tudakul and Kumazar Reservoirs Important Bird Area (IBA) located 20 kms to the southwest of the Project sites, which includes the Tudakul lake, which is saline, and the Kuyu-Mazar freshwater reservoir.

Botanical surveys completed in March 2020 confirmed that the Project sites largely support wormwood steppe. This relatively sparse vegetation community is dominated by the wormwood species (*Artemisia* spp.), frequent Isirik (*Peganum harmala*), frequent grasses (e.g., *Poa* spp.) and occasional thistles (*Asteraceae*). Although the sites are used for grazing by domestic animals, it is largely Natural Habitat as defined in PS6, although localized areas within the northern and eastern boundaries of the site have been modified by historic cultivation. A single IUCN-declared Endangered bird species, the Steppe Eagle (*Aquila nipalensis*), was observed migrating northwards over the sites in March 2020. The sponsor consulted the Emirate Centre for the Conservation of the Houbara (ECCH) in Uzbekistan regarding the Asian Houbara (*Chlamydotis macqueenii*), and ECCH confirmed that there was potential for this species to occur in the area, although it was not observed during surveys completed in March and June 2020. There are no wetlands on the Project sites that might attract congregations of water birds. The only other threatened species confirmed on-site is the IUCN-declared Vulnerable Central Asian Tortoise (*Testudo horsfieldii*).

The ESIAs provide a short list of likely biodiversity-related mitigation measures (e.g., pre-construction survey, relocation of fauna, prohibition of hunting), as well as operational-phase monitoring of bird collisions with solar panels and overhead transmission lines. As specified in ESAP Action Item 22, the Companies will develop a construction-phase Biodiversity Management Plan (BMP) prior to commencement of civil works at each site that will be designed to achieve no net loss of Natural Habitat (as defined in PS6) and associated threatened species (including the Central Asian Tortoise), with appropriate mitigation measures to preserve the integrity of topsoil and existing natural vegetation on-site and restore natural vegetation in areas disturbed during construction.

As per ESAP Action Item 23, the Companies will develop operational-phase BMPs that include long-term management of Natural Habitat on-site and control of invasive alien species. The BMPs will also include biodiversity monitoring plans for the operational phase, including for monitoring of bird and tortoise fatalities, and to ensure no net loss of Natural Habitat.

As per ESAP Action Item 24, if the Companies install any new, or makes modifications to existing, overhead transmission lines, they will install and maintain bird flight deflectors for the life of the Project.

(j) Stakeholder Engagement

Stakeholder engagement activities undertaken for the Project thus far consist of the first round of stakeholder-engagement activities (from an E&S perspective), which took place in Samarkand and Jizzakh regions on 8-20 September 2020. This engagement was undertaken by a technical consultant appointed by IFC to provide advisory services, including the development of an E&S Scoping Study, as part of IFC's Uzbekistan Scaling Solar 2 Advisory Project.

Further stakeholder engagement activities took place on 16-19 September 2021 as part of preparation of the ESIAs. This engagement was undertaken by the Companies' ESIA consultant.

Finally, engagement with stakeholders during the LRP phase comprises the following exercises:

• Disclosure of the LRP preparation process to the community: 8-9 August 2022

• Socioeconomic census of households with leaseholds impacted by the Project: 29 August – 2 September 2022

• Stakeholder interviews and workshops: 22-23 September 2022

• Valuation meetings with households with assets or crops impacted by the Project: 15 September 2022 – Ongoing

This engagement consists of a series of face-to-face meetings with a range of stakeholders including:

- Affected stakeholders (including land users, i.e., farmers)
- Institutional stakeholders (public agencies concerned with any of the Project activities)
- Other interested stakeholders (mahalas, khokimiyats, national and international non-governmental organizations (NGOs) and other civil-society organizations).

The consultants reported during these rounds of engagement that Project-affected persons/farmers "are concerned about obtaining alternative land plots from the Administration and or the lost compensation." Further engagement activities with farmers are currently envisioned as part of the finalization of LRPs and as required during later phases of the Project. It is the Companies' understanding that the regional governments will be concluding the process of land compensation by mid-February 2023.

During consultations with herders, Mahalas, the Khokimiyat, and other community members in Jizzakh region, the following community support measures were discussed:

- Some shelters could be provided to the north of the Project site to allow herders and their herds to shelter overnight or just for rest periods in the day. This option will be discussed for future benefits once the Project is operational and access has been impacted.
- The road between three villages and the bridges located over the floodway could be improved. It will require significant expenditure; therefore, it could be considered as part of a wider community development program once the Project is in operation.
- Training is to be provided to all herders and other community members who wish to attend. The training will focus on animal veterinary health, including how to identify and treat common diseases and injuries.
- Another issue raised was the lack of mobile-phone signal at Jizzakh. Given that is likely that the Developer will need a mobile-phone signal for communication during Project construction, it would be beneficial to both the Project and the community to install a new mobile-phone mast in the area to boost signal. This has been standard practice in other Masdar projects, and it would be feasible to leave this mast in place for continued use by the community after construction.
  There is a general lack of employment in Jizzakh, particularly for women. There is limited seasonal farming work unlike in other areas of Uzbekistan. Training on meat, milk, and wool production is recommended to be conducted for women in affected households (herders and farmers).

During engagements with stakeholders in Samarkand in November 2022, further feedback was obtained from farmers along the OTL, herders and historical farmers about their preferences for livelihood restoration. Several options were discussed with each PAP. They agreed on the following:

• All historic farmers expressed an interest in both the land enhancement grant (with technical advice) and employment opportunities by the project. Historic farmers all expressed an interest in using their grant to buy sheep.

• Farmers along the OTL and their spouses expressed an interest in training on business and finance, increasing agricultural yields, veterinary training and also meat, milk and wool processing (land enhancement grants were not an option presented to farmers along the OTL due to the very small impact).

• Herders expressed an interest in access to the Feed Bank at the Solar PV Site and also the provision of materials for storage of feed.

Stakeholder Engagement Plans (SEPs) have been developed for each Project site. The SEPs set out the process for undertaking engagement and consultations with stakeholders, including definition of roles and responsibilities, outlining of the regulatory policy framework, and establishment of processes for stakeholder identification and analysis, stakeholder engagement, and information management, and grievance mechanisms.

The SEPs are live documents which will be periodically revised by the Project Companies in the course of the Project lifecycle as required. The current versions cover engagement activities to be undertaken during the development of the national Environmental Impact Assessment (EIA) and finalization of site-specific E&S Impact Assessments (ESIA) and LRPs.

As described under PS1, subsequent to the delivery of the ESIAs, the Companies will update and further revise the SEPs prior to commencement of the construction phase. The SEPs will be fully aligned with WB requirements. Due to the COVID-19 restrictions on face-to-face meetings, the SEPs should also comply with the World Bank's Technical Note: Public Consultations and Stakeholder Engagement in World Bank-supported Operations when there are constraints on conducting public meetings, dated March 20, 2020.

It should be noted that stakeholder engagement is to be understood as a continuous process, and it will be carried out by the Companies throughout the life cycle of the Project or until the Companies transfer ownership of the respective subproject to the public off-taker once the respective PPA expires. Stakeholder engagement responsibilities will not be transferred to contractors (EPC or O&M).

<u>Community Grievance Redress Mechanisms (GRMs)</u>

As described in the SEPs, community Grievance Redress Mechanisms (GRM) have been developed for the Project sites. The GRMs have been established as accessible and cost-free platforms that offer prompt and effective response to aggrieved entities in a manner that is transparent, culturally appropriate, free from manipulation, interference, coercion, intimidation, and retribution, and that are also open to judicial or administrative recourse. The GRMs will be implemented by two Community Liaison Officers (CLOs) who are fluent in Uzbek and Russian. As required under ESAP Action Item 19, the Companies will ensure that these grievance mechanisms are updated as required and adequate tools and resources are allocated to ensure that they are fully operational prior to commencing construction activities. The grievance mechanisms allow for complaints to be filed in several ways: by telephone; email; direct messages (text or telegram); online - by email and/or through an online form; complaints boxes at the Project site gates; through mahalla leaders; and during meetings with the CLOs.

The CGMs are aligned with PS1 and are robust, accessible, and anonymous for all manner of complaints that may come in from the community. The CGMs will receive and address complaints about GBV/SEA/SH between employees/contractors and members of the community. The GRMs will include arrangements to ensures confidentiality and, where possible, effective referrals to support services or the provision of support services. The GRMs must identify specific response methods for any complaints involving GBV/SEA/SH.

ENVIRONMENTAL AND SOCIAL ACTION PLAN (ESAP)

Performance Standards	Due Dates	Indicators for completion
PS1: Assessment and Management of Social and Environmental Risks and Issues:		
1. The Companies and their contractors will implement the Environmental and Social Management Systems (ESMSs) developed in accordance with the general requirements of PS1 and in line with the objectives of ISO14001 (certification is not required). The ESMSs identify, manage and control risks related to Project activities and its requirements will be applicable to any project sub-contractors.	Prior to Construction	Submit to Lenders the reports on the site specific ESMS implementation based on the adopted adequate policies, procedures, practices and management programs that are commensurate with the anticipated impacts and risks of the project.
2. The Companies will: (a) Develop a Construction Environmental & Social Management Plan (CESMP) for each site consistent with the site-specific ESIAs, PS and national requirements. The CESMPs will consist of a suite of sub-plans, which will include but not be limited to waste management; pollution prevention; water management; hazardous materials; biodiversity management; emergency response; community health, safety & security; road safety & traffic; worker accommodation; local labor force recruitment, labor influx management; SEA/SH prevention and response; cultural heritage (including chance finds); environmental monitoring; stakeholder engagement (including grievance management); contractor management. (b) Require their EPC contractors to develop, implement and maintain their own ESMPs and associated procedures aligned with project requirements and their scope of work.	Prior to Construction Prior to Construction	CESMPs & associated sub-plans submitted and acceptable to Lenders EPC ESMPs & associated sub-plans submitted and acceptable to Lenders
3. The Companies will: (c) Develop an Operations Environment & Social Management Plan (OESMP) for each site acceptable to Bank, which are consistent with the site specific ESIAs, World Bank Performance Standards in line with OP4.03 and national requirements. The OESMPs will consist of a suite of sub-plans, which will	Prior to Commercial Operations	OESMPs & associated sub-plans submitted and acceptable to Lenders

include but not be limited to: waste management; pollution prevention; water management; hazardous materials; emergency response; community health, safety & security; biodiversity management; environmental monitoring; stakeholder engagement (including grievance management). (d) Require their O&M contractors develop, implement and maintain their own ESHS MPs and associated procedures aligned with project requirements and their scope of work.	Prior to Commercial Operations	O&M Contractors ESHS MPs & associated sub-plans submitted and acceptable to Lenders
4. The Companies and their contractors will develop an Occupational Health & Safety (OHS) MS aligned (but not necessarily accredited) with OHSAS18001 / ISO 45001 and of a scale appropriate to their scope of work. The OHS MSs will identify, manage and control risks related to Project activities and its requirements will be applicable to any project sub-contractors.	Prior to Construction	Submit to Lenders the Project OHS MS documentation and evidence of the existence of adequate policies, procedures, practices and management programs that are commensurate with the anticipated impacts and risks of the project.
5. The Companies will: (a) Develop an OHS Management Plan for each project site, aligned with PS 2 and national requirements, which will outline minimum health & safety requirements for the project. (b) Ensure that their EPC & O&M contractors develop OHS Management Plans and set of sub-plans / procedures to ensure that all applicable health and safety legislation and relevant sections of the World Bank PS2 and the WBG General EHS Guidelines are met.	Prior to Construction Prior to Commercial Operations	OHS Management Plans submitted and acceptable to Lenders EPC OHS & associated sub-plans submitted and acceptable to Lenders
6. Develop and include EHS and labor provisions and compliance conditions in their EPC and O&M contracts which will provide contractors and third part service Project Management Consultant (PMC) providers (including security agencies) with clear guidelines on labor performance.	Prior to Construction	Companies to provide contract provisions to Lenders for review and approval

7. The Companies will (a) Establish an EHS&S Management Structure including an overall organizational chart detailing the Environmental, Social, Health and Safety teams, structure and relationship at the corporate and project site levels, including PMC, Contractors and sub-Contractors. At a corporate level, the client will appoint a qualified E&S and OHS representative to be the lenders focal point throughout the duration of the construction & operational phases of the project. (b) At a site level, appoint a suitably qualified stakeholder liaison manager. (c) Appoint through PMC suitably qualified project E&S and OHS manager, and human resource management support.	Prior to Construction	Provide an Overarching EHS&S Org Chart Provide Lenders with CVs & job descriptions for approval prior to hire
8. The Companies will require their PMC, EPC & O&M Contractors (including their sub-contractors) to appoint a suitably qualified environmental and OHS team to manage their scope of work. The team will have adequate qualifications & experience including the knowledge of international requirements & best practice.	Prior to Construction / Commercial Operations	Provide Lenders job descriptions and proof of hire.
9. The Companies will hire independent environmental & social consultants (IESC) acceptable to all lenders involved in the project. The scope of the IESC shall include review and approval of key E&S and OHS Company and contractor deliverables (CESMP, OESMP, etc.), and monitor efficiency of E&S management plans implementation quarterly during construction phase and annually in the first three years of the operations.	Prior to Construction	Engage IESC acceptable to Lenders. Provide IESC Monitoring Reports acceptable to Lenders
10. The Companies will ensure they have adequate proof of the voluntary return of land to the project by Project-Affected People	Condition of Disbursement	Provide Lenders documentation that land return is voluntary and legal.
PS2: Labor and Working Conditions		
11. The Companies will develop and implement project specific Human Resource Policies and	Within 90 days of Financial	Human Resource Policies and Procedures submitted and

Procedures, aligned with local labor codes, World Bank PS2 and International Labor Organization (ILO) requirements, for the Project which will contain an enforceable code of conduct applicable to all workers. These procedures must be made available to all employees in their language(s).	Agreement signature Throughout the project implementation	acceptable to Lenders.
The Companies will: (a) Supplement existing, publicly available, local workforce statistical data, by undertaking further stakeholder engagement to determine likely local & regional workforce availability, gender & skill levels to guide workforce planning. (b) The Companies will prepare a Local Recruitment & Employment Plan (LEP) for each site, aligned with local labor codes, World Bank PS2 and ILO requirements which outlines the projects local recruitment strategy and processes. The LEPs will outline the project's commitment to encourage & maximize local workers, vulnerable persons and women in the workforce including retention and promotion. They will include a tiered recruitment strategy aiming to maximize local, regional & national employment and will include a focus on preventing and responding to gender-based violence/sexual exploitation and abuse/sexual harassment (GBV/SEA/SH) for women construction workers, and in local communities affected by the project.	Condition of Disbursement Condition of Disbursement	Evidence of the completion of stakeholder engagement related to local & regional workforce availability Local Recruitment & Employment Plans submitted and acceptable to Lenders.
12. The Companies will ensure that all employee contracts are consistent with local labor codes, ILO and World Bank PS2 requirements. All employees shall be provided a copy of their contracts (in a language they understand). All construction phase worker contracts shall clearly describe the short-term nature of the project and provide an indication of likely employment duration. Ensure all workers have contracts and background checks including references from most recent employers.	Prior to Construction	Provide employee contract templates for Lender review

13. The Companies will: (a) Define and implement a confidential grievance reporting, referral and support systems for workers - Worker Grievance Mechanism (WGM) - in accordance with World Bank PS2. The WGM is to include specific considerations related to GBV/SEA/SH grievances. This should include multiple entry points to raise and address allegations including options to report anonymously if preferred. The WGM should be extended to all third-party workers performing activities for the project as required. WGM to be operated and management by the PMC during construction. During operations, the O&M contractors will establish and operate a WGM aligned with the Companies' MS (including training / awareness actions outlined below). (b) Provide specific trainings to key staff involved in the management of grievances (HR personal, grievance officers, senior management etc.) explaining the WGM and resolution process. Particular emphasis on identifying and appropriately managing harassment and bullying and GBV/SEA/SH (c) Provide general awareness to employees via. the site induction and ongoing toolbox talks on harassment and bullying. Engage with women employees on their concerns regarding transportation and safety. Deliver periodic mandatory training on GBV/SEA/SH to all workers, including contractors, subcontractors and core suppliers, as well as relevant consultants and clients. (d) Conduct mapping of local formal services (healthcare, counselling) and informal resources to support those who have experienced harassment.	Prior to Construction Training will be provided during site induction Trainings will be provided during site induction Prior to Construction	Workers Grievance Mechanisms submitted and acceptable to Lenders Evidence of specific gender related WGM trainings to be provided to Lenders Evidence of delivery of awareness trainings Summary of available local services
14. The EPC Contractors will develop a Workers Accommodation plan for each site in line with "Worker's accommodation: Process and	Prior to Construction	Workers' Accommodation Plans aligned with guidelines and including specific Covid-

Standards" Guidance Note by World Bank Performance Standards and EBRD. Consideration of the need for workers accommodation must take account of the Covid-19 pandemic and the potential impact of the construction workforce on the local communities. Conduct safety audits to identify settings affected by the project that might increase the risk of GBV/SEA/SH. For example, consider whether adequate measures have been put in place to manage interaction points with communities such as truck stops. Provide safe, secure and separate living spaces for male and female construction workers; including adequate lighting and segregated wash facilities		19 provisions submitted and acceptable to Lenders.
15. The Companies will require its contractors to develop and implement an 'Training Needs Analysis' and 'Training & Competency Plan' to ensure that all workers are appropriately trained, licensed / permitted & competent to undertake all tasks required of them within their role.	Prior to Construction	'Training Needs Analysis' & 'Training & Competency Plan' submitted and acceptable to Lenders
16. The Companies will develop and implement an 'E&S Supplier & Vendor Management Plan' for primary suppliers. The client will undertake supplier/vendor E&S risk assessments for primary suppliers and review potential supplier/vendor labor issues and risks including child labor, forced labor, working conditions etc. Include assessment of forced labor, gender and safety risks in bidding process for contractors.	Prior to Construction	E&S Supplier & Vendor Management Plans for primary suppliers submitted and acceptable to Lenders.
PS3: Resource Efficiency and Pollution Prevention		
17. The Companies will conduct an Alternatives Analysis for water use taking into account environmental, social and technical considerations and subsequently prepare a Water Management Plan for each site that will provide the best option for water supply during construction and operational phases in ways to avoid or minimize significant negative impacts on water supply for local community, farmers and headers	Prior to Construction	Water Management Plans submitted and acceptable to Lenders
18. The Companies and their contractors will develop site-specific Waste Management Plans and procedures for the construction and operational	Prior to Construction	Waste Management Plans submitted and acceptable to Lenders

phases for the project for safe handling, transport, and disposal of solid, liquid and hazardous material aligned with local legal requirements, World Bank PS3 and WBG EHS general guidelines. It will contain a waste inventory and will undertake an assessment of disposal & treatment facilities / options.		
PS4: Community Health, Safety and Security		
19. The Companies will: (a) Update site specific Stakeholder Engagement Plans (SEPs) (including community grievance mechanisms {CGMs}) prior to commencement of the construction phase and periodically as required throughout the Project duration. The SEPs will be fully aligned with the World Bank PS requirements including safe, confidential, and accessible grievance mechanisms for local communities. (b) Ensure that their contractors implement site-specific Community Health and Safety Management Plans (CHSMPs) and Emergency Preparedness and Response Plans (EPRPs) developed and adopted as part of the OESMP referred in ESAP 4. (c) Inform local stakeholders about the construction program, the major construction activities, and expected increase in traffic around the site. (d) Consult with local stakeholders and authorities regarding the numbers of workers that will be on site during the construction period, the potential for employment, worker accommodation, and worker welfare and health.	Prior to Construction	Stakeholder Engagement Plans submitted and acceptable to Lenders Evidence of engagement activities with local stakeholders provided to Lenders Report on the implementation status of CHSMPs and EPRPs as part of IESC Monitoring Reports acceptable to Lenders referred in ESAP 10. Evidence of engagement activities with local stakeholders provided to Lenders Evidence of engagement activities provided to Lenders
20. The Companies will develop Security MPs / Codes of Conduct for site security personnel which will be in line with the requirements of PS2, PS4 and the Voluntary Principles of Security & Human Rights. The EPC's security forces management plans will mirror that of the Company.	Prior to Construction	Security Management Plans submitted and acceptable to Lenders
PS5: Land Acquisition and Involuntary Resettlement		

21. The Companies will finalize and describe the measures to be taken by responsible government agencies to allocate replacement land plots and to compensate affected persons. If these measures do not meet the relevant requirements of this Performance Standard, the Company will develop a Resettlement Action Plan (RAP) to complement government actions. This may include additional compensation for lost assets, and additional efforts to restore lost livelihoods where applicable.	Prior to Construction	Provide Lenders with evidence that impacted land users have been provided with alternate land by the government. Develop additional RAP items as required following completion of a Gap Analysis
PS6 - Biodiversity Conservation and Sustainable Management of Living Natural Resources		
22. The Companies will develop a construction-phase Biodiversity Management Plan (BMP) that demonstrates no net loss of Natural Habitat and associated threatened species (including the Central Asian Tortoise), with appropriate mitigation measures to preserve the integrity of topsoil and existing natural vegetation on-site and restoration of natural vegetation in areas disturbed during construction.	Prior to Construction	Construction Phase Biodiversity Management Plan submitted and acceptable to Lenders.
23. The Companies will develop an operational-phase BMP that includes long-term management of Natural Habitat on-site and control of invasive alien species. The BMP should also include a monitoring plan for the operational phase, including monitoring of bird and tortoise fatalities and to determine no net loss of Natural Habitats.	Prior to Commercial Operations Date	Operations Phase Biodiversity Management Plan submitted and acceptable to Lenders.
24. If the Companies install any new, or makes modifications to, existing overhead transmission lines, they will install and maintain bird flight deflectors for the life of the project	Prior to Operations	Provide evidence of the installation of bird flight deflectors to Lenders

ANNEX 5: IMPLEMENTATION ARRANGEMENTS

UZBEKISTAN
Scaling Solar 2 Independent Power Producer (IPP) Project

The MoEF, MoE, and MIIT, on behalf of the GoU, have conducted a competitive bidding process for the selection of an investor to design, finance, construct, and operate the 440-MW Scaling Solar 2 IPP plants at two sites, Jizzakh and Samarkand. IFC Transaction Advisory Services advised the Government on structuring and tendering the Project. The Project was structured and tendered using the Scaling Solar Initiative approach — a WBG product consisting of a set of standardized documents which include an RFQ, Request for Proposal, Project Agreements, IFC financing term sheet, World Bank Guarantees term sheet, and MIGA insurance term sheet.

The Project is anticipated to follow customary arrangements for such private-sector projects and will be implemented through special-purpose companies (registered in Uzbekistan) that will have overall responsibility for the design, finance, supply, construction, testing, commissioning, and O&M of their respective power-plant assets for the duration of the PPP contractual agreements (PPAs and GSAs). Each such project company will set up appropriate management structures to undertake its respective project. The Project participants are experienced and familiar with Uzbekistan's and the Project's requirements, and, according to the Lenders' technical advisor, are expected to reduce the associated risks through the good observance of applicable norms, standards, and other relevant requirements pertaining to the Project.

The liquidity mechanisms under the Project will help the public off-taker, NEGU, meet its obligations under the PPAs to procure and provide payment securities to the IPP companies. It will provide the private IPP companies with facilities to manage any temporary interruptions in monthly PPA payments from the public off-taker. In the event the public off-taker is unable to make a monthly PPA invoice payment owed to one of the private IPP companies, the latter may draw a corresponding amount under the relevant L/C issued by Natixis (Step 1). The public off-taker first, and the Government second, are required to repay the amount drawn and reinstate the L/C at its original value (Step 2). In cases in which the public off-taker and the Government fail to reinstate the L/C at its original value within 12 months of the draw on the L/C, the L/C issuing bank, Natixis, will have recourse to the World Bank under the relevant World Bank guarantee for the outstanding L/C draw amounts and any interest accrued (Step 3). Finally, the Indemnity Agreement to be concluded between the GoU and the World Bank will oblige the GoU to reimburse the World Bank the amount paid out by the World Bank to the L/C-issuing bank under the guarantee (Step 4).

It is expected that the current implementation arrangements for the Project will be replicated for other IPP projects under IFC Advisory support in the near term. As power-sector reform in Uzbekistan continues, the institutional and implementation arrangements for future projects may differ from those of the Project. Any such changes will be assessed and included in subsequent guarantee projects.

The IPP companies, typically SPVs, will have primary responsibility for the financial management of the IPP projects. They will install and maintain adequate FM systems, including systems for accounting, reporting, auditing, and internal controls, and will employ relevantly qualified staff. The annual financial statements will be prepared in accordance with IFRS/IAS. In addition, they will be audited in accordance with ISA. The copy of authorized audit reports and Management Letters will be provided to the World Bank within six months after the end of the relevant reporting period. The financial management-related requirements of the World Bank Policy on Investment Project Financing do not apply to private-sector parties involved in a project supported by a Bank guarantee, such as this one.

The *World Bank Procurement Regulations for IPF Borrowers* do not apply to a project supported by a Bank guarantee, such as this one.

The current Uzbek legal framework for public procurement has been adopted only recently and is set out mainly in the following legislation: (a) Public Procurement Law (PPL), (b) Public Procurement Resolution No. 3550, (c) Regulation No. 3016, and (d) Decree of the President No. PP-4544. Before adoption of the PPL, there was no comprehensive regulation providing for thorough governance of public procurement procedures, which resulted in many problems, uncertainties, and abuses. Presently, with an established public procurement policy, the Government aims to eliminate unfair competition and increase the efficiency and transparency of the expenditures of public purchasers. The PPL and Regulations regulate the procurement of goods, works, and services carried out by any government agency, parastatal body, or any other body or unit established and mandated by the Government to carry out procurement (such entities referred to under the PPL as 'procuring entities') using public funds.

The PPL introduces two general categories of public purchasers, as follows: (a) state-budget-financed purchasers and (b) corporate purchasers, which include (i) SOEs; (ii) entities in which more than 50 percent of the shares are held by the state; and (iii) entities in which more than 50 percent of the shares are held by entities with more than 50 percent of the shares held by the state.

Among the above categories, the PPL determines a special category of public purchasers, the so-called 'strategic public purchasers', the list of which is approved under Strategic Purchasers Resolution No. 3487 and revised in Decree No. PP-4544. In accordance with the PPL (Article 1), the PPL does not govern procurement carried out by strategic public purchasers. Currently, the list of 'strategic public purchasers' includes NEGU, the 'Thermal Power Plants JSCs, and Regional Electric Networks JSC.

ANNEX 6: IMPLEMENTATION SUPPORT PLAN

UZBEKISTAN
Scaling Solar 2 Independent Power Producer (IPP) Project

Strategy and Approach for Implementation Support

1. The Implementation Support Plan described in Table 6 below explains how the World Bank team will supervise and monitor the proposed Project, the Project risks, and the Project indicators. It is also linked to the results identified in the Results Framework. Supervision arrangements will ensure adequate monitoring, evaluation of risks, and escalation to manage the risk of any call on the guarantees.

Implementation Support Plan and Resource Requirements

2. The level of technical support needed includes staff with energy-sector knowledge and expertise, specialized project-based guarantee project expertise (including legal counsel and financial experts), environment and social specialists, and power engineering as well as monitoring and evaluation expertise. The responsibility for this support lies with the energy-sector co-task team leaders and the guarantee-specialist co-task team leader with support from other experts. The main focus in terms of support during implementation is summarized below in Table 6.

Table 6. Implementation Support and Corporate Monitoring Plan

Time	Focus	Skills	Resource Estimate	Partner Role
Months 0–12	Effectiveness, financial closure, selection of L/C bank, environment and social, construction progress, and political developments.	• Sector • Guarantee/commercial • Financial • Legal • Environment and Social • Engineering • Country team	US$100,000	n.a.
Months 12–24	• Review of progress in construction and generation by the IPPs • Review of sector technical and financial performance • Review of environment	• Sector • Guarantee/commercial • Financial • Legal • Environment • Social • Monitoring and evaluation	US$50,000	n.a.

Time	Focus	Skills	Resource Estimate	Partner Role
	and social compliance • Review of progress of the sector and the IPPs • Review of status of completion against indicators and PDO			
Through end of guarantee effectiveness period	Corporate monitoring of legal covenants and risks that could lead to a possible call on any of the signed IBRD guarantees	• Sector • Guarantee/commercial • Financial • Legal	US$30,000 per year, including US$20,000 of staff cost and US$10,000 of travel (one trip of two staff per year).	

Table 7. Skills Mix Required

Skills Needed	Number of Staff Weeks (weeks per year)	Number of Trips	Comments
Team leader	3–5	2–3 per year	To be adjusted annually depending on available supervision budget
Energy specialist, co-task team leader	3–5	2–3 per year	
Guarantee specialist, co- task team leader	3–5	2–3 per year	
Energy specialist	3-5	Located in Uzbekistan	
Legal specialist	2–3	Depending on needs	
Financial analyst	1–2	2 per year	
Power engineer	1–2	1 per year	
Social	1–2	2 per year	
Environmental	1–2	Local staff	
Monitoring	1–2	1 per year	
Procurement	1–2	Local staff	
FM	1–2	Local staff	